

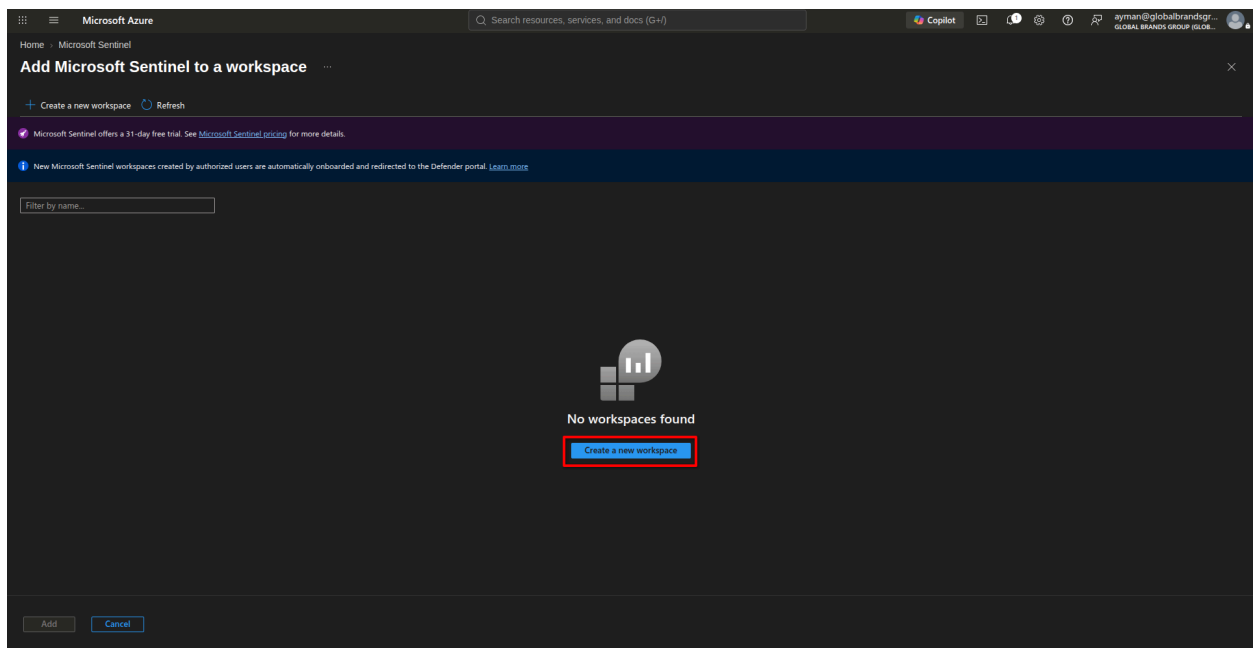
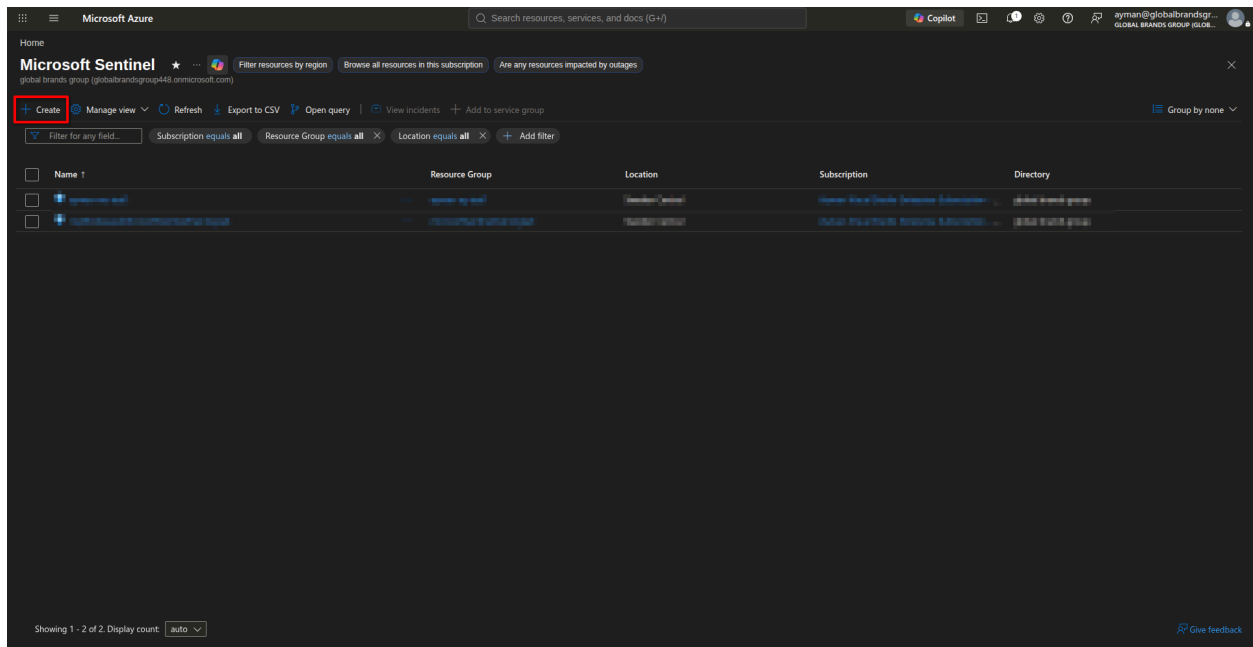


Microsoft Sentinel Use-Case Implementation Part 1

This is a build-order checklist for implementing the three use cases (Threat Activity, Unwanted Mail, Multiple Login Attempts) from the runbook. Work top to bottom — later steps assume earlier ones are done.

Phase 0: Environment Prerequisites (do once, before any use case)

- 1. Confirm/create the Sentinel workspace** — verify Log Analytics workspace exists and Sentinel is enabled on it.



Microsoft Azure

Home > Microsoft Sentinel > Add Microsoft Sentinel to a workspace

Create Log Analytics workspace

Basics Tags Review < Create

Basics

A Log Analytics workspace is the basic management unit of Azure Monitor Logs. There are specific considerations you should take when creating a new Log Analytics workspace. [Learn more](#)

With Azure Monitor Logs you can easily store, retain, and query data collected from your monitored resources in Azure and other environments for valuable insights. A Log Analytics workspace is the logical storage unit where your log data is collected and stored.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription  Ayman-Visual Studio Enterprise Subscription - MPN

Resource group  (New) rg-useCases

[Create new](#)

Instance details

Name  ws-useCases

Region  Sweden Central

[Review < Create](#) [Previous](#) [Next > Tags](#)

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Create Log Analytics workspace

Basics Tags Review < Create

Basics

 Log Analytics workspace by Microsoft

Basics

Subscription Ayman-Visual Studio Enterprise Subscription - MPN

Resource group rg-useCases

Name ws-useCases

Region Sweden Central

Pricing

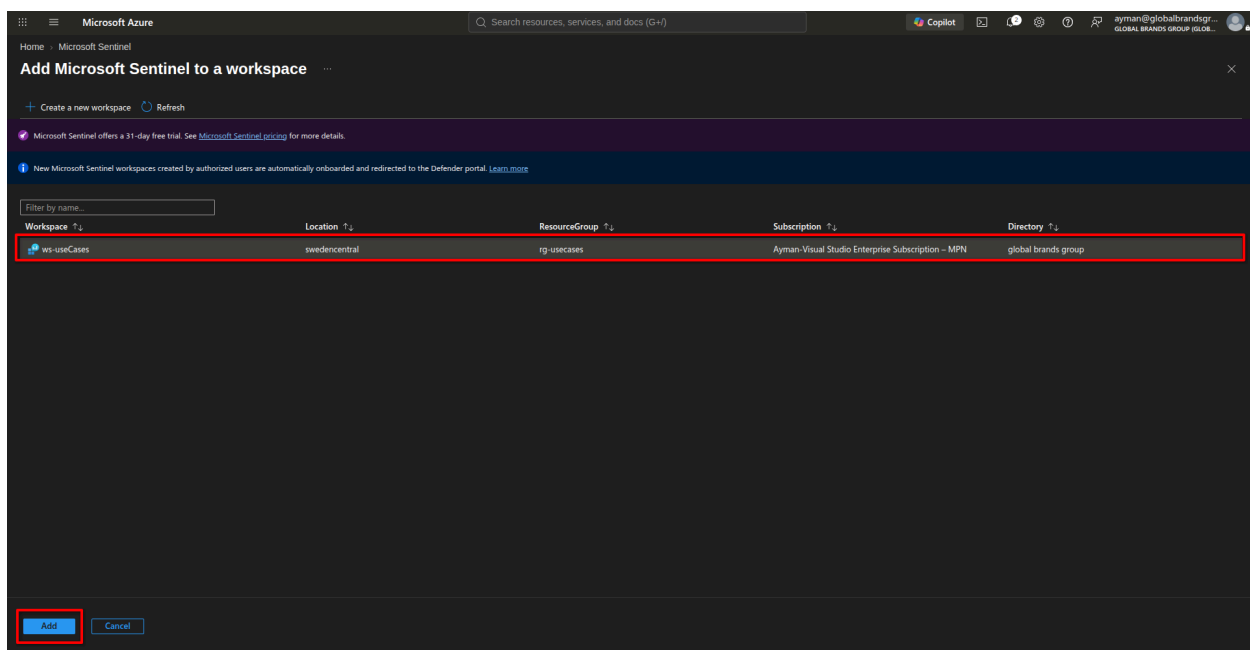
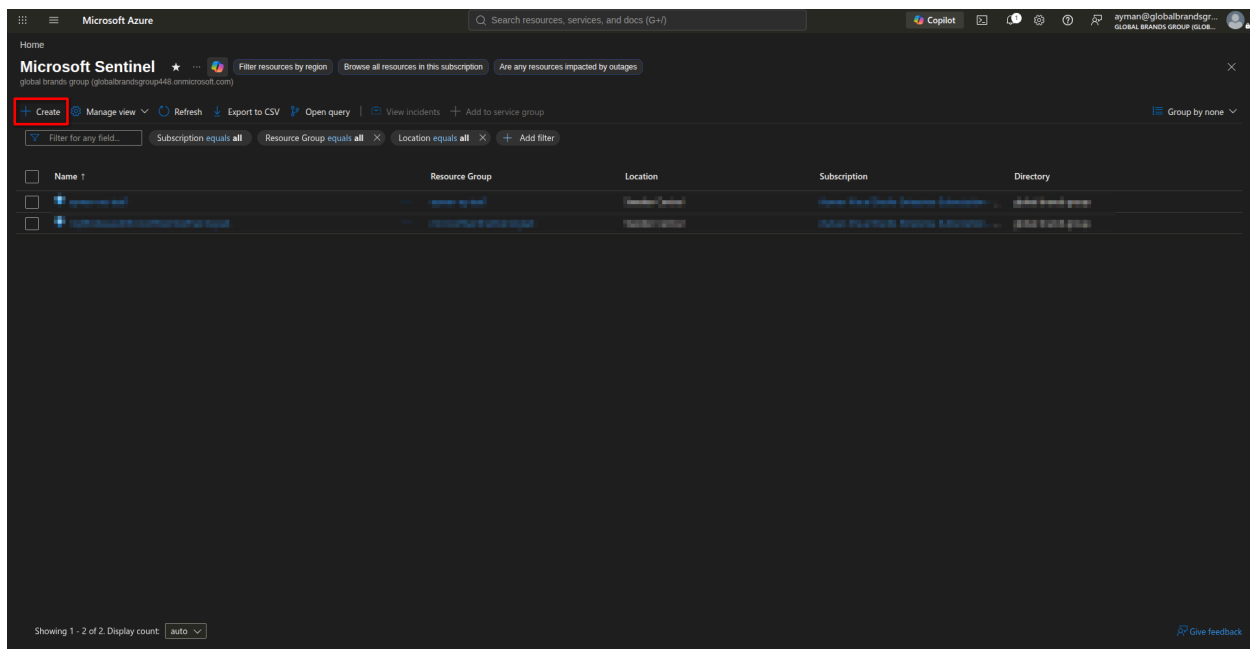
Pricing tier Pay-as-you-go (Per GB 2018)

The cost of your workspace depends on the volume of data ingested and how long it is retained. Regional pricing details are available on the [Azure Monitor pricing page](#). You can change to a different pricing tier after the workspace is created. [Learn more](#) about Log Analytics pricing models.

Tags

None

[Create](#) [Previous](#) [Download a template for automation](#)



1. Enable data connectors:

- Microsoft Entra ID (Sign-in logs + Audit logs)
 - In order to export Sign-in data, your organization needs Microsoft Entra ID P1 or P2 license.

Microsoft Defender

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Content hub

501 Solutions | 372 Standalone contents | 1 Installed | 0 Updates

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Search: Microsoft Entra ID

Status: All | Content type: All | Support: All | Provider: All | Category: All | Content sources: All

Install/Update | Delete

Content title	Status	Content source	Provider	Support
Microsoft Entra ID	Not installed	Solution	Microsoft	Microsoft
Microsoft Entra ID PowerShell accessing non E...	Not installed	Solution	Microsoft	Microsoft
Microsoft Entra ID Sign-in logs	Not installed	Solution	Microsoft	Microsoft
Microsoft Entra ID Audit logs	Not installed	Solution	Microsoft	Microsoft
Microsoft Entra ID Role Management Permissio...	Not installed	Solution	Microsoft	Microsoft
Distributed Password cracking attempts in Micr...	Not installed	Solution	Microsoft	Microsoft
Brute force attack against an Entra-authenticat...	Not installed	Solution	Microsoft	Microsoft
External guest invitation followed by Microsoft ...	Not installed	Solution	Microsoft	Microsoft

Previous | Page 1 of 1 | Next | Showing 1 to 18 of 18 results.

Microsoft Entra ID

Microsoft Provider | Support | Version 3.3.16

Description

Note: Please refer to the following before installing the solution:

- Review the solution [Release Notes](#)

The Microsoft Entra ID solution for Microsoft Sentinel enables you to ingest Microsoft Entra ID Audit, Sign-in, Provisioning, Risk Events and Risk User/Service Principal logs using Diagnostic Settings into Microsoft Sentinel.

Data Connectors: 1, Workbooks: 3, Analytic Rules: 73, Watchlists: 1, Playbooks: 11

[Learn more about Microsoft Sentinel](#) | [Learn more about Solutions](#)

Content type

- 73 Analytics rule
- 1 Data connector
- 11 Playbook
- 1 Watchlist
- 3 Workbook

Category

Identity, Security - Automation (SOAR)

Pricing

Free

Install | View details (5)

Microsoft Defender

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Connector details > Data connectors

Refresh | Guides | Workspace: ws-uscases

2 Connectors | 1 Connected | More content at Content Hub

Search by name or provider | Providers: Microsoft

Data Types: All | Status: All

Status	Connector name	Updates
Microsoft	Microsoft 365 Insider ...	...
Microsoft	Microsoft Entra ID	...

Microsoft Entra ID

Disconnected Status | Microsoft Provider | Last Log Received

Description

Gain insights into Microsoft Entra ID by connecting Audit and Sign-in logs to Microsoft Sentinel to gather insights around Microsoft Entra ID scenarios. You can learn about app usage, conditional access policies, legacy auth relate details using our Sign-in logs. You can get information on your Self Service Password Reset (SSPR) usage, Microsoft Entra ID Management activities like user, group, role, app management using our Audit logs table.

Last data received

Content source: Microsoft Entra ID | Version: 1.0.0

Author: Microsoft | Supported by: Microsoft Corporation | Email

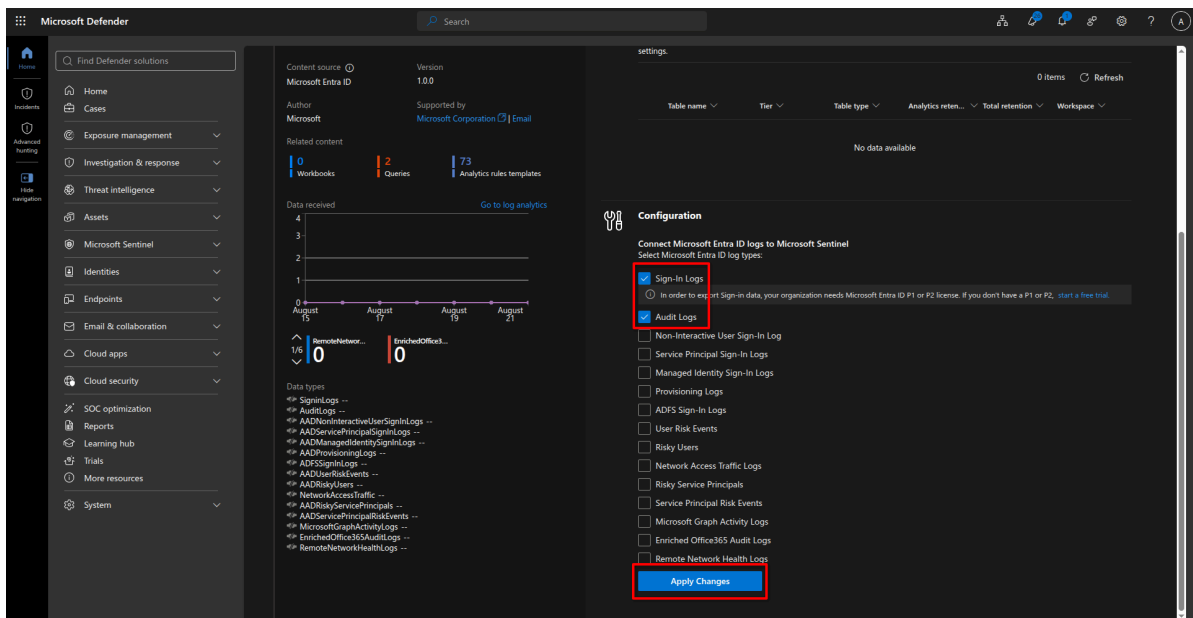
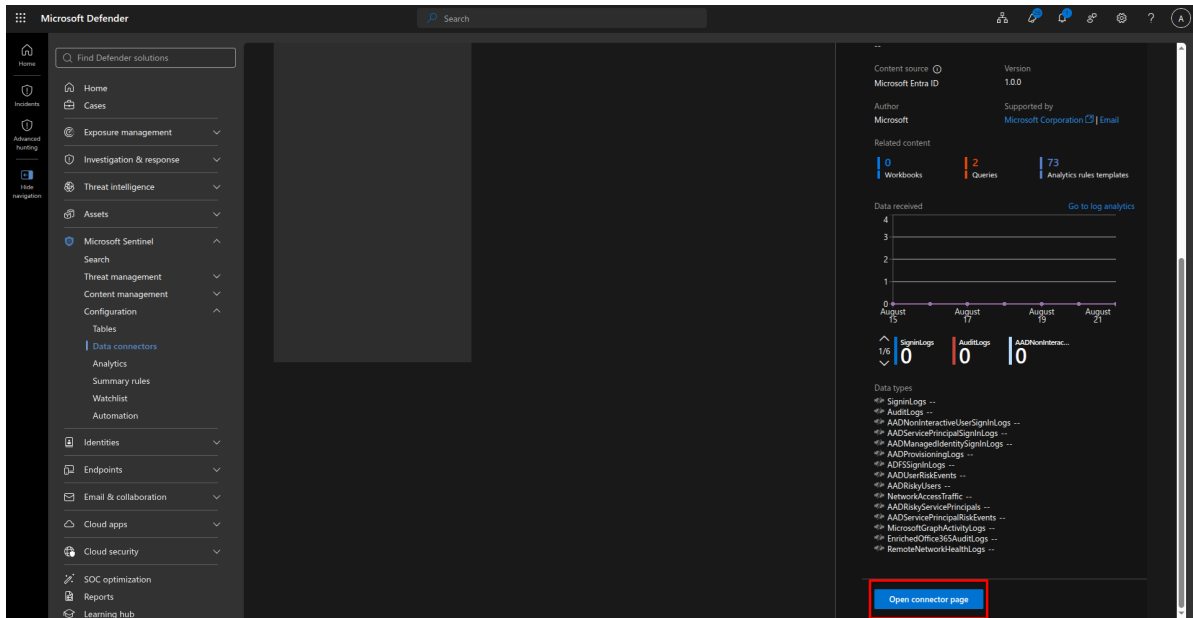
Related content

2 Workbooks | 2 Queries | 73 Analytics rules templates

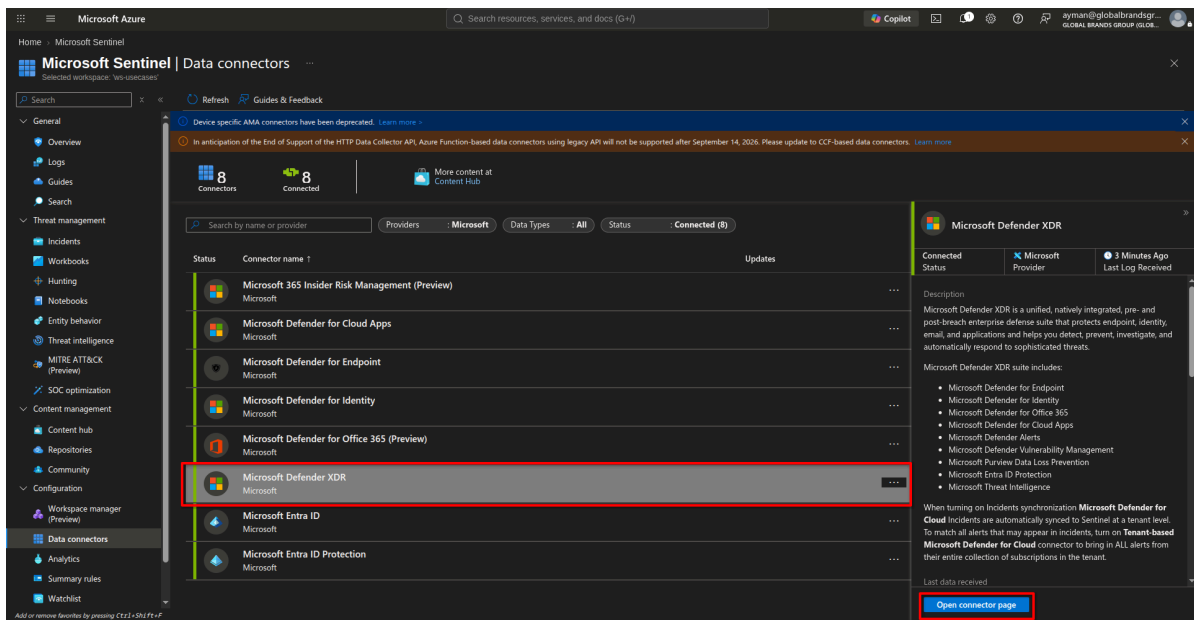
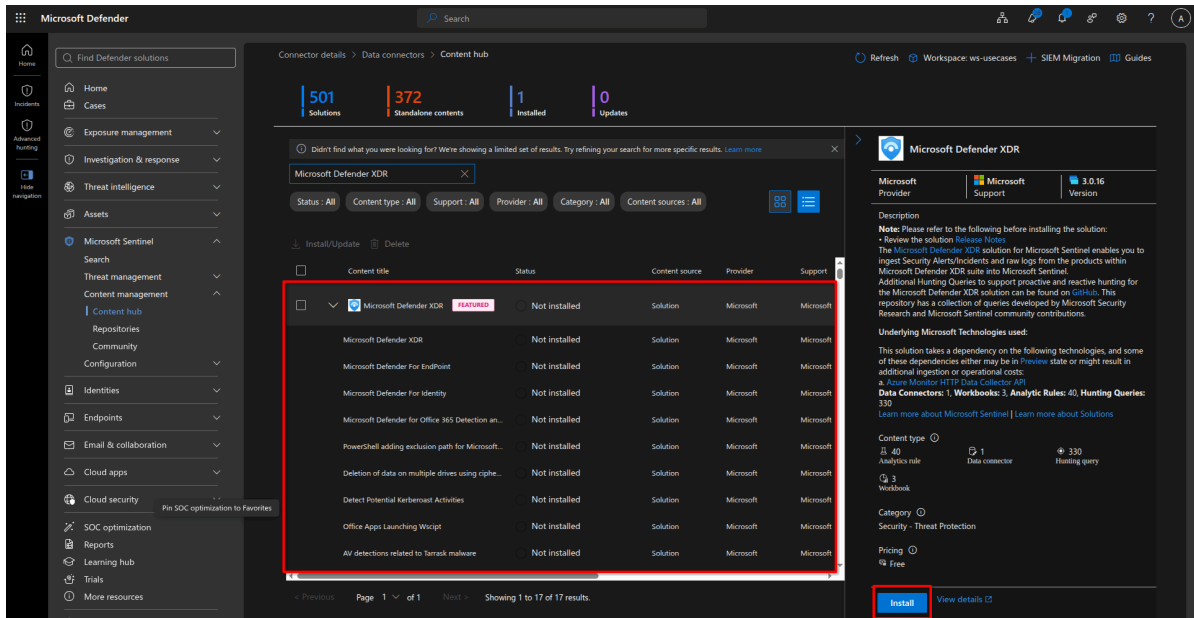
Data received

4
3
2
1

[Go to log analytics](#)



- Microsoft Defender XDR



Microsoft Azure

Home > Microsoft Sentinel | Data connectors

Microsoft Defender XDR

Connected Status **Microsoft** Provider 2 minutes ago Last Log Received

Description
Microsoft Defender XDR is a unified, natively integrated, pre- and post-breach enterprise defense suite that protects endpoint, identity, email, and applications and helps you detect, prevent, investigate, and automatically respond to sophisticated threats.

Microsoft Defender XDR suite includes:

- Microsoft Defender XDR
- Microsoft Defender for Endpoint
- Microsoft Defender for Identity
- Microsoft Defender for Office 365
- Microsoft Defender for Cloud Apps
- Microsoft Defender Alerts
- Microsoft Purview Data Loss Prevention
- Microsoft Entra ID Protection

When turning on Incidents synchronization **Microsoft Defender for Cloud** incidents are automatically synced to Sentinel at a tenant level. To match all alerts that may appear in incidents, turn on **Tenant-based Microsoft Defender for Cloud** connector to bring in ALL alerts from their entire collection of subscriptions in the tenant.

Last data received: 08/23/26, 09:24 AM

Content source: Microsoft Defender XDR Version: 1.0.0

Author: Microsoft Supported by: Microsoft Corporation | Email

Related content: 0 Workbooks, 4 Queries, 40 Analytics rules templates

Data received: 08/23/26, 09:24 AM

Instructions

Microsoft Defender for Endpoint (10/10 connected)

Name	Description
☒ DeviceInfo	Machine information (including OS information)
☒ DeviceNetworkInfo	Network properties of machines
☒ DeviceProcessEvents	Process creation and related events
☒ DeviceNetworkEvents	Network connection and related events
☒ DeviceFileEvents	File creation, modification, and other file system events
☒ DeviceRegistryEvents	Creation and modification of registry entries
☒ DeviceLoginEvents	Sign-ins and other authentication events
☒ DeviceImageLoadEvents	DLL loading events
☒ DeviceEvents	Additional events types

Microsoft Azure

Home > Microsoft Sentinel | Data connectors

Microsoft Defender XDR

Connected Status **Microsoft** Provider 2 minutes ago Last Log Received

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Last data received: 08/23/26, 09:24 AM

Content source: Microsoft Defender XDR Version: 1.0.0

Author: Microsoft Supported by: Microsoft Corporation | Email

Related content: 0 Workbooks, 4 Queries, 40 Analytics rules templates

Data received: 08/23/26, 09:24 AM

Instructions

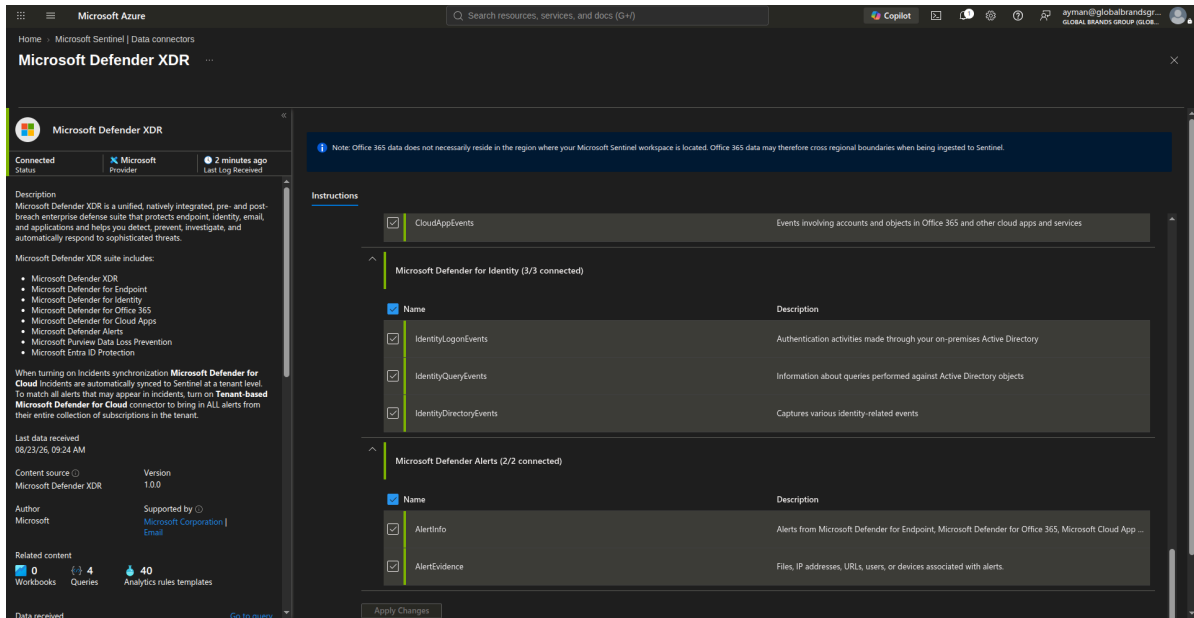
DeviceFileCertificateInfo

Name	Description
☒ EmailEvents	Office 365 email events, including email delivery and blocking events
☒ EmailUrlInfo	Information about URLs in Office 365 emails
☒ EmailAttachmentInfo	Information about files attached to Office 365 emails
☒ EmailPostDeliveryEvents	Security events that occur post-delivery, after Office 365 has delivered an email message to the recipient
☒ UrlClickEvents	Events involving URLs clicked, selected, or requested on Microsoft Defender for Office 365

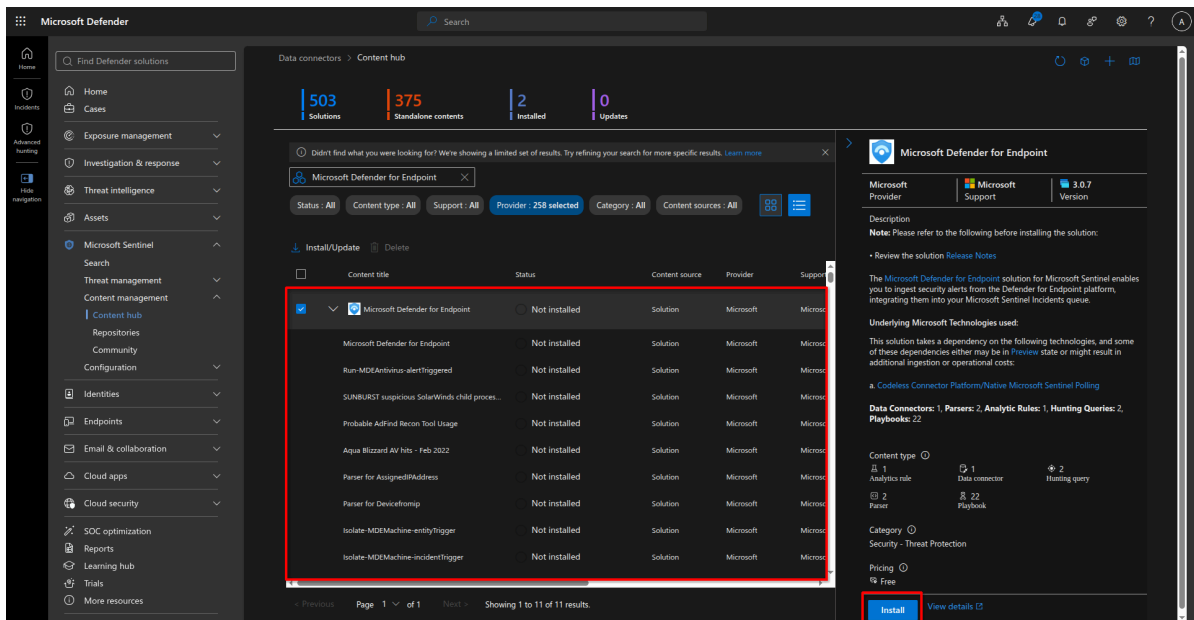
Microsoft Defender for Office 365 (5/5 connected)

Microsoft Defender for Cloud Apps (1/1 connected)

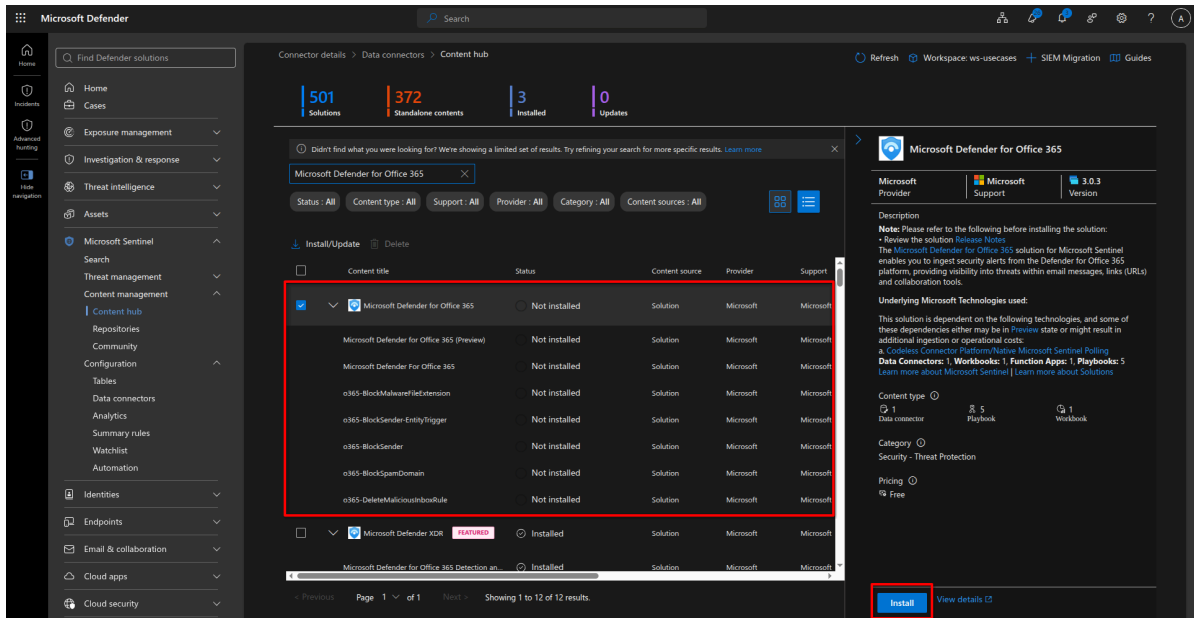
Name	Description
☒ CloudAppEvents	Events involving accounts and objects in Office 365 and other cloud apps and services



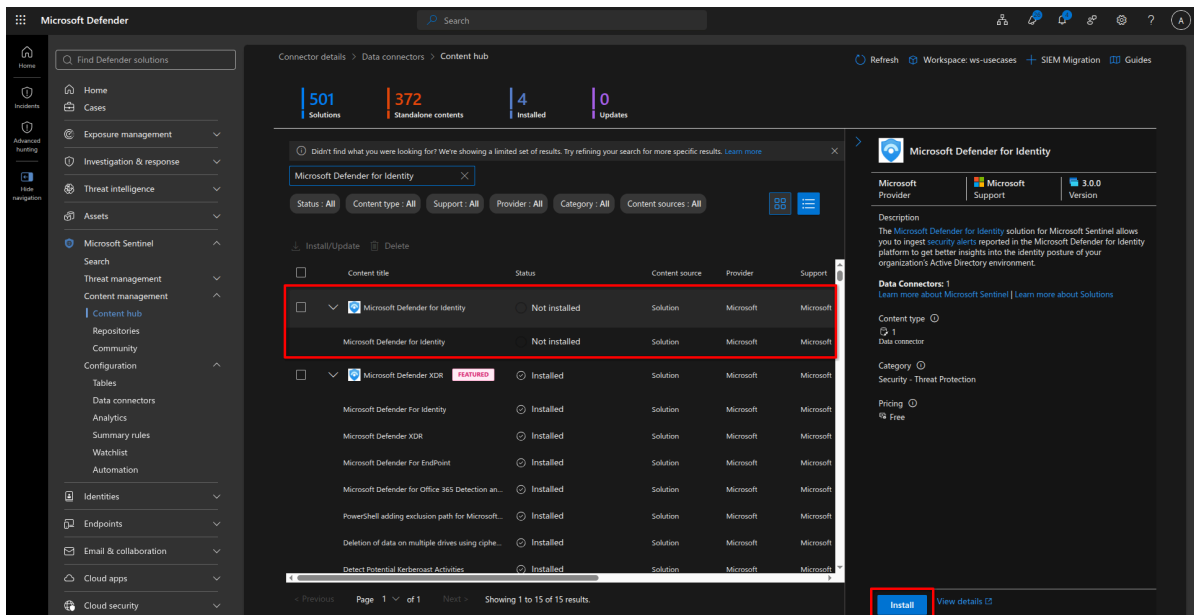
- Microsoft Defender for Endpoint



- Microsoft Defender for Office 365



- Microsoft Defender for Identity



2. **Validate entity mapping** — run a test alert from each connector and confirm **Account**, **Host**, **IP**, **URL**, **File**, and **Email** entities populate correctly on the incident.
3. **Create the automation identity** — set up a managed identity or automation account for Logic Apps. Scope permissions per-playbook (least privilege), not tenant-wide.

4. **Connect the ticketing system** (ServiceNow / Jira / internal ITSM) — confirm the Logic Apps connector or API credentials work with a test ticket.
 5. **Set up the notification channel** (email / Teams / webhook) — send a test message end-to-end.
 6. **Build the allowlist** — a maintained list (e.g., Sentinel watchlist) of approved admins, service accounts, scanners, VPN gateways, and security tooling. Every playbook will check against this before acting.
 7. **Define severity + approval policy** — document which actions require analyst approval (ransomware response, account disabling) vs. which can run fully automated.
 8. **Prepare test assets** — a test user, test device, a safe phishing simulation template, and a way to generate controlled sign-in failures.
-

Phase 1: Case 1 — Threat Activity (malware / spyware / suspicious connection / ransomware)

Step 1 — Detection

1. Confirm Defender for Endpoint + Defender XDR connectors are live.
 - a. I have a machine that's onboarded to MDE. Follow the steps in the Microsoft documentation: <https://learn.microsoft.com/en-us/defender-endpoint/onboarding>
2. Create/enable analytics rules for: high-confidence malware, ransomware, spyware, suspicious connection, endpoint detection alerts.
 - a. You can use these KQL queries from GitHub: <https://github.com/andranglin/RootGuard/blob/master/detection-engineering/threat-detection/attack-triage-playbooks-kql-triage/ransomware-detection-playbook.md>
 - b. I will test on ransomware.
 - c. Here is the KQL I will use:

```

let TimeWindow = 15m;
let RenameThreshold = 30;
let MinDistinctFolders = 3;
let RareExtensions = dynamic([".encrypted", ".locked", ".enc", ".crypt", ".crypted", ".locky", ".cerber", ".zzz", ".xyz", ".ryk", ".conti", ".lockbit", ".wcry", ".wncry"]);
DeviceFileEvents
| where Timestamp > ago(1h)
| where ActionType == "FileRenamed"
| where InitiatingProcessFileName !in~ (ExcludedProcesses)
| extend OldExt = tostring(split(PreviousFileName, ".")[-1])
| extend NewExt = tostring(split(FileName, ".")[-1])
| where NewExt != OldExt
| summarize RenameCount = count(), DistinctFolders = dcount(FolderPath), DistinctExtensionsTouched = dcount(OldExt), SampleFiles = make_set(FileName, 10), ExtensionsSeen = make_set(NewExt, 10), StartTimeUtc = min(Timestamp), EndTimeUtc = max(Timestamp) by DeviceName, DeviceId, InitiatingProcessAccountName, InitiatingProcessAccountSid, InitiatingProcessFileName, InitiatingProcessSHA256, bin(Timestamp, TimeWindow)
| where RenameCount >= RenameThreshold and DistinctFolders >= MinDistinctFolders
| extend IsKnownRansomExt = set_has_element(RareExtensions, tostring(ExtensionsSeen[0]))
| extend AccountName = InitiatingProcessAccountName
| extend SHA256Algorithm = "SHA256"
| project StartTimeUtc, EndTimeUtc, DeviceName, DeviceId, AccountName, InitiatingProcessAccountSid, InitiatingProcessFileName, InitiatingProcessSHA256, SHA256Algorithm, RenameCount, DistinctFolders, DistinctExtensionsTouched, ExtensionsSeen, SampleFiles, IsKnownRansomExt
| order by RenameCount desc

```


Description: Detects a high volume of file renames involving extension changes across multiple folders on a single device within a short time window - a strong behavioural indicator of ransomware encrypting a victim's files. Extension changes are compared old-to-new per rename (not a loose regex match) to reduce false positives from routine save/versioning operations, and the detection requires spread across multiple folders to distinguish mass encryption from a single app renaming its own working files. Covers only the encryption/impact stage of the attack chain (MITRE T1486) - deploy alongside persistence, lateral movement, and exfiltration detections for full kill-chain coverage.

d. and here is the ARM file to deploy it:

```
{
  "$schema": "https://schema.management.azure.com/schemas/2019-04-01/deploymentTemplate.json#",
  "contentVersion": "1.0.0.0",
  "parameters": {
    "workspace": {
      "type": "String"
    }
  },
  "resources": [
    {
      "id": "[concat(resourceId('Microsoft.OperationalInsights/workspaces/providers', parameters('workspace'), 'Microsoft.SecurityInsights'), '/alertRules/be346c45-098c-4569-9613-bf532bc09df6')]",
      "name": "[concat(parameters('workspace'), '/Microsoft.SecurityInsights/be346c45-098c-4569-9613-bf532bc09df6')]",
      "type": "Microsoft.OperationalInsights/workspaces/providers/alertRules",
      "kind": "Scheduled",
      "apiVersion": "2023-12-01-preview",
      "properties": {
```

```

        "displayName": "Ransomware - Mass File Extension Change Detected",
        "description": "Detects a high volume of file renames involving extension changes across multiple folders on a single device within a short time window - a strong behavioural indicator of ransomware encrypting a victim's files. Extension changes are compared old-to-new per rename (not a loose regex match) to reduce false positives from routine save/versioning operations, and the detection requires spread across multiple folders to distinguish mass encryption from a single app renaming its own working files. Covers only the encryption/impact stage of the attack chain (MITRE T1486) - deploy alongside persistence, lateral movement, and exfiltration detections for full kill-chain coverage.",
        "severity": "High",
        "enabled": true,
        "query": "let TimeWindow = 15m;\nlet RenameThreshold = 30;\nlet MinDistinctFolders = 3;\nlet ExcludedProcesses = dynamic([\"OneDrive.exe\", \"Dropbox.exe\", \"backup.exe\"]);\nlet RareExtensions = dynamic([\".encrypted\", \".locked\", \".enc\", \".crypt\", \".crypted\", \".locky\", \".cerber\", \".zzz\", \".xyz\", \".ryk\", \".conti\", \".lockbit\", \".wcry\", \".wncry\"]);\nDeviceFileEvents\n| where Timestamp > ago(1h)\n| where ActionType == \"FileRenamed\"\n| where InitiatingProcessFileName !in~ (ExcludedProcesses)\n| extend OldExt = tostring(split(PreviousFileName, \".\")[ -1 ])\n| extend NewExt = tostring(split(FileName, \".\")[ -1 ])\n| where NewExt != OldExt\n| summarize RenameCount = count(), DistinctFolders = dcount(FolderPath), DistinctExtensionsTouched = dcount(OldExt), SampleFiles = make_set(FileName, 10), ExtensionsSeen = make_set(NewExt, 10), StartTimeUtc = min(Timestamp), EndTimeUtc = max(Timestamp) by DeviceName, DeviceId, InitiatingProcessAccountName, InitiatingProcessAccountSid, InitiatingProcessFileName, InitiatingProcessSHA256, bin(Timestamp, TimeWind

```

```

ow)\n| where RenameCount >= RenameThreshold and DistinctFo
lders >= MinDistinctFolders\n| extend IsKnownRansomExt = s
et_has_element(RareExtensions, toString(ExtensionsSeen
[0]))\n| extend AccountName = InitiatingProcessAccountName
\n| extend SHA256Algorithm = \"SHA256\"\n| project StartTi
meUtc, EndTimeUtc, DeviceName, DeviceId, AccountName, Init
iatingProcessAccountSid, InitiatingProcessFileName, Initia
tingProcessSHA256, SHA256Algorithm, RenameCount, DistinctF
olders, DistinctExtensionsTouched, ExtensionsSeen, SampleF
iles, IsKnownRansomExt\n| order by RenameCount desc",
    "queryFrequency": "PT15M",
    "queryPeriod": "PT1H",
    "triggerOperator": "GreaterThan",
    "triggerThreshold": 0,
    "suppressionDuration": "PT6H",
    "suppressionEnabled": false,
    "startTimeUtc": null,
    "tactics": [
        "Impact"
    ],
    "techniques": [
        "T1486"
    ],
    "subTechniques": [],
    "alertRuleTemplateName": null,
    "incidentConfiguration": {
        "createIncident": true,
        "groupingConfiguration": {
            "enabled": true,
            "reopenClosedIncident": false,
            "lookbackDuration": "PT15M",
            "matchingMethod": "AllEntities",
            "groupByEntities": [
                "Host"
            ],
            "groupByAlertDetails": [],

```

```

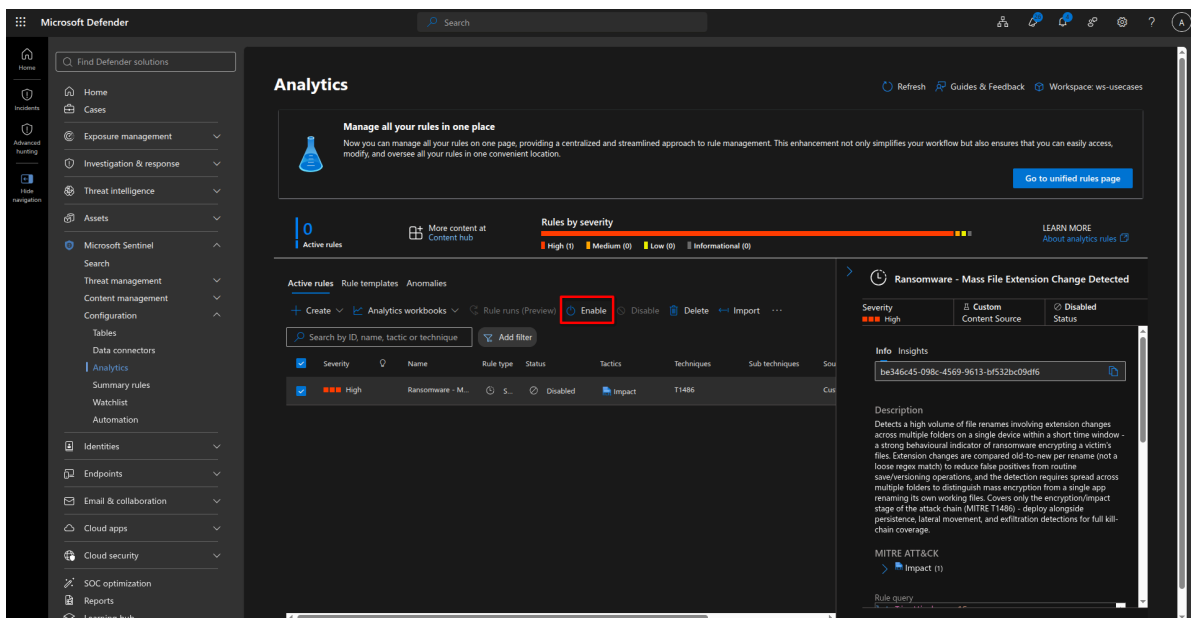
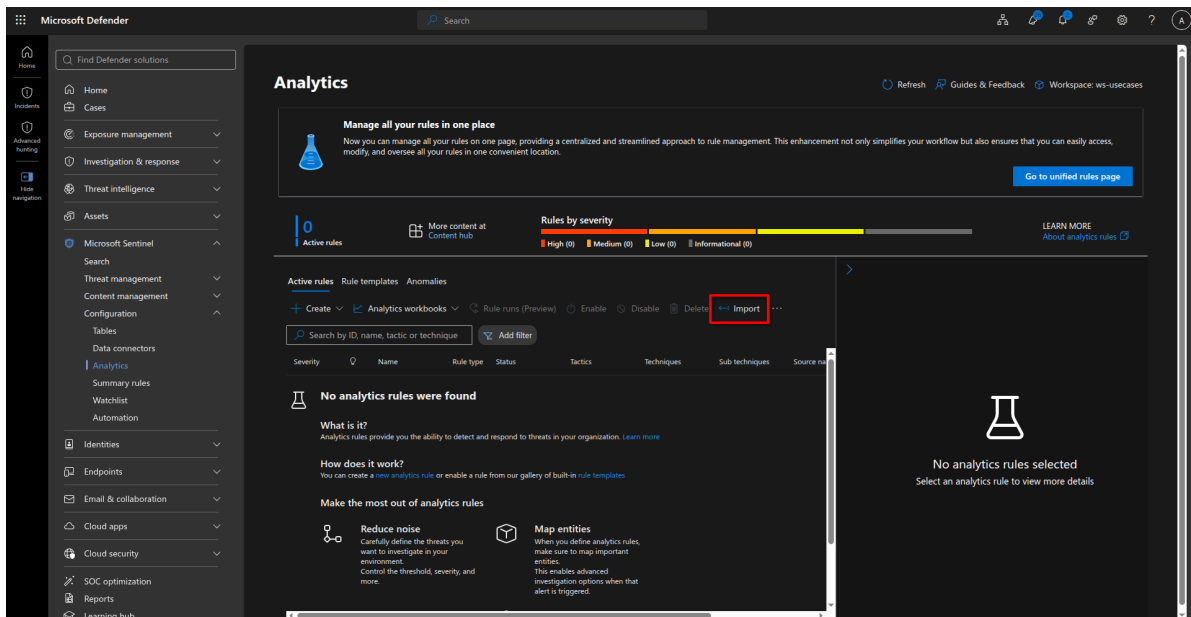
        "groupByCustomDetails": []
    },
    "eventGroupingSettings": {
        "aggregationKind": "SingleAlert"
    },
    "alertDetailsOverride": {
        "alertDisplayNameFormat": "Mass file rename on {{DeviceName}} by {{AccountName}}",
        "alertDescriptionFormat": "{{RenameCount}} files renamed across {{DistinctFolders}} folders on {{DeviceName}} within 15 minutes."
    },
    "customDetails": {
        "RenameCount": "RenameCount",
        "DistinctFolders": "DistinctFolders",
        "InitiatingProcess": "InitiatingProcessFileName",
        "KnownRansomExtension": "IsKnownRansomExt"
    },
    "entityMappings": [
        {
            "entityType": "Host",
            "fieldMappings": [
                {
                    "identifier": "FullName",
                    "columnName": "DeviceName"
                },
                {
                    "identifier": "AzureID",
                    "columnName": "DeviceId"
                }
            ]
        }
    ],
    {

```

```

        "entityType": "Account",
        "fieldMappings": [
            {
                "identifier": "Name",
                "columnName": "AccountNam
e"
            },
            {
                "identifier": "Sid",
                "columnName": "InitiatingP
rocessAccountSid"
            }
        ]
    },
    {
        "entityType": "FileHash",
        "fieldMappings": [
            {
                "identifier": "Algorithm",
                "columnName": "SHA256Algor
ithm"
            },
            {
                "identifier": "Value",
                "columnName": "InitiatingP
rocessSHA256"
            }
        ]
    }
],
"sentinelEntitiesMappings": null,
"templateVersion": null
}
}

```



You can find ransomware samples that this analytic rule is intended to detect in the MalwareBazaar signature database:

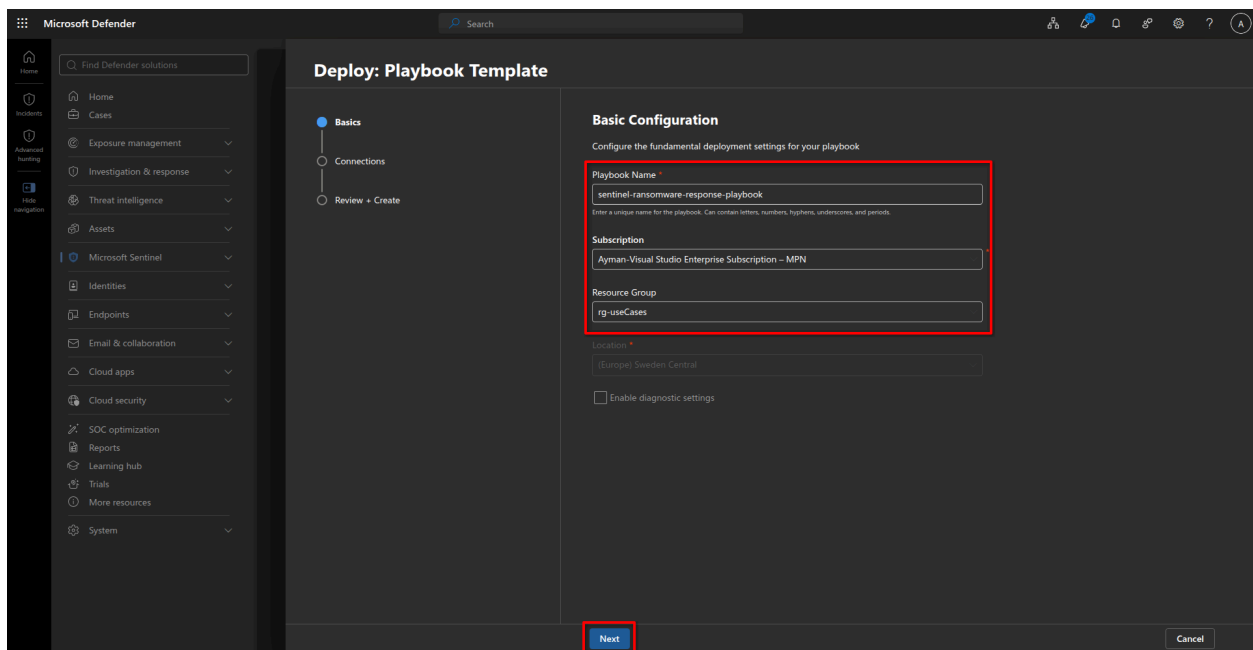
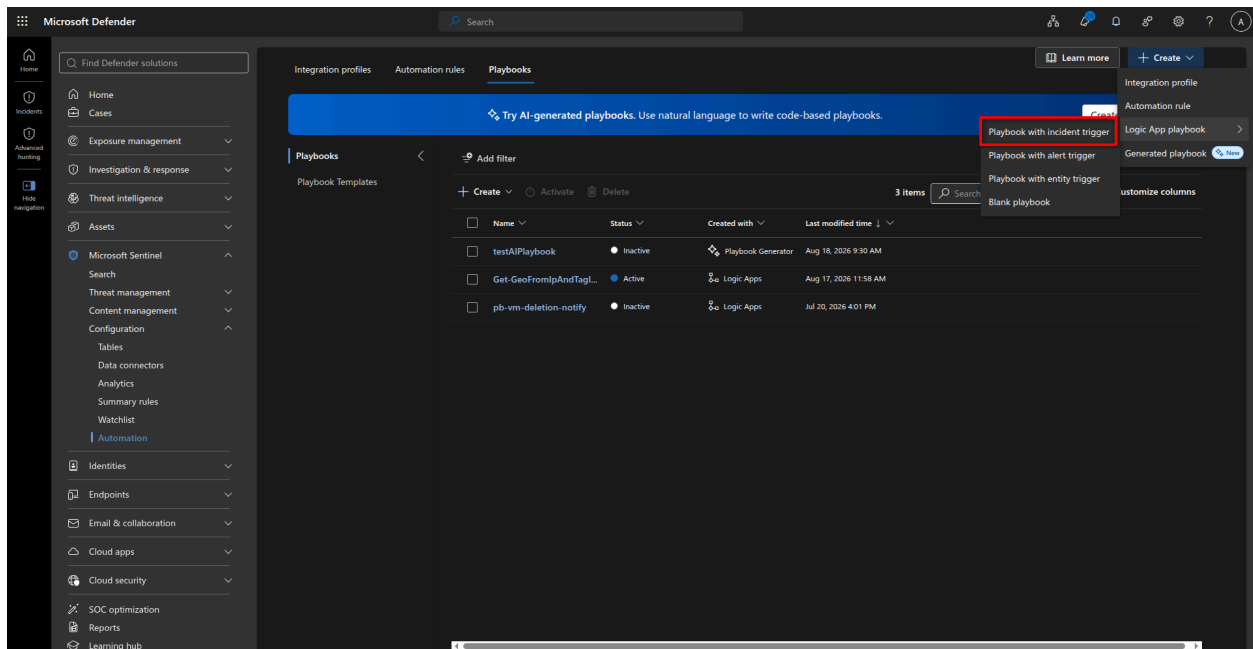
<https://bazaar.abuse.ch/browse/signature/LockBit/>

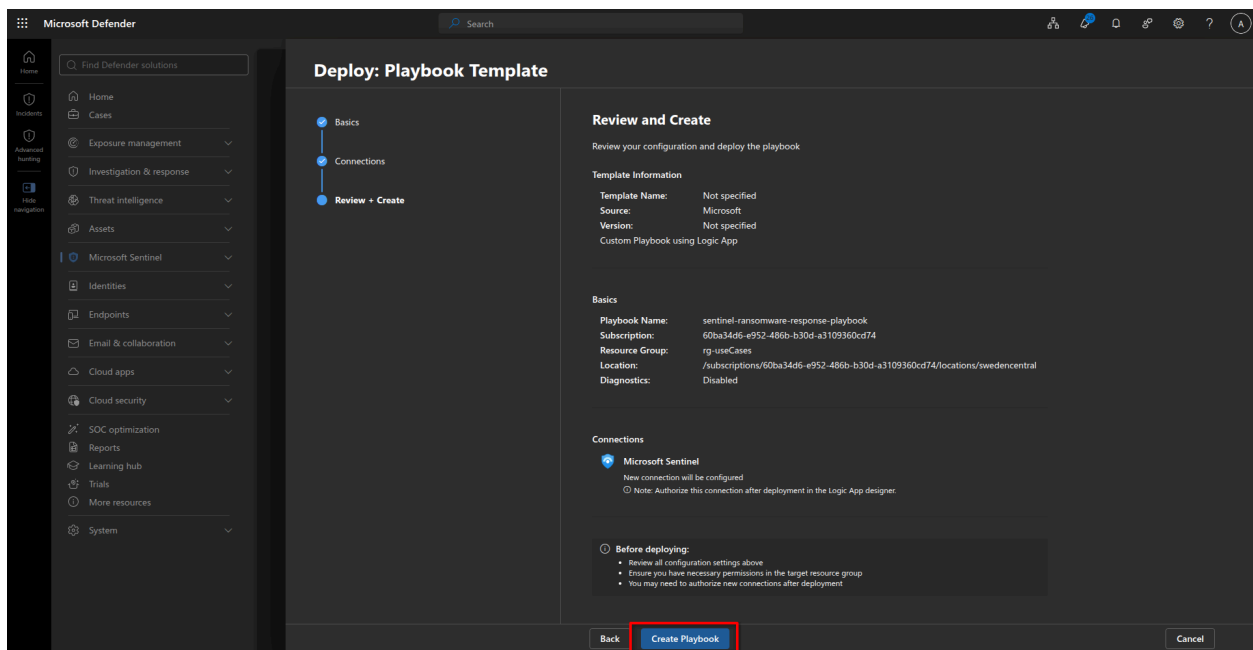
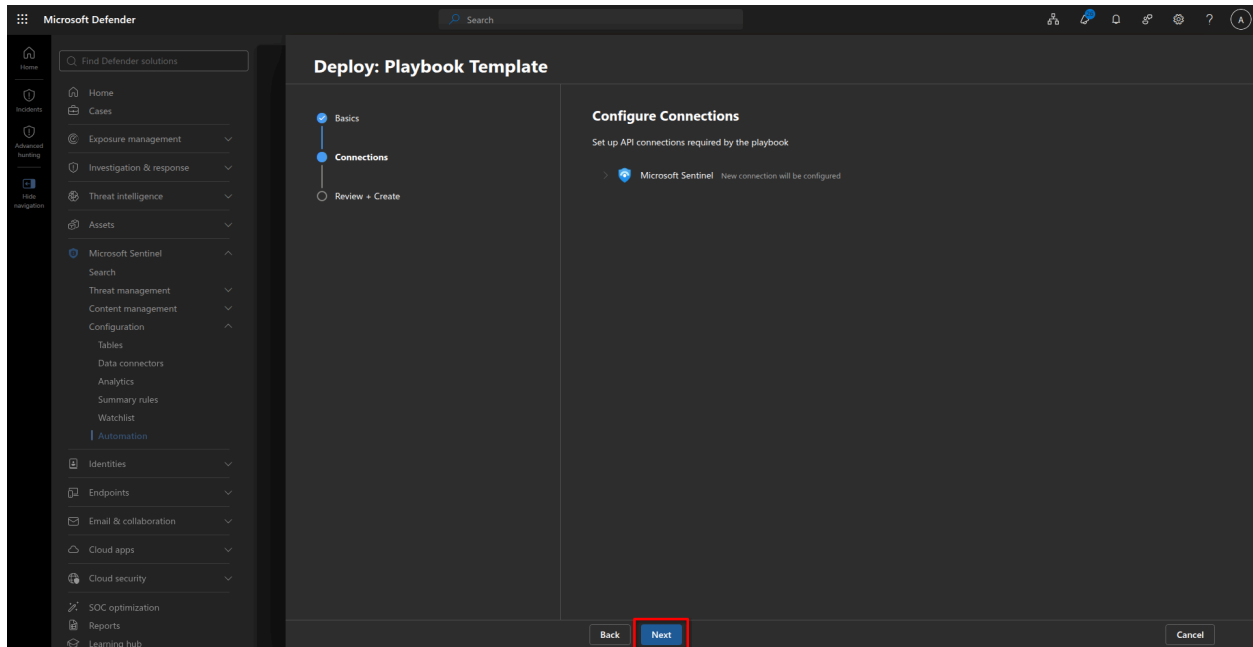
MALWARE bazaar				
Browse Upload Hunting Alerts Access Data Resources Login				
Malware Samples				
The table below shows all malware samples that have been identified by MalwareBazaar as lockbit (max 1000).				
Show 50 entries		Search:		
Firstseen (UTC)	SHA256 hash	Tags	Reporter	
2026-08-19 09:11:13	99b41a3ffe3e2d26ad47...	exe lockbit Ransom	Anonymous	
2026-02-05 15:43:26	47c6ba872eea70c5f923...	exe lockbit	adrian_luca	
2025-12-19 11:58:10	a61687dca6e71baa451...	BlackMatter exe lockbit	NDA0E	
2025-12-06 17:31:25	637be4d4df565ad9299b...	exe lockbit	juroots	
2025-12-01 12:07:20	032690f113289d50b672...	exe github-vitamp lockbit	skocherhan	
2025-11-09 08:51:23	bfe5a9f95eabd077326e...	lockbit Nebula NebulaDropper yes	Anonymous	
2025-09-20 01:27:29	4dc06ecee904b9165fa6...	elf linux lockbit lockbit 5.0 lockbit Ransomware	Lacortie1337	
2025-09-20 01:27:01	90b06f07eb75045ea3d4...	elf exe lockbit lockbit Ransomware	Lacortie1337	
2025-09-20 01:26:23	180e93a0918ab584a82...	chwingdong exe lockbit lockbit 5.0 lockbit signed	Lacortie1337	
2025-09-15 06:07:35	7ea5afbc166c4e23498a...	exe lockbit lockbit Ransomware signed	Neiki	
2025-06-25 08:04:17	25fba0e92a00184dde66...	exe lockbit lockbit lockbit Ransomware	TheRavenFile	
2025-06-25 08:02:33	7a42a2591858a6901ad...	exe lockbit lockbit lockbit Ransomware	TheRavenFile	
2025-06-25 08:01:22	f63f84ac24e0ae43cd5b...	exe lockbit lockbit lockbit Ransomware	TheRavenFile	
2025-06-25 08:00:12	33e5396c26b5f35264d...	exe lockbit lockbit lockbit Ransomware	TheRavenFile	
2025-06-21 23:20:57	392f4f64305a84bc145...	exe lockbit	Anonymous	
2025-06-08 06:50:51	771a06e3a1bdb42a724...	exe lockbit Ransomware	admIn_usa32	
2025-06-01 23:54:53	f734bc911b2662cc5fad...	exe lockbit Thor Ransomware	rhm0x	
2025-04-22 08:32:11	9a007e70a934bc61793...	exe lockbit	SecureInfoCom	
2025-04-22 06:33:45	7ed131e9f7d787b0c7...	exe lockbit	SecureInfoCom	

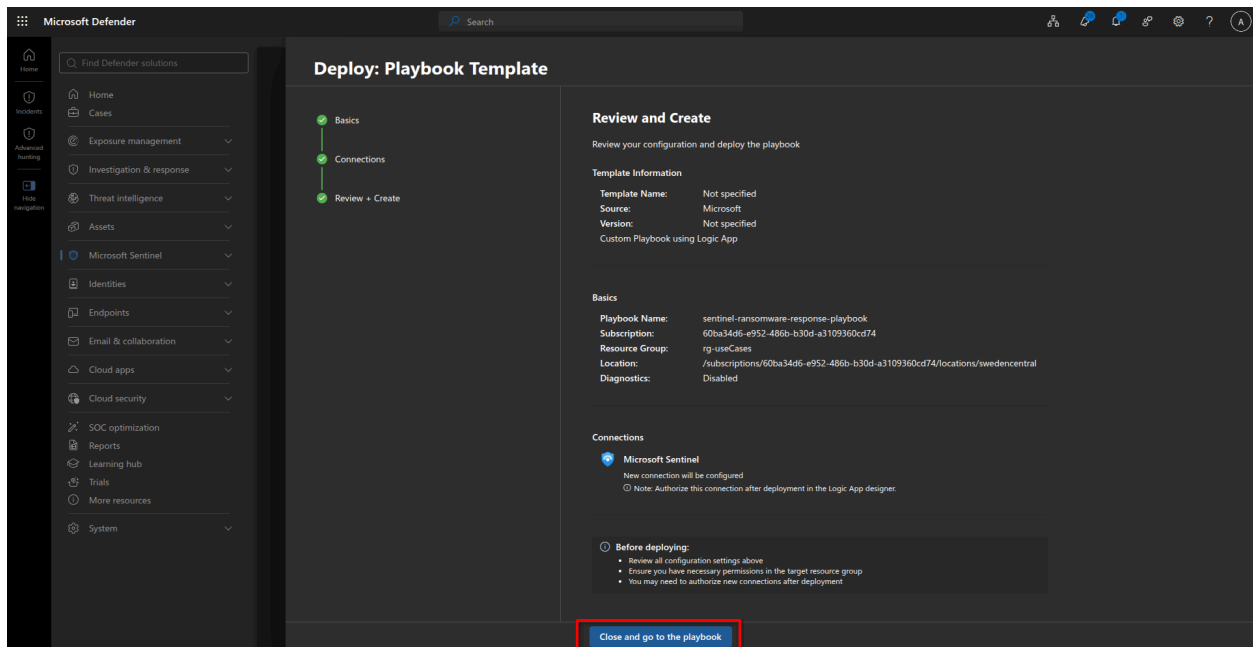
- Configure alert correlation so related alerts (same account/device/IP/alert ID) merge into one incident.
- Map device and user as incident entities in the rule.
- Set severity: **High** for confirmed malware, **Critical** for confirmed ransomware or active spread.
- Add allowlist exclusion condition (security testing, malware research, admin tooling) to the rule logic.

Step 2.1 — Build the playbook

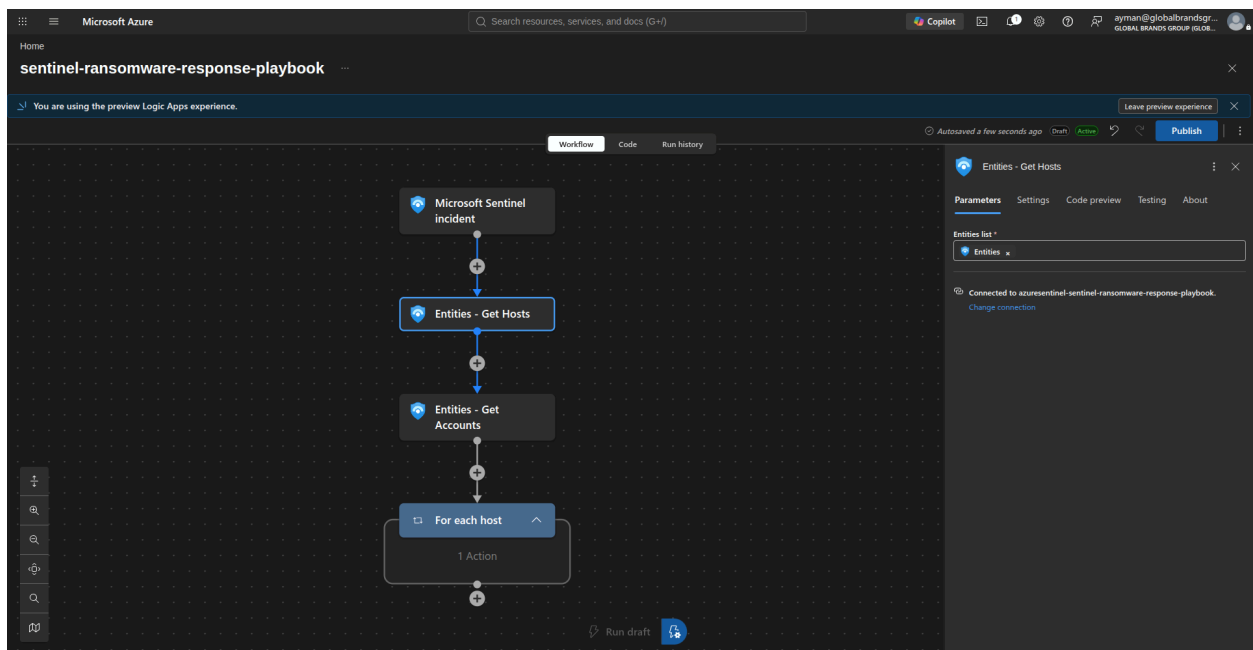
- Trigger: Sentinel incident created.

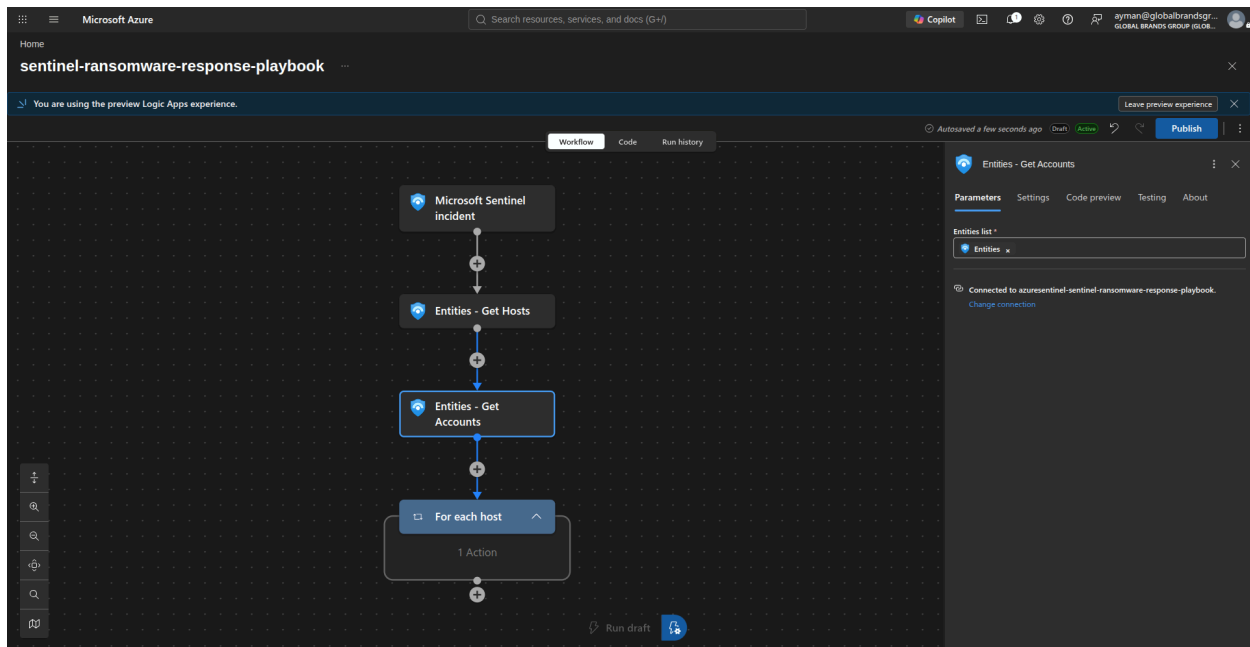






2. Parse incident entities → extract device ID, hostname, UPN, account ID, alert ID, file/hash.





Automated MDE Device Isolation Playbook

Small Story Before We Begin

Somewhere in GBG's network, 2:47 AM.

A workstation named `WKSTN01` starts behaving strangely. A process is encrypting files rapidly. Microsoft Sentinel detects the pattern and creates a new incident.

The playbook responds automatically. It identifies every host in the incident, resolves the Microsoft Defender for Endpoint device ID, isolates the device, starts a full antivirus scan, and records the result in the Sentinel incident.

Chapter 1: The Wake-Up Call: `Microsoft_Sentinel_incident`

This is the doorbell.

When Sentinel creates the incident, an automation rule calls the Logic App webhook. The Logic App receives the incident through `triggerBody()`, including:

```
Incident ID
Incident number
Related alerts
Related entities
```

The related entities can include machines, user accounts, and file hashes.

An automation rule can run immediately after incident creation. At that point, Sentinel may not yet have added the MDE device ID to every host entity.

Chapter 2: Reading the Case File

The device path starts with these actions:

```
Entities_-_Get_Hosts  
Entities_-_Get_Accounts  
Entities_-_Get_FileHashes
```

The incident's related entities arrive as one mixed collection. The Sentinel entity actions separate the collection into categories. This playbook uses the host results for device containment.

For example, Sentinel may return a host like this:

```
{  
  "HostName": "WKSTN01",  
  "azureID": "0d6de596704a3f93129ed07a57f55e08c5ba7a9d"  
}
```

The `azureID` value is used as an MDE `DeviceId` only when it is present and exactly 40 characters long.

Chapter 3: One Incident Can Contain Several Devices: `For_each_host`

An incident can contain one compromised machine or many. The `For_each_host` loop processes every host independently:

```
Host 1 -> resolve and contain if appropriate  
Host 2 -> resolve and contain if appropriate  
Host 3 -> resolve and contain if appropriate
```

Each host follows the same safety checks and containment sequence.

Chapter 4: First Choice: Does Sentinel Already Know the MDE Device ID?

Action:

```
Condition_-_MDE_Device_ID_Present
```

Defender for Endpoint isolates devices using its internal 40-character device identifier:

```
DeviceId: 0d6de596704a3f93129ed07a57f55e08c5ba7a9d
```

The condition checks:

```
Is azureID present?  
Is azureID exactly 40 characters long?
```

If both checks pass, the playbook uses the value directly. This is the preferred route because it avoids hostname matching.

Chapter 5a: Direct Containment Using Sentinel's Device ID

Action:

```
Isolate_Machine_By_MDE_Device_ID
```

The Logic App calls the Defender for Endpoint isolation API:

```
POST <https://api.securitycenter.microsoft.com/api/machines/{DeviceId}/isolate>
```

For example:

```
POST <https://api.securitycenter.microsoft.com/api/machines/0d6de596704a3f93129ed07a57f55e08c5ba7a9d/isolate>
```

The request body is:

```
{  
  "Comment": "Isolated automatically via Sentinel playbook du
```

```
e to incident #1234",  
  "IsolationType": "Full"  
}
```

Full isolation blocks normal network connectivity while preserving the Defender management path needed for investigation and response.

Chapter 5b: Start a Full Antivirus Scan

After isolation succeeds, the playbook runs:

```
Run_AV_Scan_By_MDE_Device_ID
```

The request is:

```
POST <https://api.securitycenter.microsoft.com/api/machines/{DeviceId}/runAntiVirusScan>
```

The request body is:

```
{  
  "Comment": "Full AV scan triggered automatically via Sentin  
el playbook due to incident #1234",  
  "ScanType": "Full"  
}
```

The scan action is chained after isolation. After the scan request succeeds, **Add_comment_to_incident_-_Host_Response** records that the machine was isolated and a full Defender Antivirus scan was started.

The successful HTTP action means Defender accepted the command. Isolation and scanning may continue asynchronously on the endpoint.

Chapter 6: The Automation-Rule Timing Problem

When an analyst manually runs the playbook from the Sentinel incident page, Sentinel has usually finished enriching the host entity. The **azureID** is often available, so direct isolation works.

When an automation rule runs immediately after incident creation, Sentinel may provide only:

```
HostName: WKSTN01
```

and not yet provide:

```
azureID: {MDE DeviceId}
```

Without the condition, the playbook could construct an unsafe URL such as:

```
POST <https://api.securitycenter.microsoft.com/api/machines//isolate>
```

`Condition_-_MDE_Device_ID_Present` prevents that request.

Chapter 7: Fallback Investigation: `Run_Microsoft_Graph_Hunting_Query`

If Sentinel does not provide a usable `azureID`, the playbook queries Microsoft Graph Advanced Hunting by hostname.

The KQL query is:

```
DeviceInfo
| where DeviceName =~ 'WKSTN01'
| where OnboardingStatus == 'Onboarded'
| summarize arg_max(Timestamp, *) by DeviceId
| top 1 by Timestamp desc
| project DeviceId, DeviceName, Timestamp
```

In plain language:

```
Find the newest onboarded Defender device named WKSTN01.
Return its MDE Device ID.
```

The action calls:

```
POST <https://graph.microsoft.com/v1.0/security/runHuntingQuery>
```

This fallback is useful when the incident host entity is incomplete during automation-rule execution.

Chapter 8: Reading the Hunting Result

Action:

```
Parse_Microsoft_Graph_Hunting_Response
```

Microsoft Graph returns a response similar to:

```
{
  "results": [
    {
      "DeviceId": "0d6de596704a3f93129ed07a57f55e08c5ba7a9d",
      "DeviceName": "WKSTN01",
      "Timestamp": "2026-08-25T11:17:55Z"
    }
  ]
}
```

The Parse JSON action makes the returned `DeviceId` available to the isolation and scan actions.

Chapter 9: Second Safety Check: `Condition_-_Microsoft_Graph_Machine_Found`

The playbook checks whether Advanced Hunting returned at least one result:

```
Device found?
-> Yes: isolate and scan it.
-> No: do not isolate anything.
```

This prevents a containment request from using an empty or unresolved device ID.

Chapter 10: Fallback Containment and Scan

When a device is found, the playbook uses:

```
Isolate_Machine_By_Microsoft_Graph_Device_ID  
Run_AV_Scan_By_Microsoft_Graph_Device_ID
```

The isolation request uses the device ID returned by Advanced Hunting:

```
POST <https://api.securitycenter.microsoft.com/api/machines/{ResolvedDeviceId}/isolate>
```

After isolation succeeds, the full antivirus scan starts:

```
POST <https://api.securitycenter.microsoft.com/api/machines/{ResolvedDeviceId}/runAntiVirusScan>
```

The scan is chained after isolation. After the scan request succeeds, `Add_comment_to_incident_-_Graph_Machine_Found` adds a Sentinel comment containing:

```
The host name  
The resolved MDE DeviceId  
Confirmation that isolation was requested  
Confirmation that a full Defender Antivirus scan was started
```

The incident comment is therefore added only after both Defender actions have returned success.

Chapter 11: No Device Found

If neither Sentinel's `azureID` nor Advanced Hunting provides an MDE device ID, the playbook does not attempt isolation.

Action:

```
Add_comment_to_incident_-_Device_ID_Not_Found
```

The incident receives a message similar to:

```
Could not resolve an MDE device ID for host WKSTN01.  
Manual isolation may be required.
```

This is an intentional fail-safe. It is better to require manual investigation than to send a containment request to an unknown device.

Final Story Flow

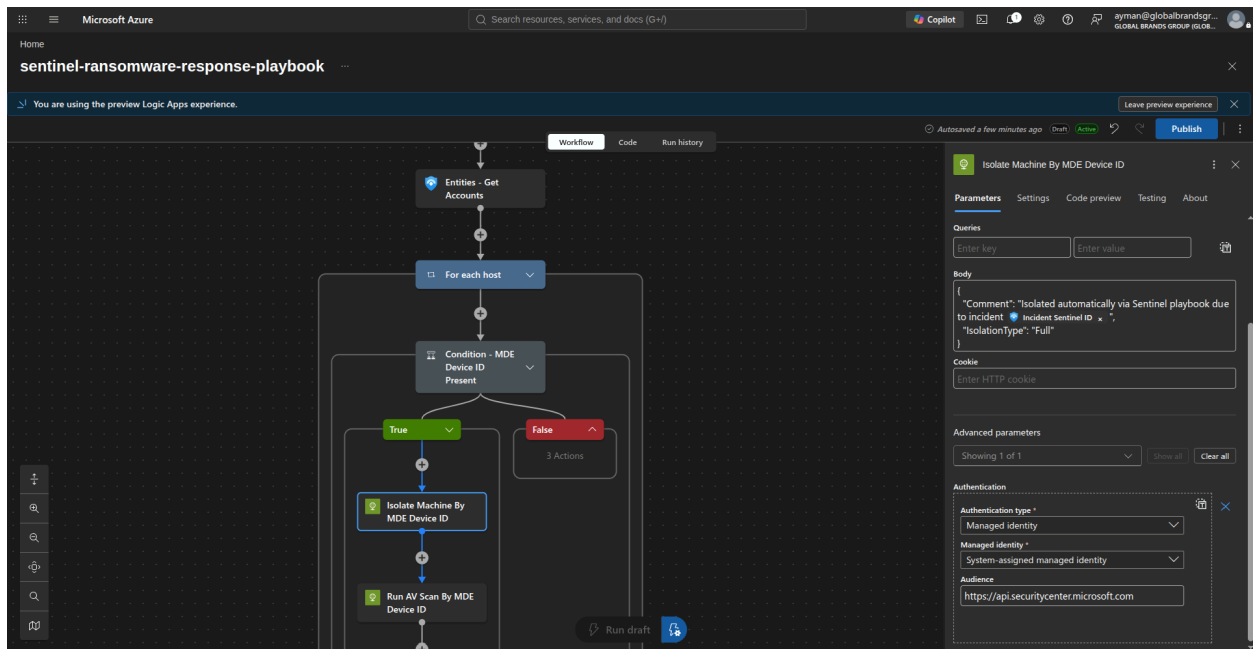
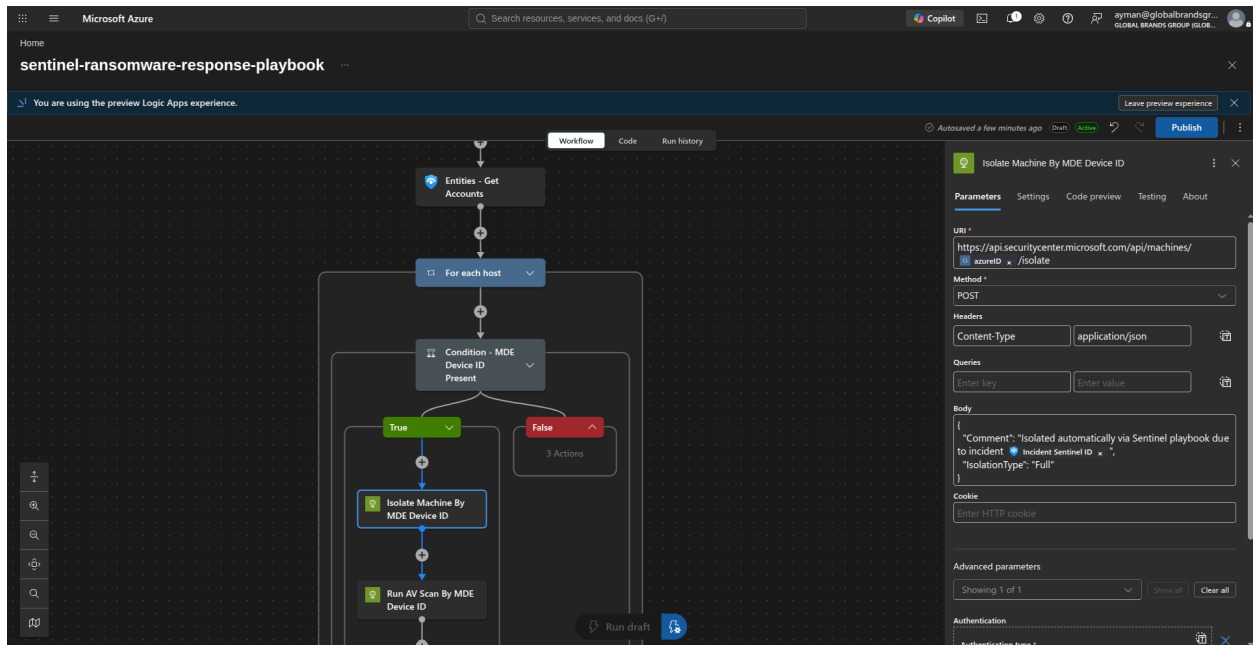
```
Sentinel incident is created  
-> Extract host entities  
-> For each host:  
  
    Does Sentinel provide a valid MDE Device ID in azureID?  
    -> Yes:  
        Isolate through Defender for Endpoint  
        Start a full antivirus scan  
        Add a Sentinel incident comment  
  
    -> No:  
        Query Microsoft Graph Advanced Hunting by hostname  
  
        Device found?  
        -> Yes:  
            Isolate using the returned MDE DeviceId  
            Start a full antivirus scan  
            Add a Sentinel incident comment with the resolved DeviceId  
  
        -> No:  
            Add a manual-investigation comment  
            Do not isolate anything
```

This supports both execution paths:

```
Manual run:  
    Sentinel commonly already provides azureID  
    -> Direct isolation and scan  
  
Automation-rule run:  
    azureID may be missing during incident creation  
    -> Microsoft Graph resolves DeviceId  
    -> Isolation and scan still occur automatically
```

3. Call Defender for Endpoint → isolate/quarantine device.

4. Start a Defender for Endpoint full antivirus scan.



Microsoft Azure

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sentinel-ransomware-response-playbook

You are using the preview Logic Apps experience.

Workflow Code Run history

For each host

Condition - MDE Device ID Present

True

Isolate Machine By MDE Device ID

Run AV Scan By MDE Device ID

Add comment to incident - Host Response

False 3 Actions

Run AV Scan By MDE Device ID

Parameters Settings Code preview Testing About

URI *

<https://api.securitycenter.microsoft.com/api/machines/{assetId}/runAntiVirusScan>

Method *

POST

Headers

Content-Type application/json

Queries

Enter key Enter value

Body

```
{
  "Comment": "Full AV scan triggered automatically via Sentinel playbook due to incident {Incident Sentinel ID}",
  "ScanType": "Full"
}
```

Cookie

Enter HTTP cookie

Advanced parameters

Showing 1 of 1 Show all Clear all

Authentication

Authentication type *

Microsoft Azure

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Workflow Code Run history

For each host

Condition - MDE Device ID Present

True

Isolate Machine By MDE Device ID

Run AV Scan By MDE Device ID

Add comment to incident - Host Response

False 3 Actions

Run AV Scan By MDE Device ID

Parameters Settings Code preview Testing About

Queries

Enter key Enter value

Body

```
{
  "Comment": "Full AV scan triggered automatically via Sentinel playbook due to incident {Incident Sentinel ID}",
  "ScanType": "Full"
}
```

Cookie

Enter HTTP cookie

Advanced parameters

Showing 1 of 1 Show all Clear all

Authentication

Authentication type *

Managed identity *

System-assigned managed identity

Audience

<https://api.securitycenter.microsoft.com>

Microsoft Azure

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You are using the preview Logic Apps experience.

Workflow Code Run history

For each host

Condition - MDE Device ID Present

True

Isolate Machine By MDE Device ID

Run AV Scan By MDE Device ID

Add comment to incident - Host Response

False

3 Actions

Run draft

Add comment to incident - Host Response

Parameters Settings Code preview Testing About

Incident ARM id *

Incident comment message *

Machine Hosts Hostname was isolated and a full Defender Antivirus scan was started.

Connected to azuresentinel-sentinel-ransomware-response-playbook. Change connection

Microsoft Azure

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sentinel-ransomware-response-playbook

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Workflow Code Run history

For each host

Condition - MDE Device ID Present

True

Isolate Machine By MDE Device ID

Run AV Scan By MDE Device ID

Add comment to incident - Host Response

False

Run Microsoft Graph Hunting Query

Parse Microsoft Graph Hunting Response

Condition - Microsoft Graph Machine Found

Run draft

Run Microsoft Graph Hunting Query

Parameters Settings Code preview Testing About

URI *

https://graph.microsoft.com/v1.0/security/runHuntingQuery

Method *

POST

Headers

Content-Type application/json

Queries

Enter key Enter value

Body

{ "Query": "concat(...)" }

Cookie

Enter HTTP cookie

Advanced parameters

Showing 1 of 1 Show all Clear all

Authentication

Authentication type *

Managed identity

Managed identity *

Microsoft Azure

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Workflow Code Run history

For each host

Condition - MDE Device ID Present

True

Isolate Machine By MDE Device ID

Run AV Scan By MDE Device ID

Add comment to incident - Host Response

False

Run Microsoft Graph Hunting Query

Parse Microsoft Graph Hunting Response

Condition - Microsoft Graph Machine Found

Run draft

Run Microsoft Graph Hunting Query

Parameters Settings Code preview Testing About

Content-Type application/json

Queries

Enter key Enter value

Body

```
{
  "Query": "[ concat(' ' ) ]"
}
```

Cookie

Enter HTTP cookie

Advanced parameters

Showing 1 of 1 Show all Clear all

Authentication

Authentication type * Managed identity

Managed identity * System-assigned managed identity

Audience https://graph.microsoft.com

Microsoft Azure

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You are using the preview Logic Apps experience.

Workflow Code Run history

For each host

Condition - MDE Device ID Present

True

Isolate Machine By MDE Device ID

Run AV Scan By MDE Device ID

Add comment to incident - Host Response

False

Run Microsoft Graph Hunting Query

Parse Microsoft Graph Hunting Response

Condition - Microsoft Graph Machine Found

Run draft

Parse Microsoft Graph Hunting Response

Parameters Settings Code preview Testing About

Content * Body

Schema *

```
{
  "type": "object",
  "properties": {
    "results": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "deviceId": {
            "type": "string"
          },
          "deviceName": {
            "type": "string"
          },
          "timestamp": {
            "type": "string"
          }
        }
      }
    }
  }
}
```

Use sample payload to generate schema

Microsoft Azure

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You are using the preview Logic Apps experience.

Workflow Code Run history

Run AV Scan By MDE Device ID

Add comment to incident - Host Response

Parse Microsoft Graph Hunting Response

Condition - Microsoft Graph Machine Found

True

Isolate Machine By Microsoft Graph Device ID

Run AV Scan By Microsoft Graph Device ID

False

1 Action

Run draft incident - urapn

Isolate Machine By Microsoft Graph Device ID

Parameters Settings Code preview Testing About

URI *

<https://api.securitycenter.microsoft.com/api/machines/>

Method *

POST

Headers

Content-Type application/json

Queries

Enter key Enter value

Body

```
{
  "Comment": "Isolated automatically via Sentinel playbook due to incident ",
  "IsolationType": "Full"
}
```

Cookie

Enter HTTP cookie

Advanced parameters

Showing 1 of 1 Show all Clear all

Authentication

Authentication type *

Microsoft Azure

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You are using the preview Logic Apps experience.

Workflow Code Run history

Run AV Scan By MDE Device ID

Add comment to incident - Host Response

Parse Microsoft Graph Hunting Response

Condition - Microsoft Graph Machine Found

True

Isolate Machine By Microsoft Graph Device ID

Run AV Scan By Microsoft Graph Device ID

False

1 Action

Run draft incident - urapn

Isolate Machine By Microsoft Graph Device ID

Parameters Settings Code preview Testing About

Queries

Enter key Enter value

Body

```
{
  "Comment": "Isolated automatically via Sentinel playbook due to incident ",
  "IsolationType": "Full"
}
```

Cookie

Enter HTTP cookie

Advanced parameters

Showing 1 of 1 Show all Clear all

Authentication

Authentication type *

Managed identity *

System-assigned managed identity

Audience

<https://api.securitycenter.microsoft.com>

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You are using the preview Logic Apps experience.

Workflow Code Run history

Run AV Scan By MDE Device ID

Add comment to incident - Host Response

Parse Microsoft Graph Hunting Response

Condition - Microsoft Graph Machine Found

True

Isolate Machine By Microsoft Graph Device ID

Run AV Scan By Microsoft Graph Device ID

Run draft incident - urapn

Run AV Scan By Microsoft Graph Device ID

Parameters Settings Code preview Testing About

URI *

<https://api.securitycenter.microsoft.com/api/machines/>

Method *

POST

Headers

Content-Type application/json

Queries

Enter key Enter value

Body

```
{
  "Comment": "Full AV scan triggered automatically via Sentinel playbook due to incident Incident Sentinel ID ",
  "ScanType": "Full"
}
```

Cookie

Enter HTTP cookie

Advanced parameters

Showing 1 of 1 Show all Clear all

Authentication

Authentication type *

Microsoft Azure

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You are using the preview Logic Apps experience.

Workflow Code Run history

Run AV Scan By MDE Device ID

Add comment to incident - Host Response

Parse Microsoft Graph Hunting Response

Condition - Microsoft Graph Machine Found

True

Isolate Machine By Microsoft Graph Device ID

Run AV Scan By Microsoft Graph Device ID

Run draft incident - urapn

Run AV Scan By Microsoft Graph Device ID

Parameters Settings Code preview Testing About

Queries

Enter key Enter value

Body

```
{
  "Comment": "Full AV scan triggered automatically via Sentinel playbook due to incident Incident Sentinel ID ",
  "ScanType": "Full"
}
```

Cookie

Enter HTTP cookie

Advanced parameters

Showing 1 of 1 Show all Clear all

Authentication

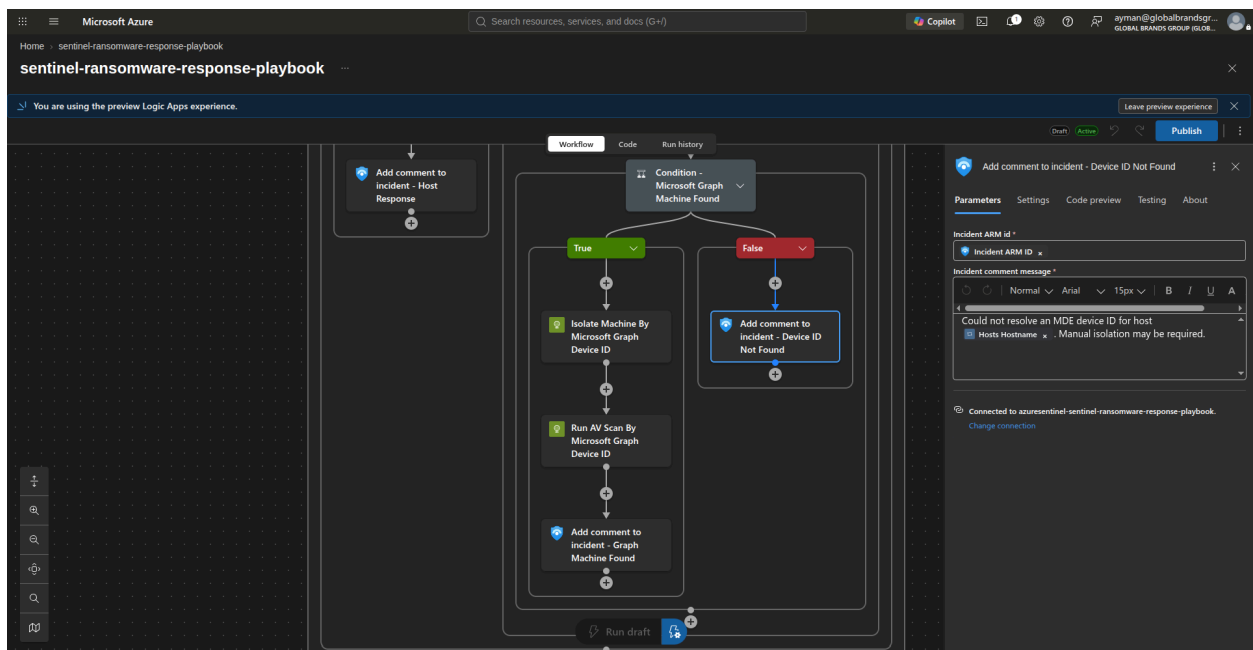
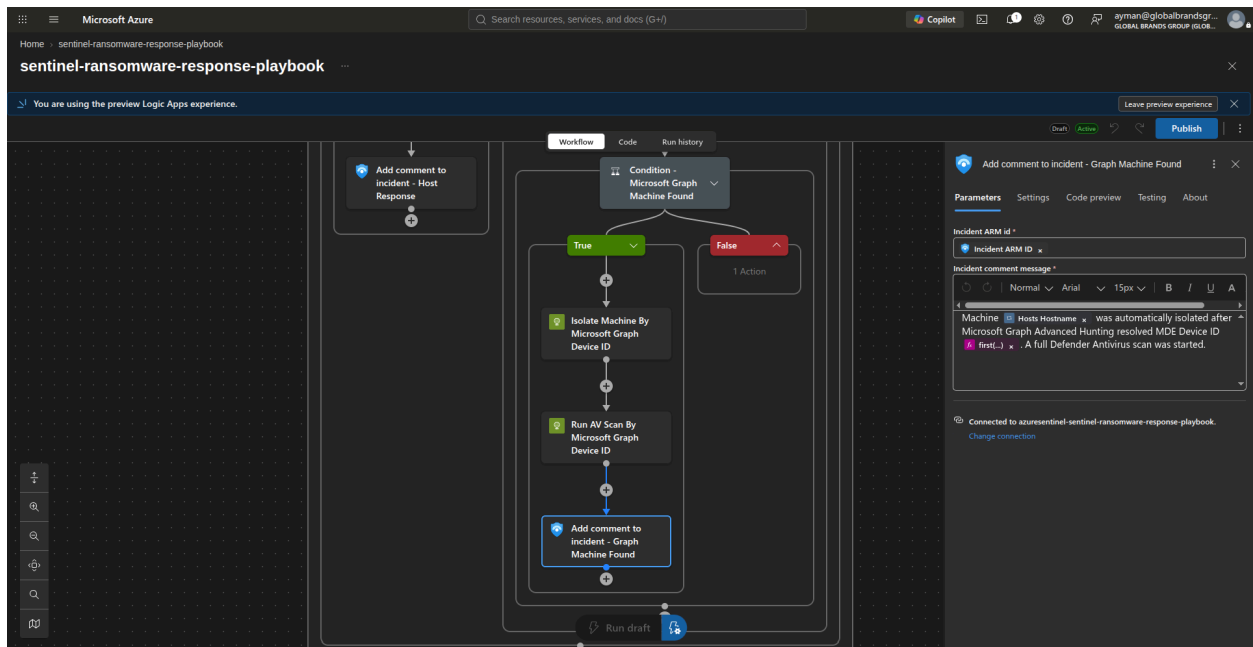
Authentication type *

Managed identity *

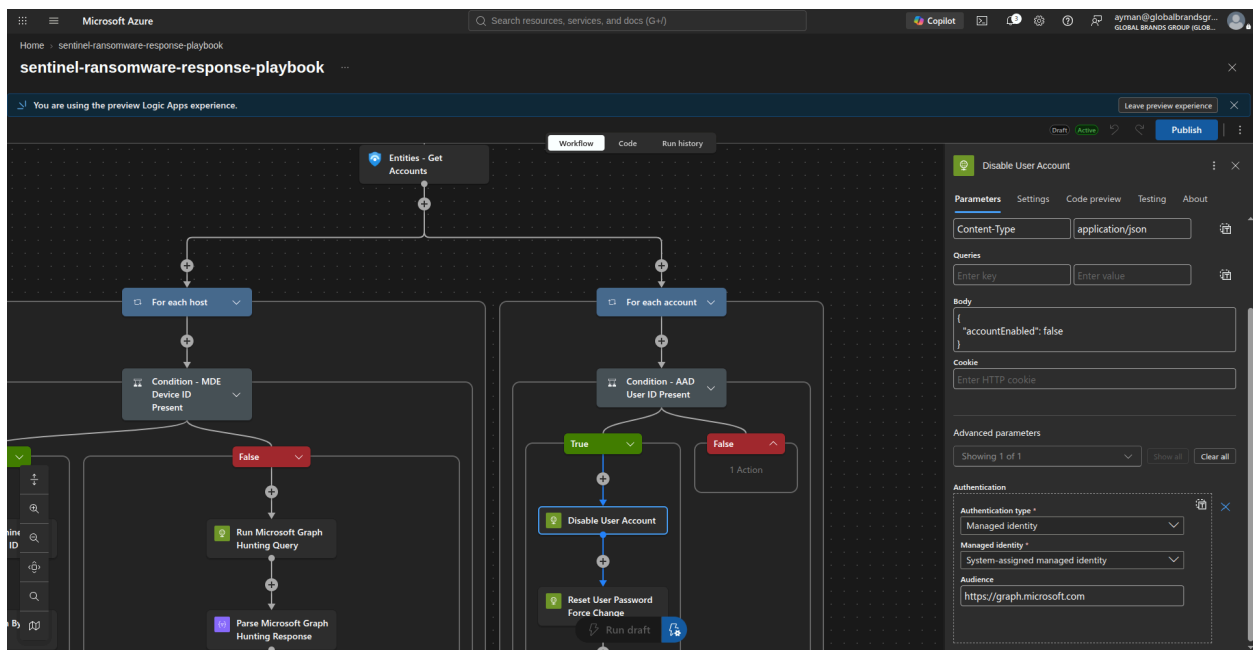
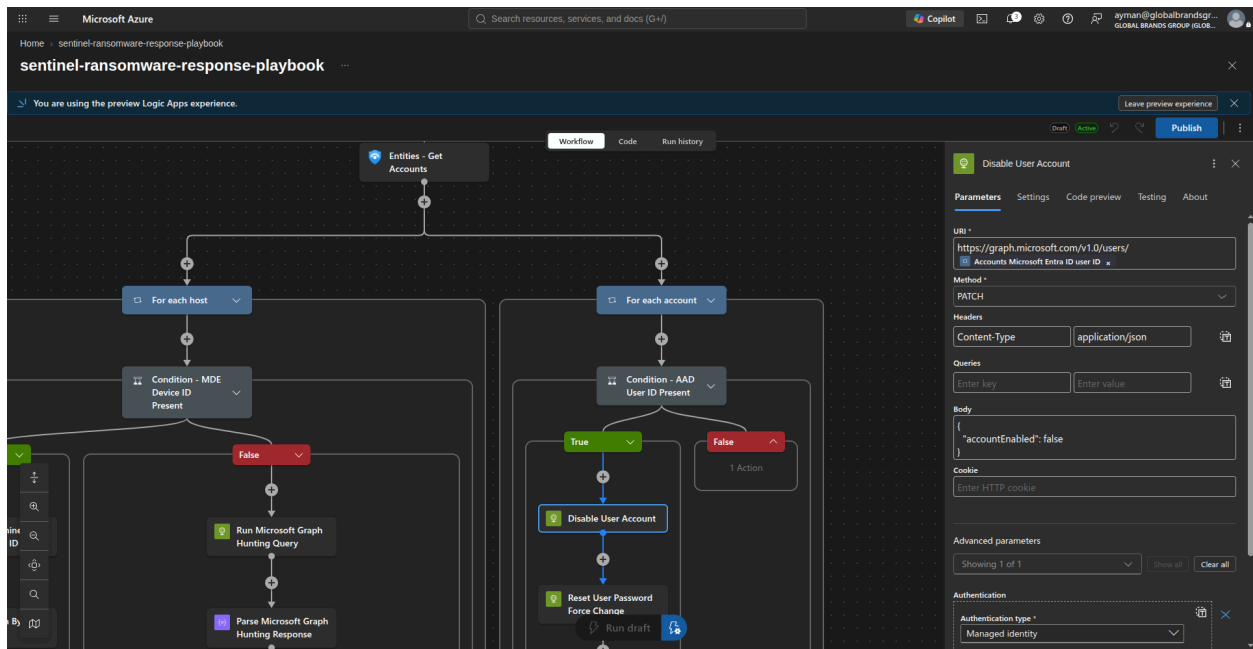
System-assigned managed identity

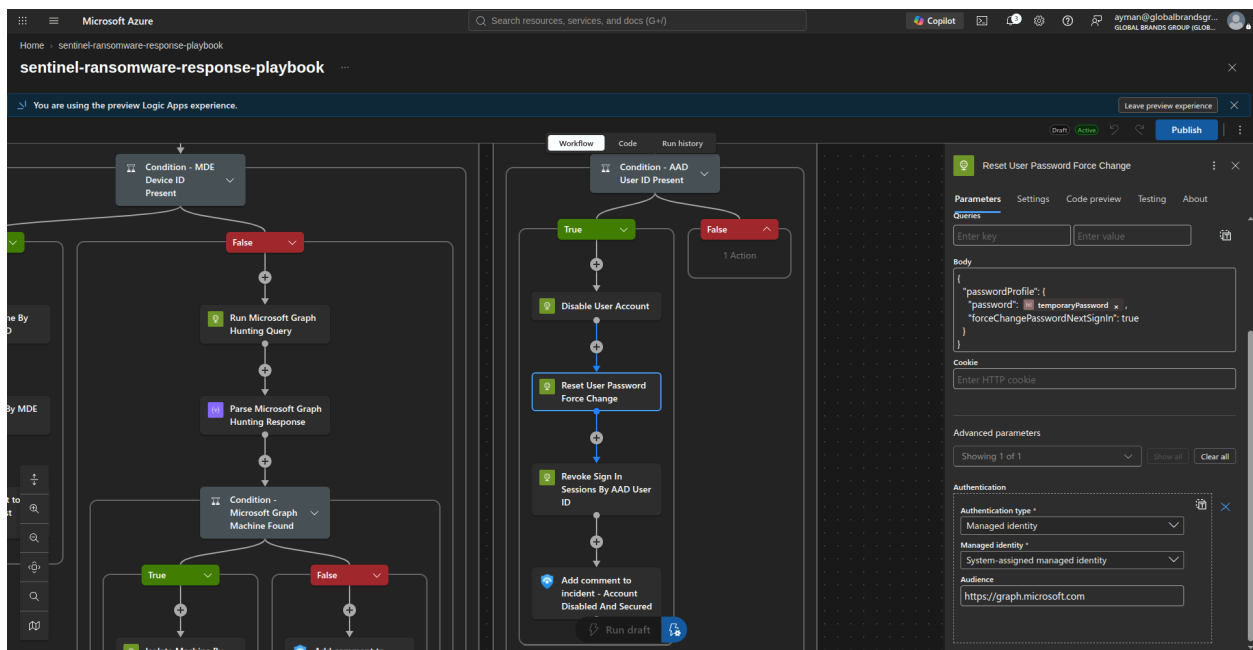
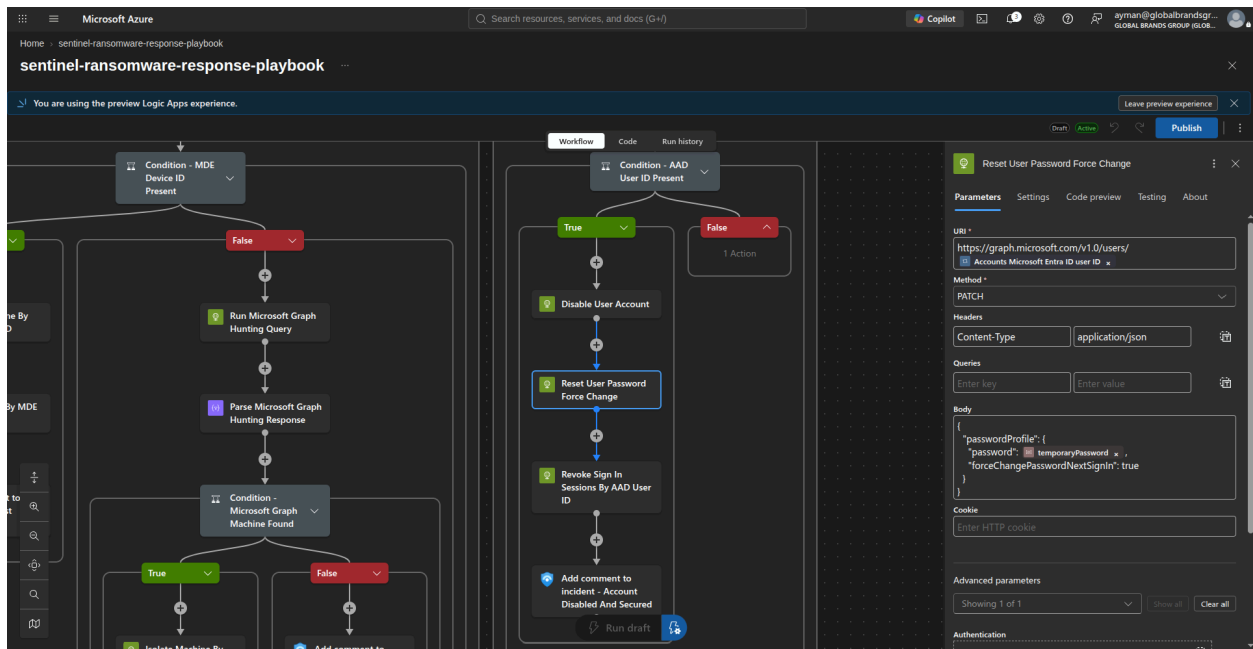
Audience

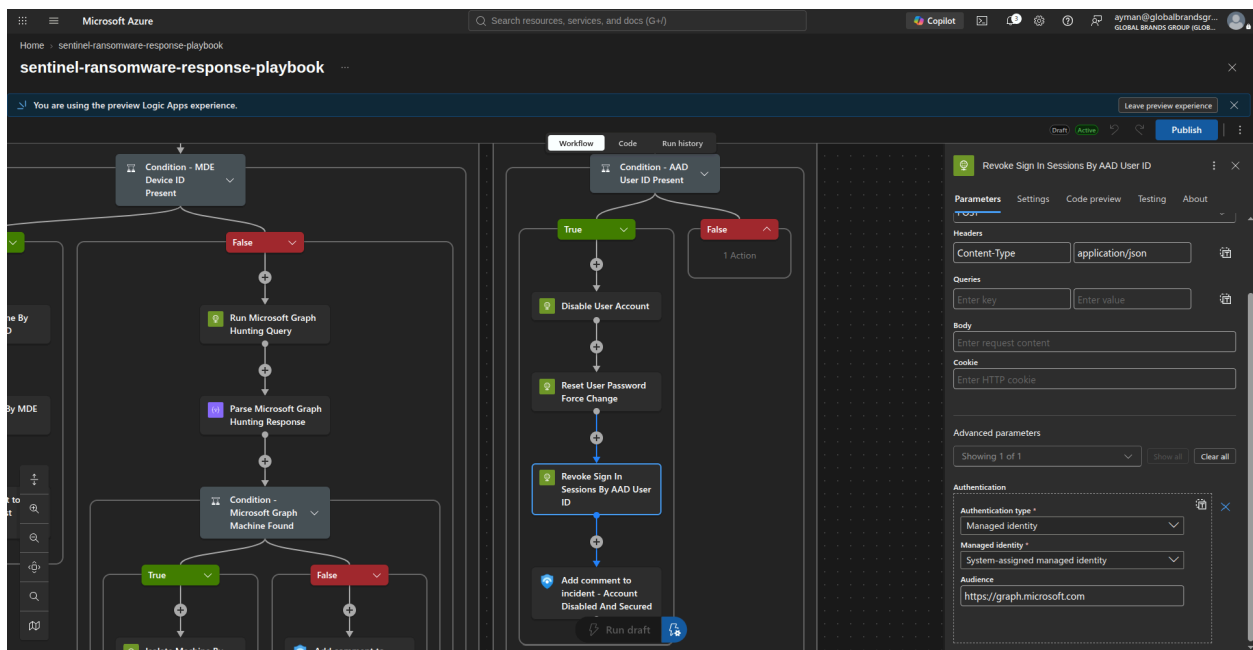
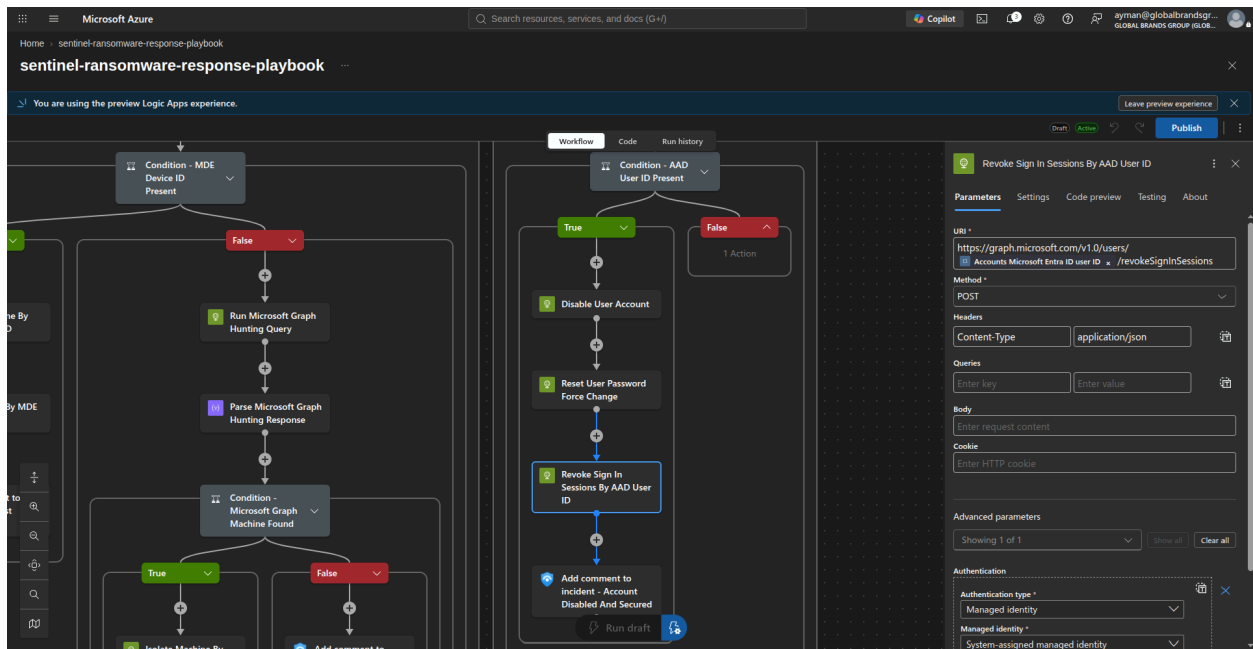
<https://api.securitycenter.microsoft.com>

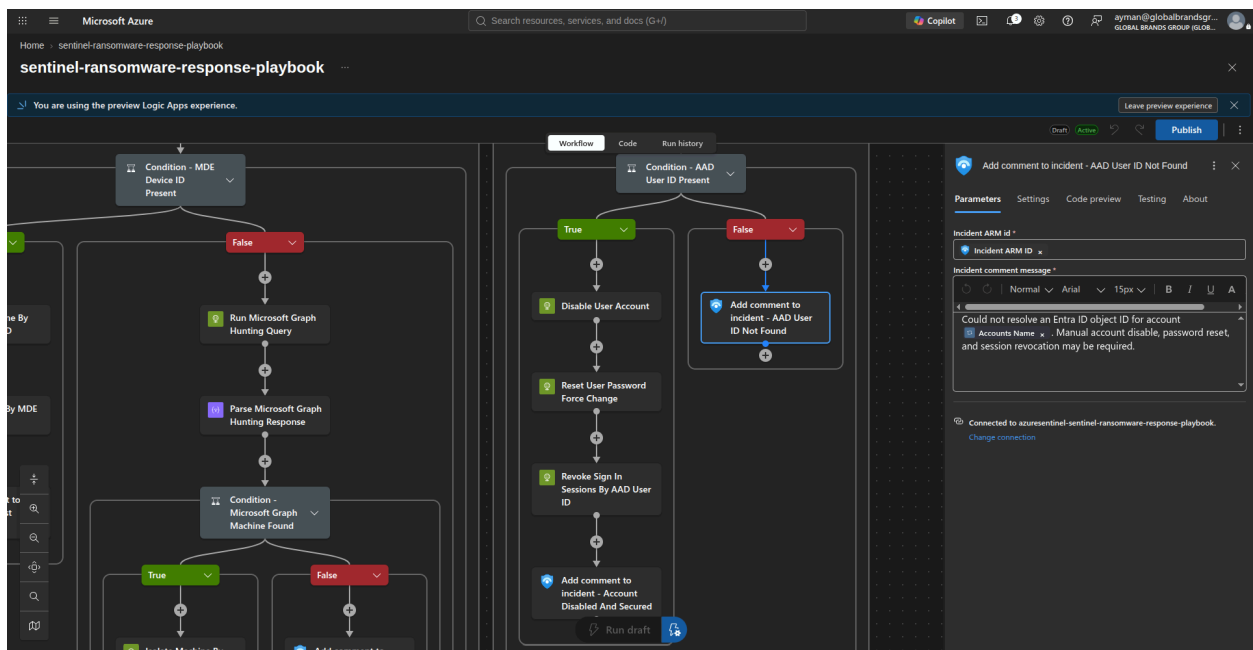
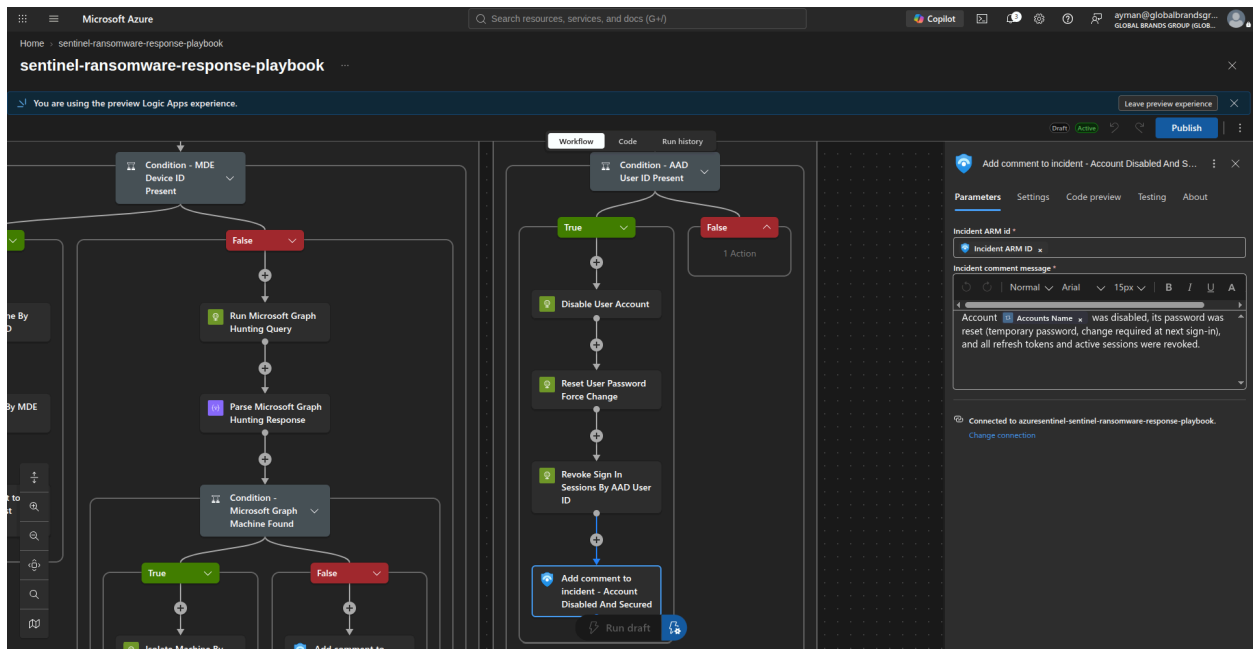


4. Disable user account.
5. Reset the user's password via the approved Entra ID action.
6. Revoke refresh tokens and active sessions.

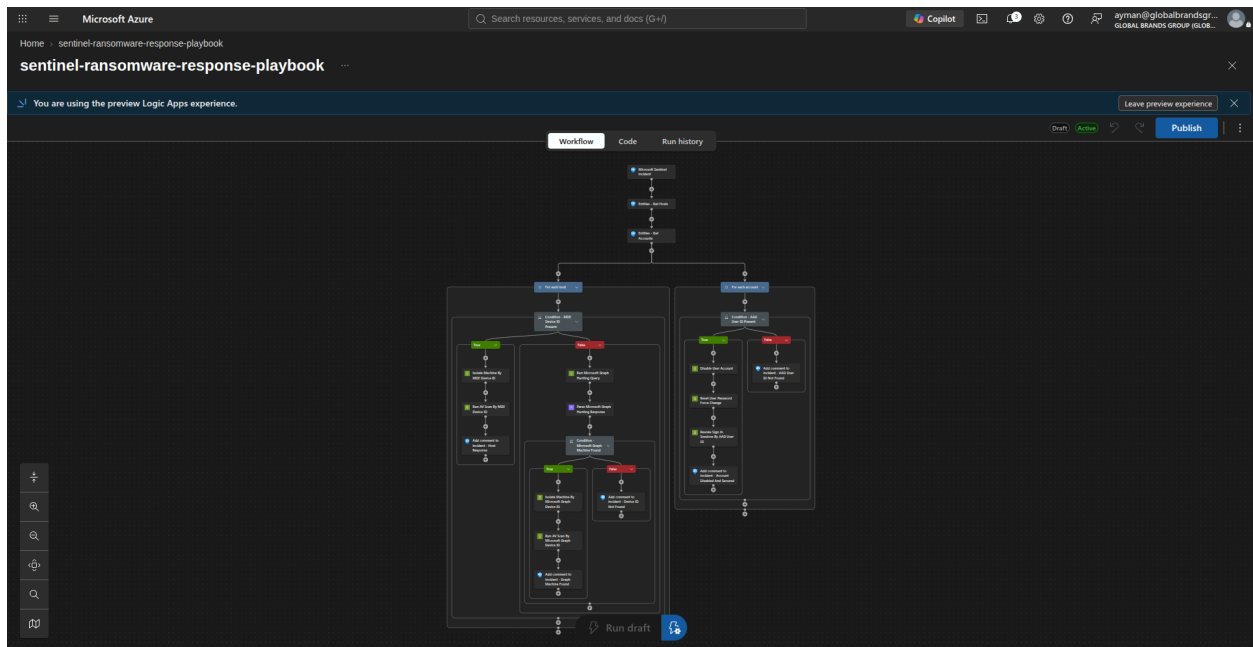








The final playbook:



The final JSON for the logic app:

```
{
  "definition": {
    "metadata": {
      "notes": {}
    },
    "$schema": "https://schema.management.azure.com/providers/Microsoft.Logic/schemas/2016-06-01/workflowdefinition.json#",
    "contentVersion": "1.0.0.0",
    "triggers": {
      "Microsoft_Sentinel_incident": {
        "type": "ApiConnectionWebhook",
        "inputs": {
          "host": {
            "connection": {
              "name": "@parameters('$connections')['azuresentinel']['connectionId']"
            }
          }
        }
      }
    }
  },
```

```

        "body": {
            "callback_url": "@{listCallbackUrl
()}"
        },
        "path": "/incident-creation"
    }
}
},
"actions": {
    "Entities_-_Get_Hosts": {
        "type": "ApiConnection",
        "inputs": {
            "host": {
                "connection": {
                    "name": "@parameters('$connection
s')['azuresentinel']['connectionId']"
                }
            },
            "method": "post",
            "body": "@triggerBody()?['object']['prop
erties']['relatedEntities']",
            "path": "/entities/host"
        },
        "runAfter": {}
    },
    "Entities_-_Get_Accounts": {
        "type": "ApiConnection",
        "inputs": {
            "host": {
                "connection": {
                    "name": "@parameters('$connection
s')['azuresentinel']['connectionId']"
                }
            },
            "method": "post",
            "body": "@triggerBody()?['object']['prop

```

```

    entities' ]?['relatedEntities' ]",
        "path": "/entities/account"
    },
    "runAfter": {
        "Entities_-_Get_Hosts": [
            "Succeeded"
        ]
    }
},
    "For_each_host": {
        "type": "Foreach",
        "foreach": "@body('Entities_-_Get_Hosts')?['H
osts']",
        "actions": {
            "Condition_-_MDE_Device_ID_Present": {
                "type": "If",
                "expression": "@and(not(empty(items
('For_each_host')?['azureID'])), equals(length(items('For_eac
h_host')?['azureID']), 40))",
                "actions": {
                    "Isolate_Machine_By_MDE_Device_I
D": {
                        "type": "Http",
                        "inputs": {
                            "uri": "https://api.secur
itycenter.microsoft.com/api/machines/@{items('For_each_hos
t')?['azureID']}/isolate",
                            "method": "POST",
                            "headers": {
                                "Content-Type": "appl
ication/json"
                            },
                            "body": {
                                "Comment": "Isolated
automatically via Sentinel playbook due to incident @{trigger
Body()?['object']?['properties']?['incidentNumber']}"

```



```

        "IsolationType": "Full",
    },
    "authentication": {
        "type": "ManagedServiceIdentity",
        "audience": "https://api.securitycenter.microsoft.com"
    },
    "Run_AV_Scan_By_MDE_Device_ID": {
        "type": "Http",
        "inputs": {
            "uri": "https://api.securitycenter.microsoft.com/api/machines/@{items('For_each_host')}['azureID']}/runAntiVirusScan",
            "method": "POST",
            "headers": {
                "Content-Type": "application/json"
            },
            "body": {
                "Comment": "Full AV scan triggered automatically via Sentinel playbook due to incident @{triggerBody()}['object'] ['properties'] ['incidentNumber']}",
                "ScanType": "Full"
            }
        },
        "authentication": {
            "type": "ManagedServiceIdentity",
            "audience": "https://api.securitycenter.microsoft.com"
        }
    },
}

```



```

        "else": {
            "actions": {
                "Run_Microsoft_Graph_Hunting_
Query": {
                    "type": "Http",
                    "inputs": {
                        "uri": "https://grap
h.microsoft.com/v1.0/security/runHuntingQuery",
                        "method": "POST",
                        "headers": {
                            "Content-Type":
"application/json"
                        },
                        "body": {
                            "Query": "@{conca
t('DeviceInfo | where DeviceName =~ ''', items('For_each_hos
t')?['HostName'], '' | where OnboardingStatus == ''Onboarde
d'' | summarize arg_max(Timestamp, *) by DeviceId | top 1 by
Timestamp desc | project DeviceId, DeviceName, Timestamp)}"
                        },
                        "authentication": {
                            "type": "ManagedS
erviceIdentity",
                            "audience": "http
s://graph.microsoft.com"
                        }
                    },
                    "Parse_Microsoft_Graph_Huntin
g_Response": {
                        "type": "ParseJson",
                        "inputs": {
                            "content": "@body('Ru
n_Microsoft_Graph_Hunting_Query')",
                            "schema": {
                                "type": "object",

```

```

        "properties": {
            "results": {
                "type":
                    "array",
                    "items":
                        {
                            "type":
                                "object",
                                "properties": {
                                    "DeviceId": {
                                        "type": "string"
                                    },
                                    "DeviceName": {
                                        "type": "string"
                                    },
                                    "Timestamp": {
                                        "type": "string"
                                    }
                                }
                        }
            }
        },
        "runAfter": {
            "Run_Microsoft_Graph_
Hunting_Query": [

```

```

        "Succeeded"
    ]
}
},
"Condition_-_Microsoft_Graph_
Machine_Found": {
    "type": "If",
    "expression": "@greater(l
ength(body('Parse_Microsoft_Graph_Hunting_Response')['result
s']), 0)",
    "actions": {
        "Isolate_Machine_By_M
icrosoft_Graph_Device_ID": {
            "type": "Http",
            "inputs": {
                "uri": "http
s://api.securitycenter.microsoft.com/api/machines/@{first(bod
y('Parse_Microsoft_Graph_Hunting_Response')['results'])?['De
viceId']}/isolate",
                "method": "PO
ST",
                "headers": {
                    "Content-
Type": "application/json"
                },
                "body": {
                    "Commen
t": "Isolated automatically via Sentinel playbook due to inci
dent @{triggerBody()?['object']?['properties']?['incidentNumb
er']}",
                    "Isolatio
nType": "Full"
                },
                "authenticati
on": {
                    "type":

```

```

"ManagedServiceIdentity",
                                "audienc
e": "https://api.securitycenter.microsoft.com"
                                }
                                },
                                },
                                "Run_AV_Scan_By_Micro
soft_Graph_Device_ID": {
                                "type": "Http",
                                "inputs": {
                                    "uri": "http
s://api.securitycenter.microsoft.com/api/machines/@{first(bod
y('Parse_Microsoft_Graph_Hunting_Response')?['results'])?['De
viceId']}/runAntiVirusScan",
                                    "method": "P0
ST",
                                    "headers": {
                                        "Content-
Type": "application/json"
                                    },
                                    "body": {
                                        "Commen
t": "Full AV scan triggered automatically via Sentinel playbo
ok due to incident @{{triggerBody()?['object']?['properties']]?
['incidentNumber']}",
                                        "ScanTyp
e": "Full"
                                    },
                                    "authenticati
on": {
                                        "type":
"ManagedServiceIdentity",
                                        "audienc
e": "https://api.securitycenter.microsoft.com"
                                    }
                                },

```

```

        "runAfter": {
            "Isolate_Mach
ine_By_Microsoft_Graph_Device_ID": [
                "Succeede
d"
            ]
        },
        "Add_comment_to_incident_-_Graph_Machine_Found": {
            "type": "ApiConne
ction",
            "inputs": {
                "host": {
                    "connecti
on": {
                        "nam
e": "@parameters('$connections')['azuresentinel']['connection
Id']"
                    }
                },
                "method": "po
st",
                "body": {
                    "incident
ArmId": "@triggerBody()?['object']['id']",
                    "messag
e": "<p>Machine @items('For_each_host')['HostName'] was au
tomatically isolated after Microsoft Graph Advanced Hunting r
esolved MDE Device ID @first(body('Parse_Microsoft_Graph_Hun
ting_Response')['results'])['DeviceId']. A full Defender A
ntivirus scan was started.</p>"
                },
                "path": "/Inc
idents/Comment"
            },

```

```

        "runAfter": {
            "Run_AV_Scan_
By_Microsoft_Graph_Device_ID": [
                "Succeede
d"
            ]
        }
    },
    "else": {
        "actions": {
            "Add_comment_to_i
ncident_-_Device_ID_Not_Found": {
                "type": "ApiC
onnection",
                "inputs": {
                    "host": {
                        "conn
ection": {
                            "name": "@parameters('$connections')['azuresentinel']['connec
tionId']"
                        }
                    },
                    "method":
"post",
                    "body": {
                        "inci
dentArmId": "@triggerBody()?['object']['id']",
                        "mess
age": "<p>Could not resolve an MDE device ID for host @{items
('For_each_host')['HostName']}. Manual isolation may be requ
ired.</p>"
                    },
                    "path":
"/Incidents/Comment"

```



```

    }
  }
},
"runAfter": {
  "Parse_Microsoft_Grap
h_Hunting_Response": [
    "Succeeded"
  ]
}
}
}
}
},
"runAfter": {
  "Entities_-_Get_Accounts": [
    "Succeeded"
  ]
}
},
"For_each_account": {
  "type": "Foreach",
  "foreach": "@body('Entities_-_Get_Accounts')?
['Accounts']",
  "actions": {
    "Condition_-_AAD_User_ID_Present": {
      "type": "If",
      "expression": "@not(empty(items('For_
each_account')?['AadUserId']))",
      "actions": {
        "Disable_User_Account": {
          "type": "Http",
          "inputs": {
            "uri": "https://graph.mic
rosoft.com/v1.0/users/@{items('For_each_account')?['AadUserI

```

```

d' ]}],
    "method": "PATCH",
    "headers": {
        "Content-Type": "appl
ication/json"
    },
    "body": {
        "accountEnabled": fal
se
    },
    "authentication": {
        "type": "ManagedServi
ceIdentity",
        "audience": "https://
graph.microsoft.com"
    }
},
"Reset_User_Password_Force_Chang
e": {
    "type": "Http",
    "inputs": {
        "uri": "https://graph.mic
rosoft.com/v1.0/users/@{items('For_each_account')?['AadUserI
d' ]}],
    "method": "PATCH",
    "headers": {
        "Content-Type": "appl
ication/json"
    },
    "body": {
        "passwordProfile": {
            "password": "@par
ameters('temporaryPassword')",
            "forceChangePassw
ordNextSignIn": true

```

```

    },
    "authentication": {
        "type": "ManagedServiceIdentity",
        "audience": "https://graph.microsoft.com"
    },
    "runAfter": {
        "Disable_User_Account": [
            "Succeeded"
        ]
    }
},
"Revoke_Sign_In_Sessions_By_AAD_User_ID": {
    "type": "Http",
    "inputs": {
        "uri": "https://graph.microsoft.com/v1.0/users/@{items('For_each_account')['AadUserId']}/revokeSignInSessions",
        "method": "POST",
        "headers": {
            "Content-Type": "application/json"
        },
        "authentication": {
            "type": "ManagedServiceIdentity",
            "audience": "https://graph.microsoft.com"
        }
    },
    "runAfter": {
        "Reset_User_Password_Forcefully": [
            "Succeeded"
        ]
    }
}
}

```

```

e_Change": [
    "Succeeded"
]
},
},
"Add_comment_to_incident_-_Account_Disabled_And_Secured": {
    "type": "ApiConnection",
    "inputs": {
        "host": {
            "connection": {
                "name": "@parameters('$connections')['azuresentinel']['connectionId']"
            }
        },
        "method": "post",
        "body": {
            "incidentArmId": "@triggerBody()?['object']['id']",
            "message": "<p>Account @{items('For_each_account')['Name']} was disabled, its password was reset (temporary password, change required at next sign-in), and all refresh tokens and active sessions were revoked.</p>"
        },
        "path": "/Incidents/Comment"
    },
    "runAfter": {
        "Revoke_Sign_In_Sessions_By_AAD_User_ID": [
            "Succeeded"
        ]
    }
},
},
},

```



```

        "outputs": {},
        "parameters": {
            "temporaryPassword": {
                "defaultValue": "",
                "type": "SecureString"
            },
            "$connections": {
                "type": "Object",
                "defaultValue": {}
            }
        }
    },
    "parameters": {
        "temporaryPassword": {},
        "$connections": {
            "type": "Object",
            "value": {
                "azuresentinel": {
                    "id": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/providers/Microsoft.Web/locations/swedencentral/managedApis/azuresentinel",
                    "connectionId": "/subscriptions/60ba34d6-e952-486b-b30d-a3109360cd74/resourceGroups/rg-useCases/providers/Microsoft.Web/connections/azuresentinel-sentinel-ransomware-response-playbook",
                    "connectionName": "azuresentinel-sentinel-ransomware-response-playbook",
                    "connectionProperties": {
                        "authentication": {
                            "type": "ManagedServiceIdentity"
                        }
                    }
                }
            }
        }
    }
}

```

```
}  
}
```

Step 2.2 — Playbook permissions

Grant the Managed Identity Permissions

```
Logic App -> Settings -> Identity -> System assigned -> Object ID
```

The complete ransomware-response playbook requires five Microsoft Entra application permissions across two resource applications, plus one Microsoft Entra directory role:

```
Microsoft Defender for Endpoint: Machine.Isolate  
Microsoft Defender for Endpoint: Machine.Scan  
Microsoft Graph: ThreatHunting.Read.All  
Microsoft Graph: User-PasswordProfile.ReadWrite.All  
Microsoft Graph: User.RevokeSessions.All  
  
Microsoft Entra directory role: User Administrator
```

`Machine.Read.All` is not required by the current playbook. It is only required when a workflow uses the Defender Machines API to look up devices by `computerDnsName`. This playbook uses Sentinel `azureID` first and Microsoft Graph Advanced Hunting as its fallback.

The two user-response permissions are used by these Logic App actions:

```
Reset_User_Password_Force_Change  
Revoke_Sign_In_Sessions_By_AAD_User_ID
```

`User-PasswordProfile.ReadWrite.All` permits the managed identity to set a temporary password and `forceChangePasswordNextSignIn: true`. `User.RevokeSessions.All` invalidates refresh tokens and active browser sessions. The User Administrator directory role is required for app-only password resets for normal user accounts. Password

resets for administrator accounts may require Privileged Authentication Administrator instead.

Assign the Permissions

Run this single PowerShell block in Azure Cloud Shell as a tenant administrator who can assign enterprise-application roles. Replace `$MIGuid` with the Logic App managed identity **service principal Object ID**. Do not use the client ID or Logic App resource ID.

This uses the Microsoft Graph PowerShell SDK through `Connect-MgGraph`.

```
Import-Module Microsoft.Graph.Authentication
Import-Module Microsoft.Graph.Applications
Import-Module Microsoft.Graph.Identity.DirectoryManagement

Connect-MgGraph -Scopes @(
    "AppRoleAssignment.ReadWrite.All",
    "Application.Read.All",
    "Directory.Read.All",
    "RoleManagement.ReadWrite.Directory"
)

$MIGuid = "<Enter your managed identity Object ID>"

$MI = Get-MgServicePrincipal -ServicePrincipalId $MIGuid
if (-not $MI) {
    throw "Managed identity service principal '$MIGuid' was not found."
}

$MDEApp = Get-MgServicePrincipal `
    -Filter "appId eq 'fc780465-2017-40d4-a0c5-307022471b92'"

$GraphApp = Get-MgServicePrincipal `
    -Filter "appId eq '00000003-0000-0000-c000-000000000000'"
```



```

if (-not $MDEApp) {
    throw "Microsoft Defender for Endpoint service principal
was not found."
}

if (-not $GraphApp) {
    throw "Microsoft Graph service principal was not found."
}

function Add-ApplicationPermission {
    param(
        [Parameter(Mandatory = $true)]
        $ServicePrincipal,

        [Parameter(Mandatory = $true)]
        $ResourceServicePrincipal,

        [Parameter(Mandatory = $true)]
        [string] $PermissionValue
    )

    $AppRole = @($ResourceServicePrincipal.AppRoles | Where-Ob
ject {
        $_.Value -eq $PermissionValue -and
        $_.AllowedMemberTypes -contains "Application" -and
        $_.IsEnabled
    }) | Select-Object -First 1

    if (-not $AppRole) {
        throw "Application permission '$PermissionValue' was
not found on '$($ResourceServicePrincipal.DisplayName)'."
    }

    $Existing = @(Get-MgServicePrincipalAppRoleAssignment `
        -ServicePrincipalId $ServicePrincipal.Id `

```

```

        -All | Where-Object {
            $_.ResourceId -eq $ResourceServicePrincipal.Id -a
nd
            $_.AppRoleId -eq $AppRole.Id
        })

if ($Existing.Count -eq 0) {
    New-MgServicePrincipalAppRoleAssignment `
        -ServicePrincipalId $ServicePrincipal.Id `
        -PrincipalId $ServicePrincipal.Id `
        -ResourceId $ResourceServicePrincipal.Id `
        -AppRoleId $AppRole.Id `
        -ErrorAction Stop | Out-Null

    Write-Host "Assigned $PermissionValue"
}

if ($Existing.Count -gt 0) {
    Write-Host "Already assigned: $PermissionValue"
}
}

Add-ApplicationPermission `
    -ServicePrincipal $MI `
    -ResourceServicePrincipal $MDEApp `
    -PermissionValue "Machine.Isolate"

Add-ApplicationPermission `
    -ServicePrincipal $MI `
    -ResourceServicePrincipal $MDEApp `
    -PermissionValue "Machine.Scan"

Add-ApplicationPermission `
    -ServicePrincipal $MI `
    -ResourceServicePrincipal $GraphApp `
    -PermissionValue "ThreatHunting.Read.All"

```

```

Add-ApplicationPermission `
    -ServicePrincipal $MI `
    -ResourceServicePrincipal $GraphApp `
    -PermissionValue "User-PasswordProfile.ReadWrite.All"

Add-ApplicationPermission `
    -ServicePrincipal $MI `
    -ResourceServicePrincipal $GraphApp `
    -PermissionValue "User.RevokeSessions.All"

$UserAdministratorRoleTemplateId = "fe930be7-5e62-47db-91af-9
8c3a49a38b1"

$UserAdministratorRole = Get-MgDirectoryRole `
    -Filter "roleTemplateId eq '$UserAdministratorRoleTemplat
eId'"

if (-not $UserAdministratorRole) {
    $UserAdministratorRole = New-MgDirectoryRole `
        -BodyParameter @{
            roleTemplateId = $UserAdministratorRoleTemplateId
        }
}

$RoleMembers = Get-MgDirectoryRoleMember `
    -DirectoryRoleId $UserAdministratorRole.Id `
    -All

$AlreadyAssigned = @(
    $RoleMembers | Where-Object { $_.Id -eq $MI.Id }
).Count -gt 0

if (-not $AlreadyAssigned) {
    try {
        New-MgDirectoryRoleMemberByRef `

```

```

        -DirectoryRoleId $UserAdministratorRole.Id `
        -BodyParameter @{
            "@odata.id" = "https://graph.microsoft.com/v
1.0/directoryObjects/$( $MI.Id )"
        } `
        -ErrorAction Stop

        Write-Host "Assigned User Administrator to managed id
entity $( $MI.Id )."
    }
    catch {
        if ($_.Exception.Message -match "already exist") {
            Write-Host "Managed identity already has User Adm
inistrator."
        }
        else {
            throw
        }
    }
}

if ($AlreadyAssigned) {
    Write-Host "Managed identity already has User Administrat
or."
}

Write-Host "All application permissions and the User Administ
rator role have been processed."

```

The script is safe to run again. Existing application assignments are detected and are not duplicated. If Graph reports `Permission being assigned already exists on the object`, that permission is already assigned. The User Administrator role is activated if necessary and is also checked before assignment.

Verify the Assignments

Run this in the same Cloud Shell session:

```
$Assignments = Get-MgServicePrincipalAppRoleAssignment `
    -ServicePrincipalId $MI.Id `
    -All

$MDEPermissionIds = @{}
$MDEApp.AppRoles | ForEach-Object {
    $MDEPermissionIds[$_.Id.ToString()] = $_.Value
}

$GraphPermissionIds = @{}
$GraphApp.AppRoles | ForEach-Object {
    $GraphPermissionIds[$_.Id.ToString()] = $_.Value
}

$Assignments |
    Where-Object {
        $_.ResourceId -eq $MDEApp.Id -or
        $_.ResourceId -eq $GraphApp.Id
    } |
    ForEach-Object {
        $PermissionName = $MDEPermissionIds[$_.AppRoleId.ToSt
ring()]
        if (-not $PermissionName) {
            $PermissionName = $GraphPermissionIds[$_.AppRoleI
d.ToString()]
        }

        [PSCustomObject]@{
            Resource = $_.ResourceDisplayName
            Permission = $PermissionName
            AppRoleId = $_.AppRoleId
        }
    }
```

} |

Format-Table Resource, Permission, AppRoleId -AutoSize

The output should include:

Microsoft Defender for Endpoint	Machine.Isolate
Microsoft Defender for Endpoint	Machine.Scan
Microsoft Graph	ThreatHunting.Read.All
Microsoft Graph	User-PasswordProfile.ReadWrite.All
Microsoft Graph	User.RevokeSessions.All

The delegated scopes in `Connect-MgGraph` authorize the administrator running the setup script. They are not runtime permissions granted to the Logic App managed identity. The Logic App uses the application-role assignments and directory role created by this script.

Role propagation can take a few minutes after assignment before the playbook's calls succeed.

```
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PS /home/ayman> Import-Module Microsoft.Graph.Authentication
PS /home/ayman> Import-Module Microsoft.Graph.Applications
PS /home/ayman> Import-Module Microsoft.Graph.Identity.DirectoryManagement
PS /home/ayman> Connect-MgGraph -Scopes @('AppRoleAssignment.ReadWrite.All',
>> 'Application.Read.All',
>> 'Directory.Read.All',
>> 'RoleManagement.ReadWrite.Directory')
>>)
Welcome to Microsoft Graph!
Connected via delegated access using [redacted]
Readme: https://aka.ms/graph/sdk/powershell
SDK Docs: https://aka.ms/graph/sdk/powershell/docs
API Docs: https://aka.ms/graph/docs
NOTE: You can use the -Nowelcome parameter to suppress this message.
NOTE: Sign in by Web Account Manager (WAM) is enabled by default on Windows systems and cannot be disabled when using the default ClientId.
To disable WAM run Set-MgGraphOption -DisableLoginByWAM $true and then use a custom ClientId.
PS /home/ayman> $MIGuid = "a7b0b0b0-b0b0-b0b0-b0b0-b0b0-b0b0"
PS /home/ayman> $MI = Get-MgServicePrincipal -ServicePrincipalId $MIGuid
PS /home/ayman> if (-not $MI) {
>> throw "Managed identity service principal '$MIGuid' was not found."
>>}
PS /home/ayman> $MDEApp = Get-MgServicePrincipal `
>> -Filter "appid eq 'fc78d465-2817-4804-b8c3-387822471b92'"
PS /home/ayman> $GraphApp = Get-MgServicePrincipal `
>> -Filter "appid eq '00000003-0000-0000-c000-000000000000'"
PS /home/ayman> if (-not $MDEApp) {
>> throw "Microsoft Defender for Endpoint service principal was not found."
>>}
PS /home/ayman> if (-not $GraphApp) {
>> throw "Microsoft Graph service principal was not found."
>>}
PS /home/ayman> function Add-ApplicationPermission {
>> param(
```

```
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PS /home/ayman> function Add-ApplicationPermission {
    param(
        [Parameter(Mandatory = $true)]
        $ServicePrincipal,
        [Parameter(Mandatory = $true)]
        $ResourceServicePrincipal,
        [Parameter(Mandatory = $true)]
        [string] $PermissionValue
    )

    $AppRole = @($ResourceServicePrincipal.AppRoles | Where-Object {
        $_.Value -eq $PermissionValue -and
        $_.AllowedMemberTypes -contains 'Application' -and
        $_.IsEnabled
    }) | Select-Object -First 1

    if (-not $AppRole) {
        throw "Application permission '$PermissionValue' was not found on '$($ResourceServicePrincipal.DisplayName)'."
    }

    $Existing = @(Get-MgServicePrincipalAppRoleAssignment `
        -ServicePrincipalId $ServicePrincipal.Id `
        -All | Where-Object {
            $_.ResourceId -eq $ResourceServicePrincipal.Id -and
            $_.AppRoleId -eq $AppRole.Id
        })

    if ($Existing.Count -eq 0) {
        New-MgServicePrincipalAppRoleAssignment `
        -ServicePrincipalId $ServicePrincipal.Id `
        -PrincipalId $ServicePrincipal.Id `
        -ResourceId $ResourceServicePrincipal.Id `
        -AppRoleId $AppRole.Id `
        -ErrorAction Stop | Out-Null

        Write-Host "Assigned $PermissionValue"
    }

    if ($Existing.Count -gt 0) {
        Write-Host "Already assigned: $PermissionValue"
    }
}

PS /home/ayman>
```

```
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Switch to Bash Restart Manage files New session Editor Open in VS Code for the Web Web preview Settings Help

PS /home/ayman> Add-ApplicationPermission `
    -ServicePrincipal $MI `
    -ResourceServicePrincipal $MDEApp `
    -PermissionValue "Machine.Isolate"
Already assigned: Machine.Isolate
PS /home/ayman> Add-ApplicationPermission `
    -ServicePrincipal $MI `
    -ResourceServicePrincipal $MDEApp `
    -PermissionValue "Machine.Scan"
Already assigned: Machine.Scan
PS /home/ayman> Add-ApplicationPermission `
    -ServicePrincipal $MI `
    -ResourceServicePrincipal $GraphApp `
    -PermissionValue "ThreatHunting.Read.All"
Already assigned: ThreatHunting.Read.All
PS /home/ayman> Add-ApplicationPermission `
    -ServicePrincipal $MI `
    -ResourceServicePrincipal $GraphApp `
    -PermissionValue "User.PasswordProfile.ReadWrite.All"
Already assigned: User.PasswordProfile.ReadWrite.All
PS /home/ayman> Add-ApplicationPermission `
    -ServicePrincipal $MI `
    -ResourceServicePrincipal $GraphApp `
    -PermissionValue "User.RevokeSessions.All"
Already assigned: User.RevokeSessions.All
PS /home/ayman> $UserAdministratorRoleTemplateId = "fe938be7-5e62-47db-91af-98c3a49a38b1"
PS /home/ayman> $UserAdministratorRole = Get-MgDirectoryRole `
    -Filter "roleTemplateId eq '$UserAdministratorRoleTemplateId'"
PS /home/ayman> if (-not $UserAdministratorRole) {
    $UserAdministratorRole = New-MgDirectoryRole `
        -BodyParameter @{
            roleTemplateId = $UserAdministratorRoleTemplateId
        }
}
PS /home/ayman> $RoleMembers = Get-MgDirectoryRoleMember `
    -DirectoryRoleId $UserAdministratorRole.Id `
    -All
```

```

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Switch to Bash Restart Manage files New session Editor Open in VS Code for the Web Web preview Settings Help

PS /home/ayman> if (-not $UserAdministratorRole) {
>> $UserAdministratorRole = New-MgDirectoryRole `
>> -BodyParameter @{
>>   roleTemplateId = $UserAdministratorRoleTemplateId
>> }
>> }
PS /home/ayman> $RoleMembers = Get-MgDirectoryRoleMember `
>> -DirectoryRoleId $UserAdministratorRole.Id `
>> -All
PS /home/ayman>
PS /home/ayman> $AlreadyAssigned = @(
>> $RoleMembers | Where-Object { $_.Id -eq $MI.Id }
>> ).Count -gt 0
PS /home/ayman>
PS /home/ayman> if (-not $AlreadyAssigned) {
>> try {
>>   New-MgDirectoryRoleMemberByRef `
>>   -DirectoryRoleId $UserAdministratorRole.Id `
>>   -BodyParameter @{
>>     @odata.id = "https://graph.microsoft.com/v1.0/directoryObjects/$( $MI.Id )"
>>   } `
>>   -ErrorAction Stop
>> }
>> Write-Host "Assigned User Administrator to managed identity $( $MI.Id )."
>> }
>> catch {
>>   if ($_.Exception.Message -match "already exist") {
>>     Write-Host "Managed identity already has User Administrator."
>>   }
>>   else {
>>     throw
>>   }
>> }
>> }
Managed identity already has User Administrator.
PS /home/ayman>
PS /home/ayman> if ($AlreadyAssigned) {
>>   Write-Host "Managed identity already has User Administrator."
>> }
PS /home/ayman> Write-Host "All application permissions and the User Administrator role have been processed."
All application permissions and the User Administrator role have been processed.
PS /home/ayman>

```

```

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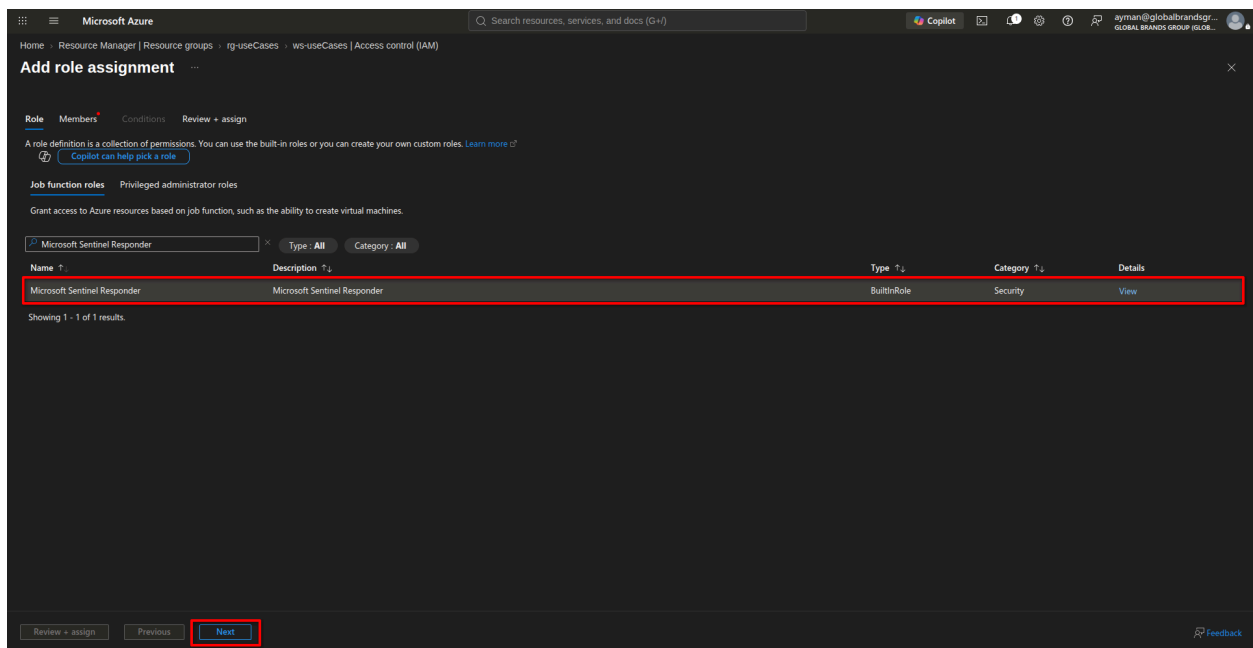
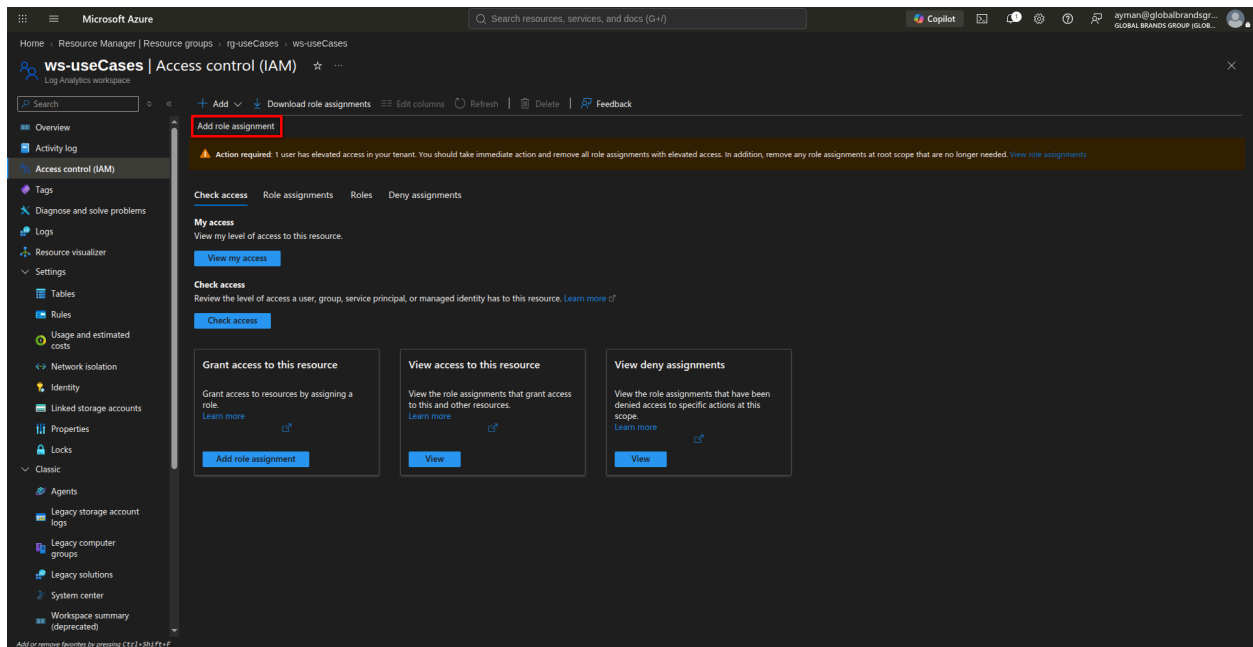
Switch to Bash Restart Manage files New session Editor Open in VS Code for the Web Web preview Settings Help

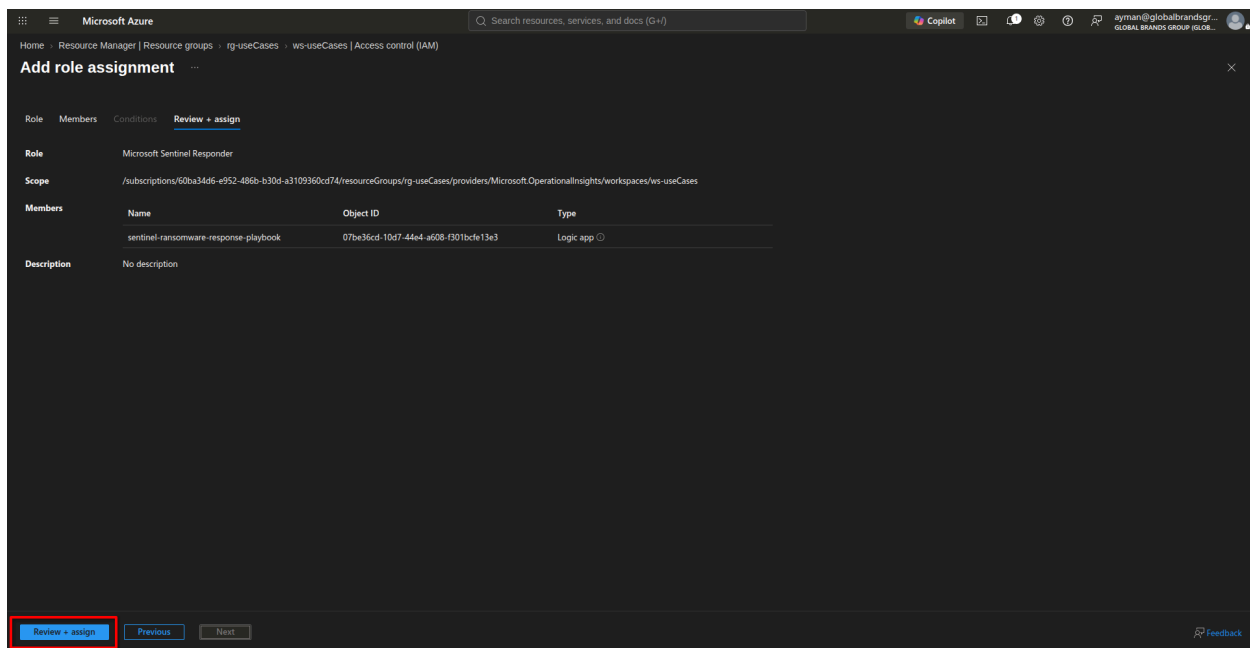
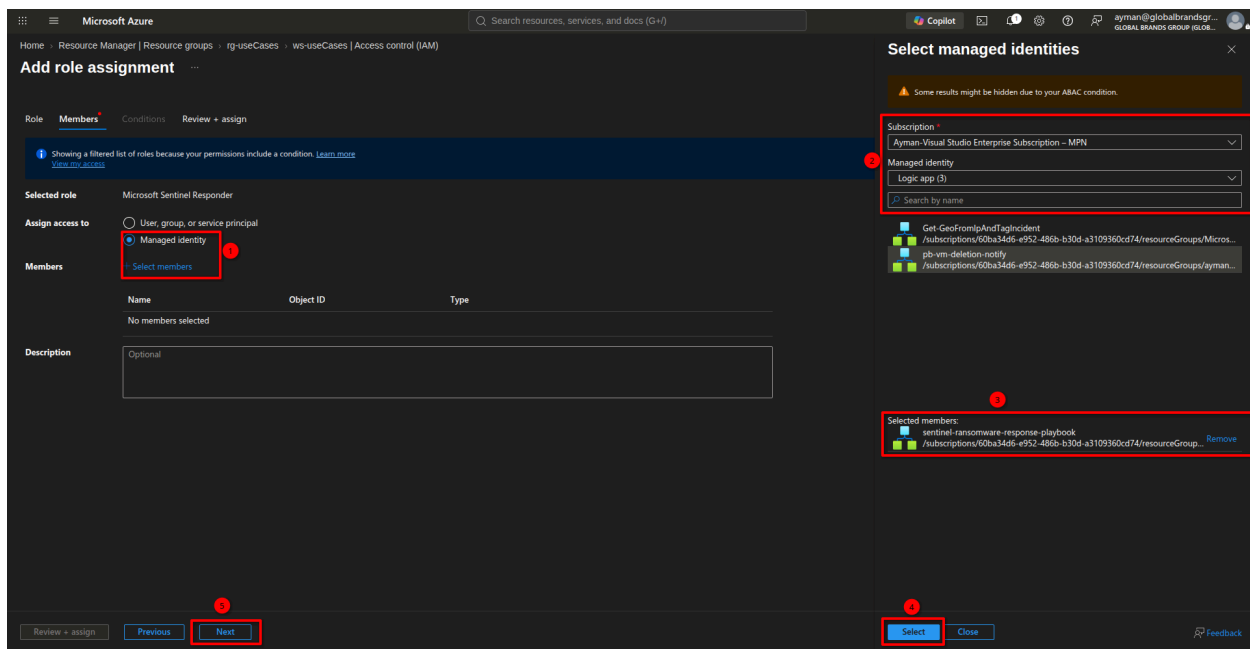
PS /home/ayman> $Assignments = Get-MgServicePrincipalAppRoleAssignment `
>> -ServicePrincipalId $MI.Id `
>> -All
PS /home/ayman>
PS /home/ayman> $MDEPermissionIds = @{}
PS /home/ayman> $MDEAppRoles | ForEach-Object {
>> $MDEPermissionIds[$_.Id.ToString()] = $_.Value
>> }
PS /home/ayman>
PS /home/ayman> $GraphPermissionIds = @{}
PS /home/ayman> $GraphAppRoles | ForEach-Object {
>> $GraphPermissionIds[$_.Id.ToString()] = $_.Value
>> }
PS /home/ayman>
PS /home/ayman> $Assignments |
>> Where-Object {
>>   $_.ResourceId -eq $MDEApp.Id -or
>>   $_.ResourceId -eq $GraphApp.Id
>> } |
>> ForEach-Object {
>>   $PermissionName = $MDEPermissionIds[$_.AppRoleId.ToString()]
>>   if (-not $PermissionName) {
>>     $PermissionName = $GraphPermissionIds[$_.AppRoleId.ToString()]
>>   }
>> }
>> [PSCustomObject]@{
>>   Resource = $_.ResourceDisplayName
>>   Permission = $PermissionName
>>   AppRoleId = $_.AppRoleId
>> }
>> } |
>> Format-Table Resource, Permission, AppRoleId -AutoSize

Resource      Permission      AppRoleId
-----
Microsoft Graph User.RevokeSessions.All 77814698-2843-4681-8185-414602479901
Microsoft Graph User.PasswordProfile.ReadWrite.All 74d0344a-4b03-4bdc-b7d1-4d484632d014
Microsoft Graph ThreatHunting.Read.All 84d334f1-3a82-46f6-a009-5d457c54afcc
WindowsDefenderATP Machine.Isolate 489c852b-e44f-401a-bcfe-df21dc9999ee
WindowsDefenderATP Machine.Scan 489c852b-e44f-401a-bcfe-df21dc9999ee
WindowsDefenderATP Machine.Read.All 489c852b-e44f-401a-bcfe-df21dc9999ee
WindowsDefenderATP AdvancedQuery.Read.All 489c852b-e44f-401a-bcfe-df21dc9999ee
PS /home/ayman>

```

Also, you should assign your Logic App's managed identity (`sentinel-ransomware-response-playbook`) the **Microsoft Sentinel Responder** role in the Sentinel workspace: **Access control (IAM) → Add role assignment → Role = Microsoft Sentinel Responder** to add comment to incident.



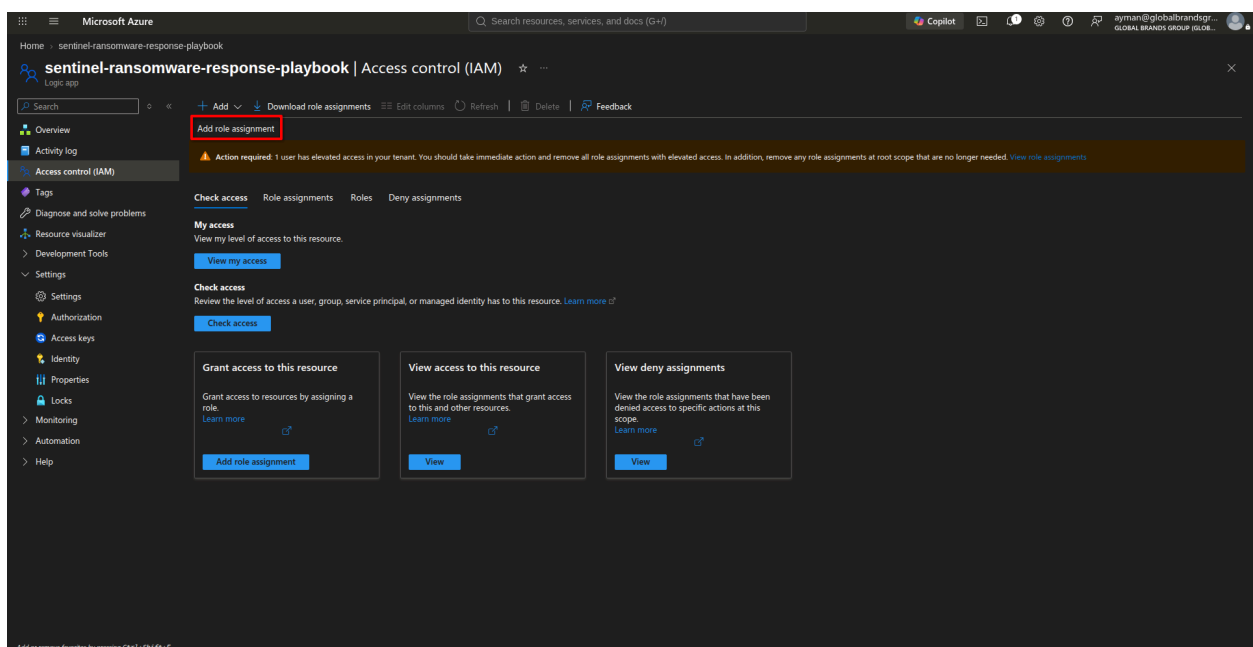


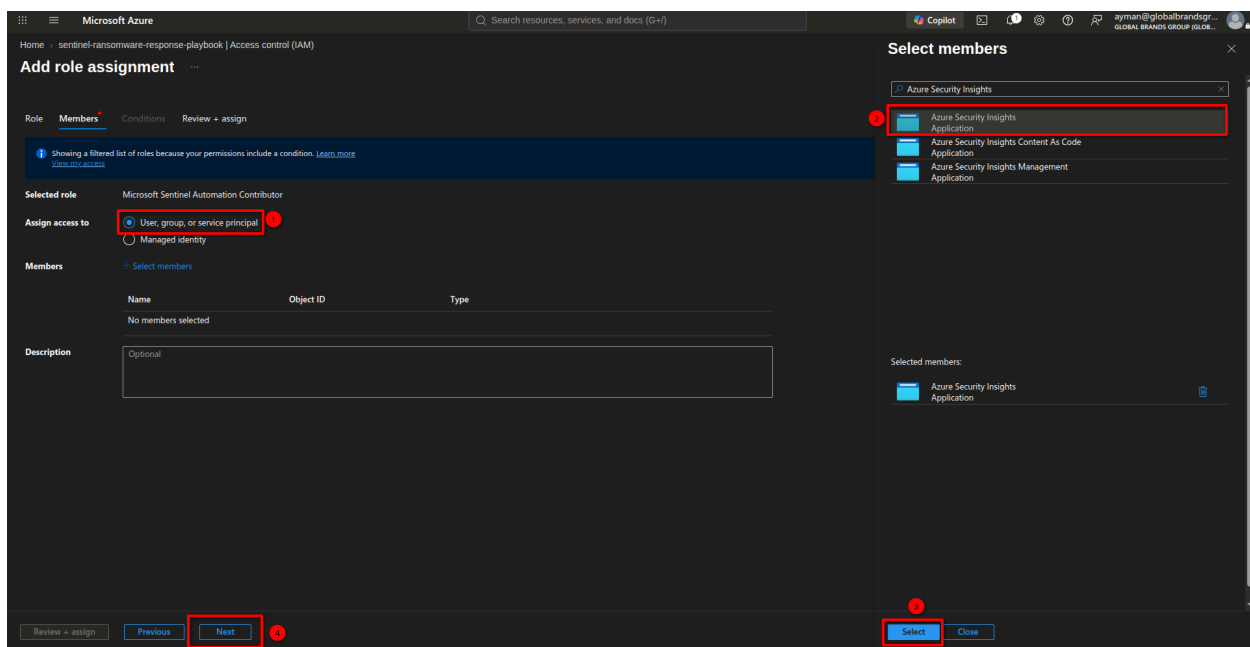
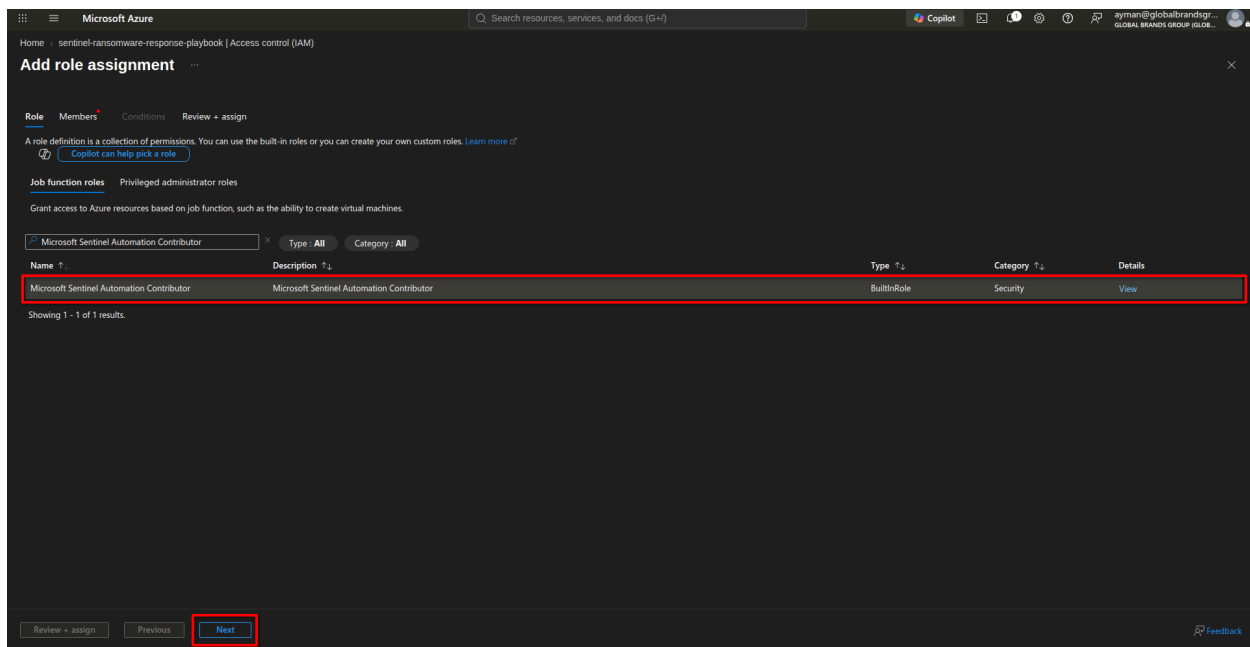
Role for Sentinel to trigger the Logic App (this is what's causing your 400 error)

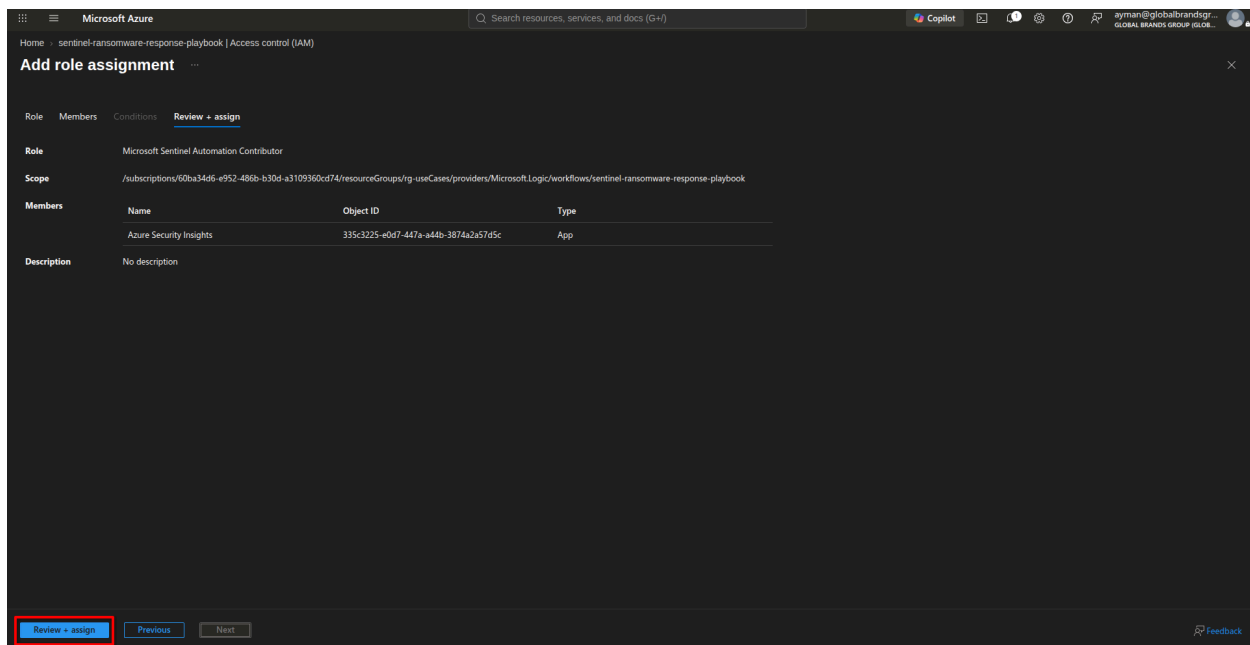
This is the opposite direction — Sentinel needs permission to *invoke* your Logic App. This is **not** assigned to the Logic App's managed identity — it's assigned to the **Microsoft Sentinel service principal** (called "Azure Security Insights" in Entra ID), scoped to the Logic App (or its Resource Group).

To fix this:

- Go to the **Logic App** → **Access control (IAM)** → **Add role assignment**
- Role: **Microsoft Sentinel Automation Contributor**
- Assign access to: **User, group, or service principal**
- Search for: **Azure Security Insights** (this is Sentinel's own service identity — not your user account, not the Logic App's managed identity)
- Select it → **Review + assign**

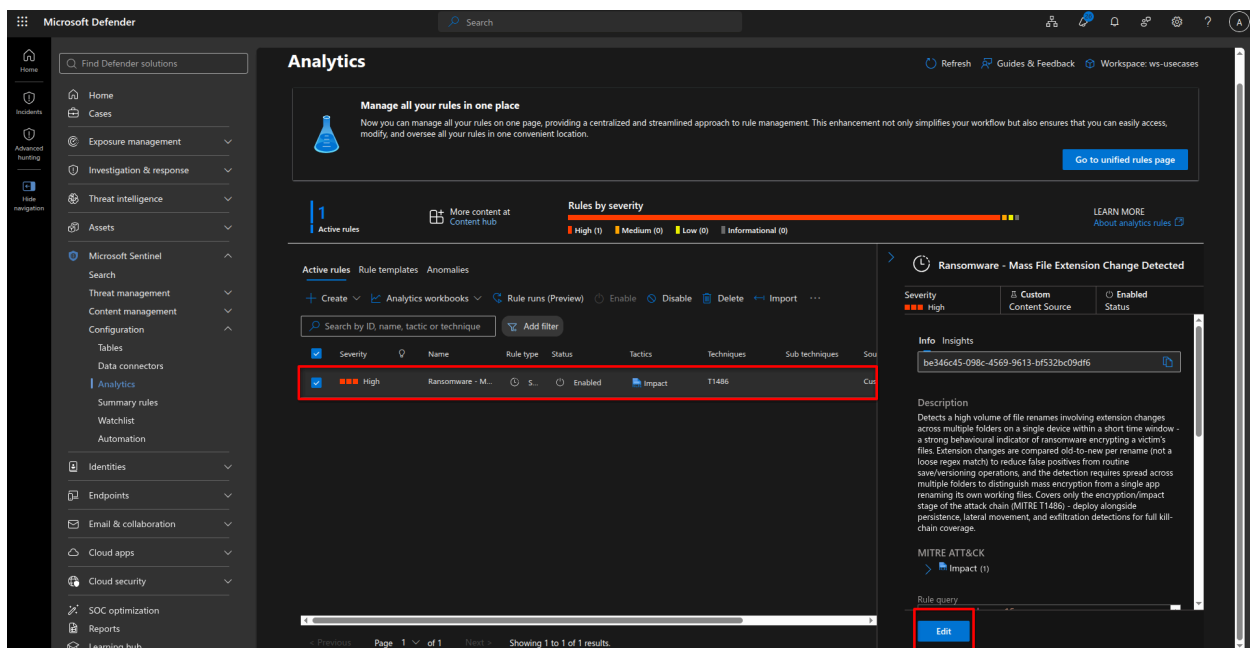


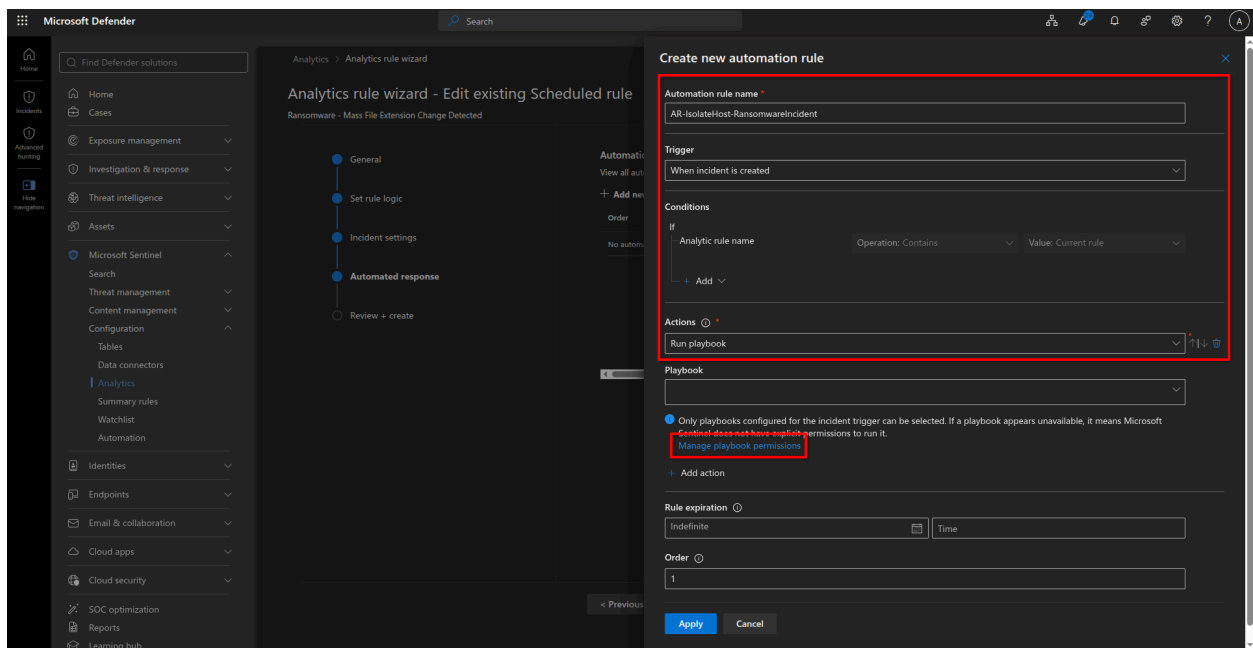
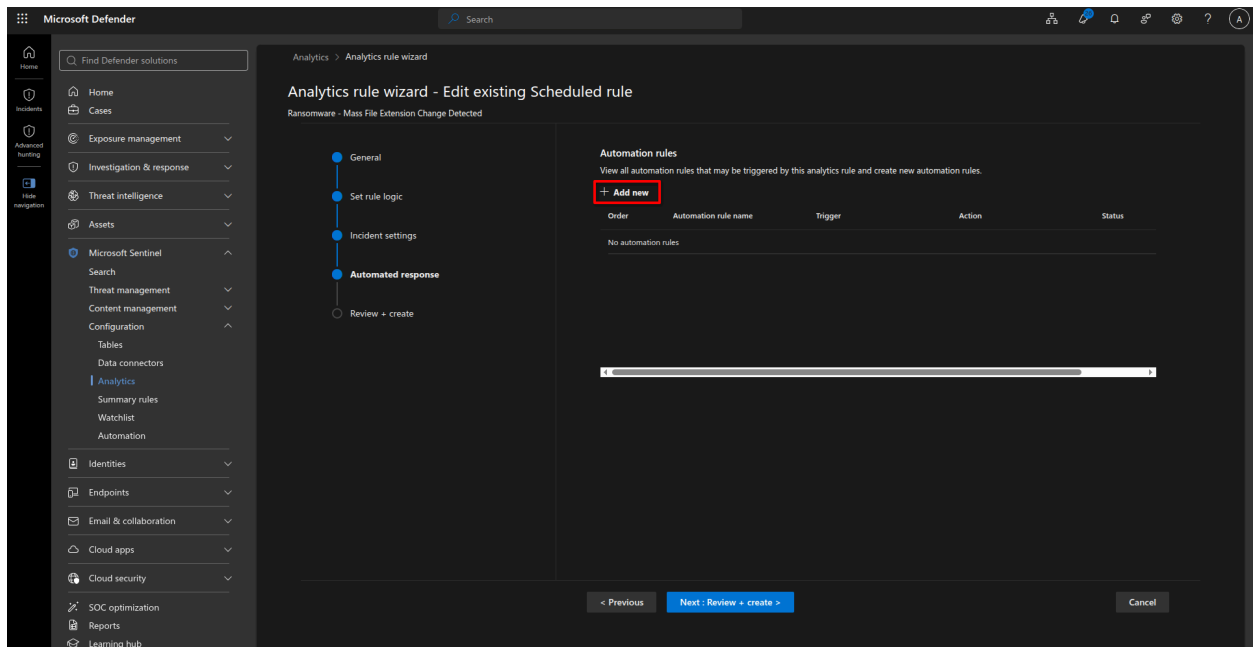


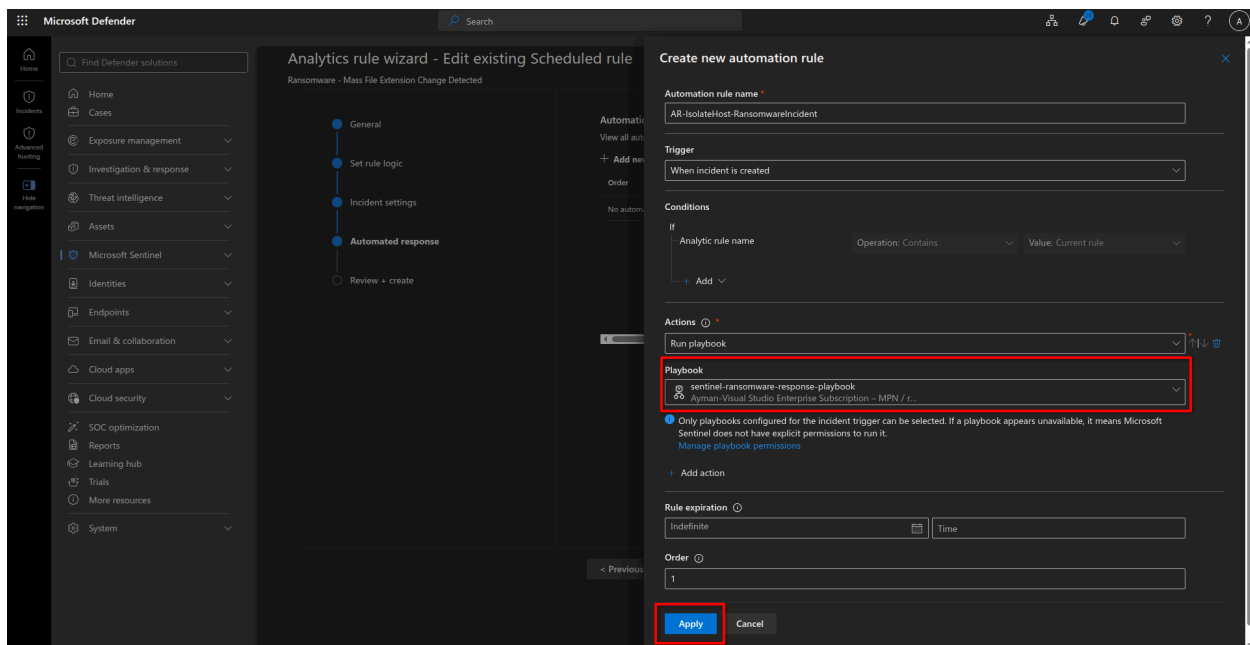
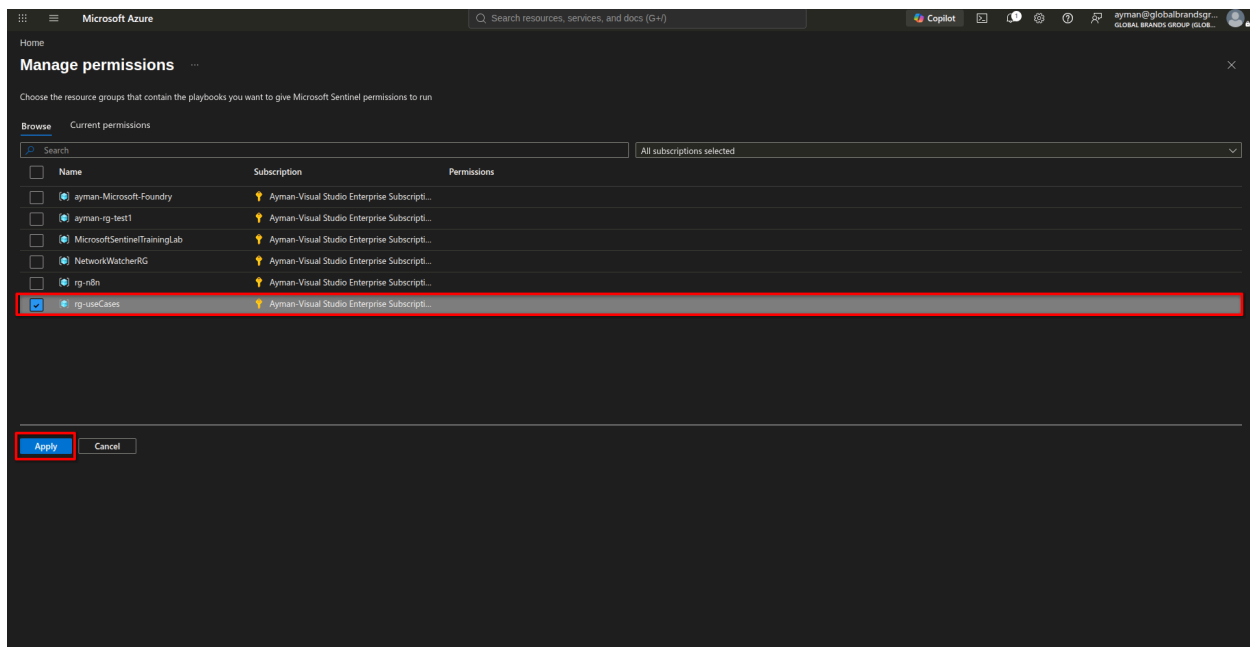


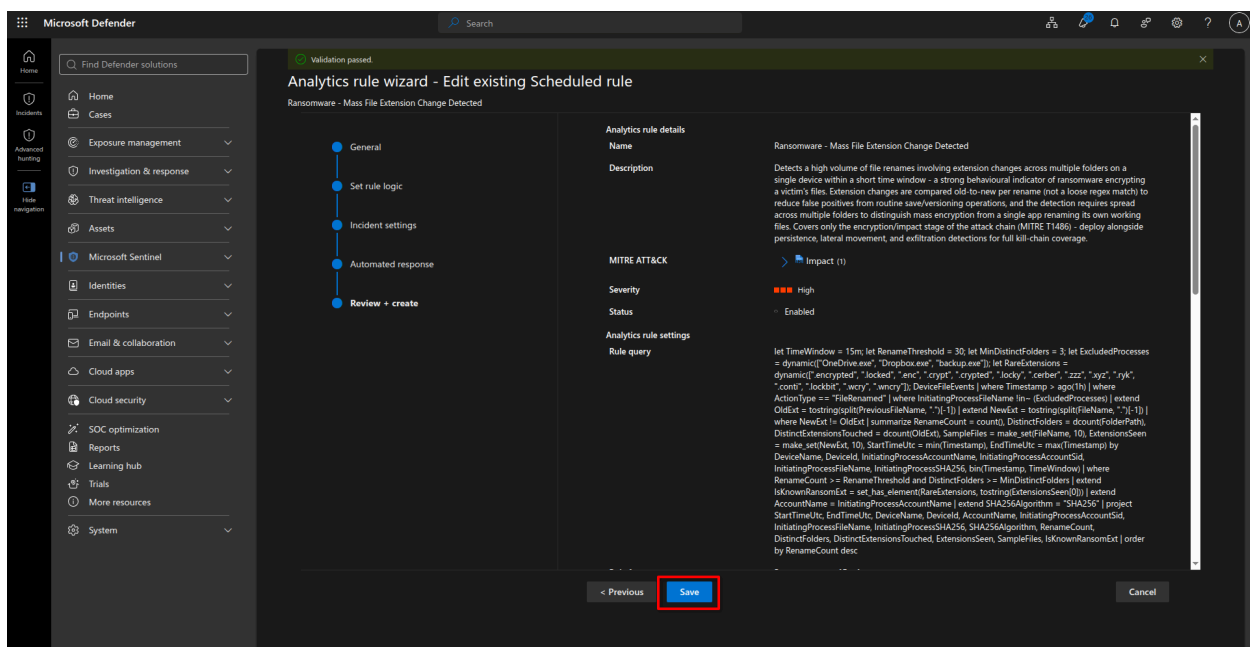
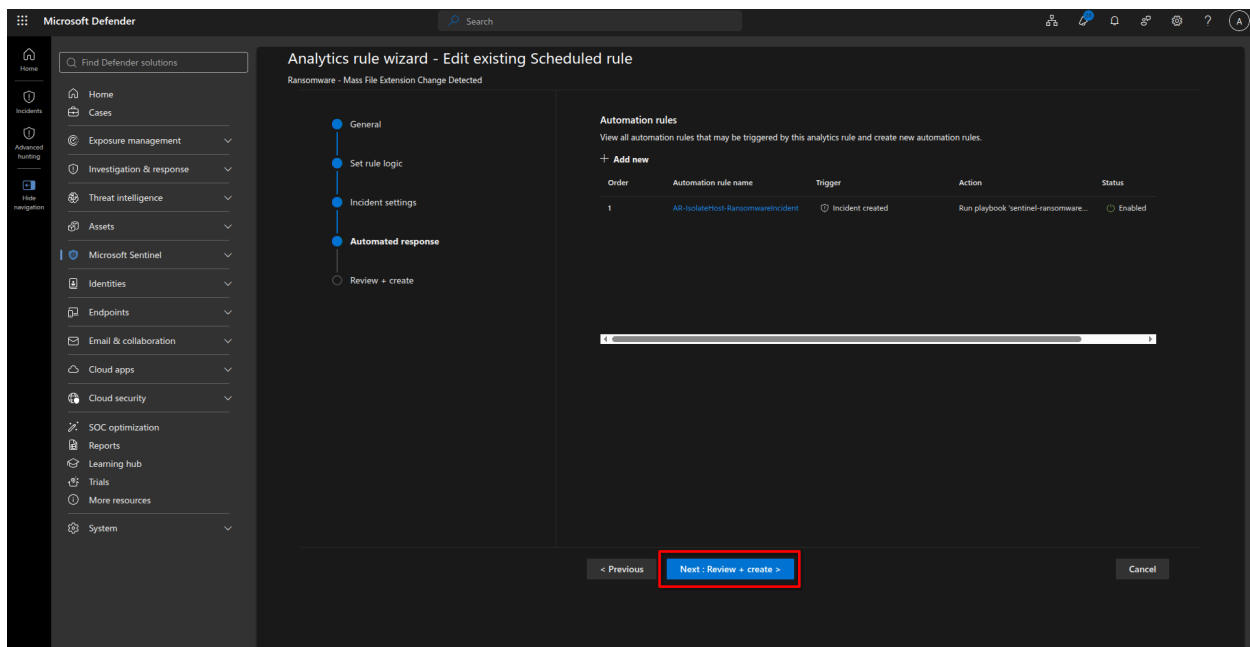
Step 3 — Automation rule

1. Create a Sentinel automation rule that fires when this incident type/severity is created.
2. Point it at a new Logic Apps playbook (build in Step 2).









Step 4 — Testing

To test the above case, I used a PowerShell script that will be detected by the analytic rule. Here is the script:

```
param(
    [ValidateRange(3, 100)]
```



```

[int]$FolderCount = 5,
[ValidateRange(1, 5000)]
[int]$FilesPerFolder = 12,
[string]$Extension = '.encrypted',
[string]$BasePath = ([Environment]::GetFolderPath('Desktop'))
)
$errorActionPreference = 'Stop'
if (-not $Extension.StartsWith('.')) { $Extension = ".$Extension" }
$runId = [guid]::NewGuid().ToString('N').Substring(0, 8)
$root = Join-Path $BasePath "run-$runId"
$sw = [System.Diagnostics.Stopwatch]::StartNew()
New-Item -ItemType Directory -Path $root -Force | Out-Null
$folders = foreach ($n in 1..$FolderCount) {
    $p = Join-Path $root ('folder{0:D2}' -f $n)
    New-Item -ItemType Directory -Path $p | Out-Null
    $p
}
$content = (1..32 | ForEach-Object { [guid]::NewGuid().ToString('N') }) -join [Environment]::NewLine
$seeded = 0
foreach ($folder in $folders) {
    foreach ($i in 1..$FilesPerFolder) {
        $name = 'doc-{0}-{1:D4}{2:D4}.txt' -f $runId, $i, (Get-Random -Maximum 10000)
        Set-Content -LiteralPath (Join-Path $folder $name) -Value $content -Encoding UTF8
        $seeded++
    }
}
$renamed = 0
$failures = 0
$files = Get-ChildItem -LiteralPath $root -Recurse -File -Filter '*.txt'
foreach ($file in $files) {

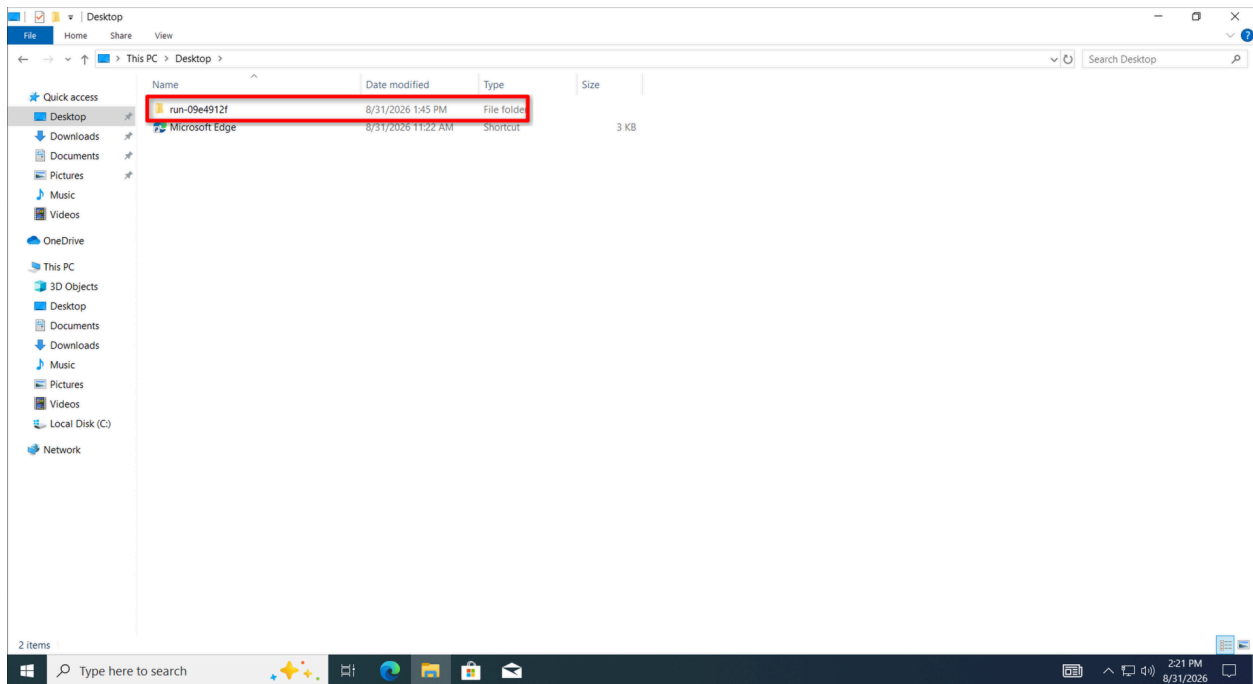
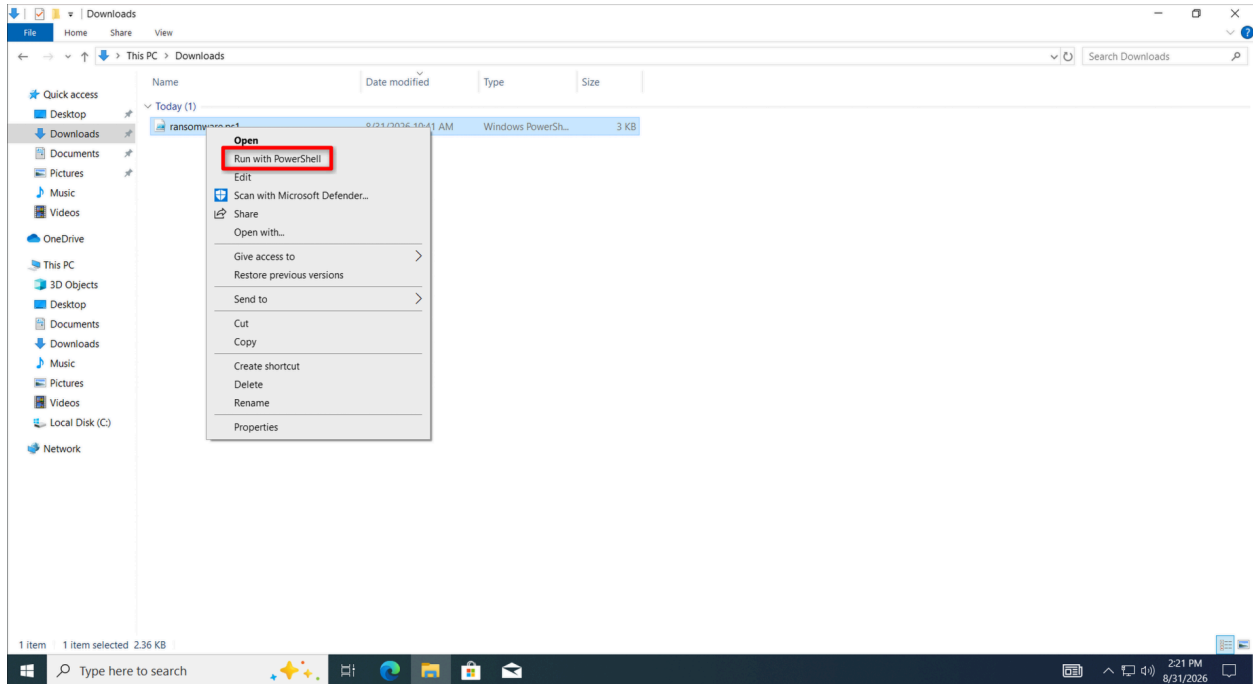
```

```

    try {
        $newName = [System.IO.Path]::ChangeExtension($file.Name, $Extension)
        Rename-Item -LiteralPath $file.FullName -NewName $newName
        $renamed++
    }
    catch {
        $failures++
        Write-Warning "Rename failed for '$($file.FullName)': $($_.Exception.Message)"
    }
}
$sw.Stop()
[pscustomobject]@{
    RunId          = $runId
    RootPath       = $root
    FoldersCreated = @($folders).Count
    FilesSeeded    = $seeded
    RenamesPerformed = $renamed
    RenameFailures = $failures
    TargetExtension = $Extension
    ElapsedSeconds = [math]::Round($sw.Elapsed.TotalSeconds, 2)
} | Format-List
Write-Host ''
Write-Host 'Expected Sentinel alert fields:' -ForegroundColor Cyan
Write-Host ("  InitiatingProcessFileName : powershell.exe")
Write-Host ("  RenameCount                  : {0}    (rule threshold: >= 30)" -f $renamed)
Write-Host ("  DistinctFolders                  : {0}    (rule threshold: >= 3)" -f @($folders).Count)
Write-Host ("  ExtensionsSeen                  : ['{0}']" -f $Extension.TrimStart('.'))
Write-Host ''

```

Write-Host 'Allow 5-20 minutes for DeviceFileEvents ingestion, then run your KQL.'



Microsoft Defender

Incidents > Mass file rename on desktop-r5b1v7 by muayman

ID 800: Mass file rename on desktop-r5b1v7 by muayman

High | Active | Unassigned | Unclassified | Last update time: Aug 31, 2026 2:01 PM

Attack story | Alerts (1) | Activities (3) | Assets (2) | Investigations (0) | Evidence and Response (0) | Summary

Filters: Severity: Any | Status: Any | Service sources: Any | Add filter

Detection & Categories

Active alerts: 1/1

Categories: 1

First activity: Aug 31, 2026 12:56:21 PM

Last activity: Aug 31, 2026 1:56:21 PM

Creation time: Aug 31, 2026 2:01:28 PM

Alerts

Aug 31, 2026 12:56 PM | New

Mass file rename on desktop-r5b1v7 by muayman

desktop-r5b1v7 | muayman

Incident graph

Layout | Group similar nodes | Entity types

desktop-r5b1v7

muayman

Communication | Association

Report incident inaccuracy

Microsoft Defender

Incidents > Mass file rename on desktop-r5b1v7 by muayman

ID 800: Mass file rename on desktop-r5b1v7 by muayman

High | Active | Unassigned | Unclassified | Last update time: Aug 31, 2026 2:01 PM

Attack story | Alerts (1) | Activities (3) | Assets (2) | Investigations (0) | Evidence and Response (0) | Summary

Refresh | Export | Add comment

Filter set: Origin: This incident | Add filter | Reset all

Activity started	Title	Type	Category	Target
Aug 31, 2026 2:01 PM	Machine desktop-r5b1v7 was automatic...	Comment	Comment	
Aug 31, 2026 2:01 PM	Account muayman was disabled, its pass...	Comment	Comment	
Aug 31, 2026 2:01 PM	Alert 'Mass file rename on desktop-r5b1...	Alert link	Correlations	

Activity Details

Activity started: Aug 31, 2026 2:01 PM

Type: Comment

Title: Comment

Category: Comment

Performed by: Playbook-sentinel-ransomware-response-playbook

Trigger: Manual

Activity status

Comments

Machine desktop-r5b1v7 was automatically isolated after Microsoft Graph Advanced Hunting resolved MDE Device ID 8c7372425b9fed412dd6a6209b05f277af6c28ee3. A full Defender Antivirus scan was started.

Microsoft Defender

Incidents > Mass file rename on desktop-ri5b1v7 by muayman

ID 800: Mass file rename on desktop-ri5b1v7 by muayman

High | Active | Unassigned | Unclassified | Last update time: Aug 31, 2026 2:01 PM

Attack story | Alerts (1) | **Activities (3)** | Assets (2) | Investigations (0) | Evidence and Response (0) | Summary

Refresh | Export | Add comment

Filter set: **Origin: This incident** | Add filter | Reset all

Activity started	Title	Type	Category	Target
Aug 31, 2026 2:01 PM	Machine desktop-ri5b1v7 was automatic...	Comment	Comment	
☒ Aug 31, 2026 2:01 PM	Account muayman was disabled, its pass...	Comment	Comment	
Aug 31, 2026 2:01 PM	Alert 'Mass file rename on desktop-ri5b1...	Alert link	Correlations	

Activity Details

Activity started: Aug 31, 2026 2:01 PM

Title: Comment

Performed by: Playbook-sentinel-ransomware-response-playbook

Activity status

Comments

Account muayman was disabled, its password was reset (temporary password change required at next sign-in), and all refresh tokens and active sessions were revoked.

Microsoft Defender

Device inventory > desktop-ri5b1v7

desktop-ri5b1v7

No known risks | Criticality: None | Active

Overview | Incidents and alerts | **Timeline** | Configuration management | Security recommendations | Inventories

Copy to clipboard | Export | Search

Event time	Event
Aug 31, 2026 2:01:42.449 PM	An inbound connection from remote IP 4.207.44.74:443 to local IP 192.168.2...
Aug 31, 2026 2:01:42.449 PM	An inbound connection from remote IP 135.18.128.208:443 to local IP 192.16...
Aug 31, 2026 2:01:41.263 PM	svchost.exe established connection with 135.18.128.208:443 (fe2crupdate.mil...
☒ Aug 31, 2026 2:01:40.713 PM	Event of type [IsolateResponse] observed on device
Aug 31, 2026 2:01:40.217 PM	mpcmdrun.exe created process conhost.exe
Aug 31, 2026 2:01:40.190 PM	msmpeng.exe created process MpCmdRun.exe
Aug 31, 2026 2:00:44.011 PM	backgroundtaskhost.exe failed to establish connection with 20.190.177.149:4...
Aug 31, 2026 2:00:44.010 PM	backgroundtaskhost.exe failed to establish connection with 20.190.177.23:44...
Aug 31, 2026 2:00:44.010 PM	backgroundtaskhost.exe failed to establish connection with 20.190.177.21:44...
Aug 31, 2026 2:00:40.779 PM	Outbound SSL connection from 192.168.228.131:51112 to 4.207.44.71:443
Aug 31, 2026 2:00:40.469 PM	An inbound connection from remote IP 4.207.44.71:443 to local IP 192.168.2...

Event of type [IsolateResponse] observed on device

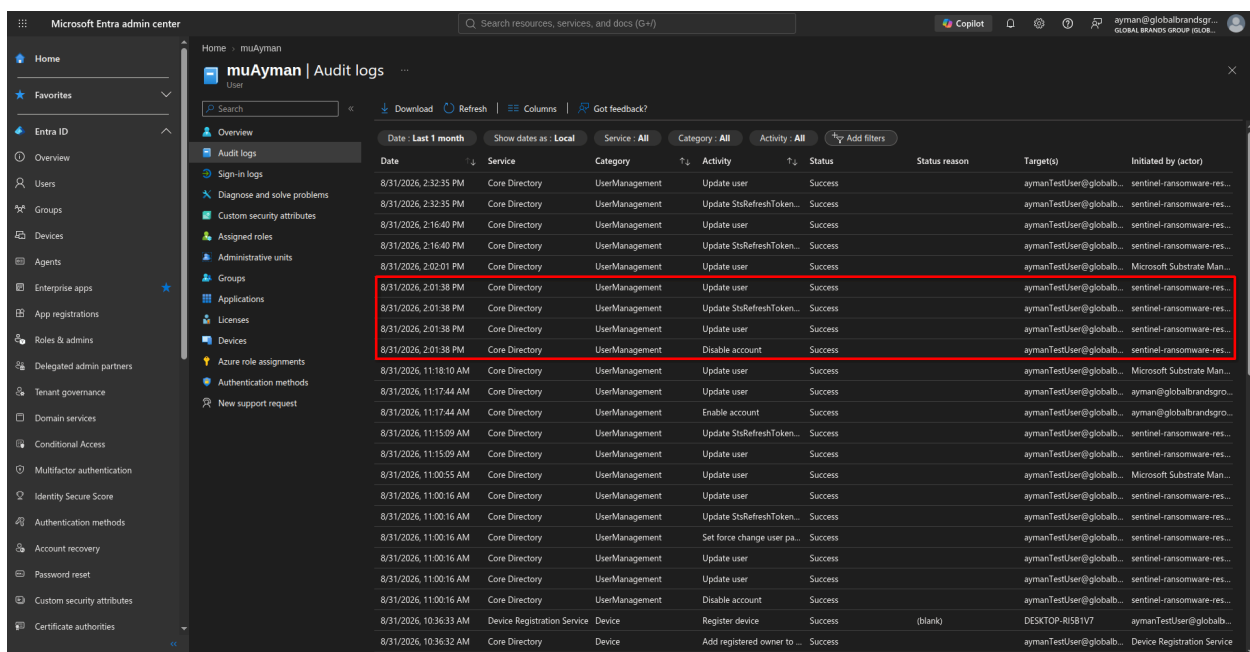
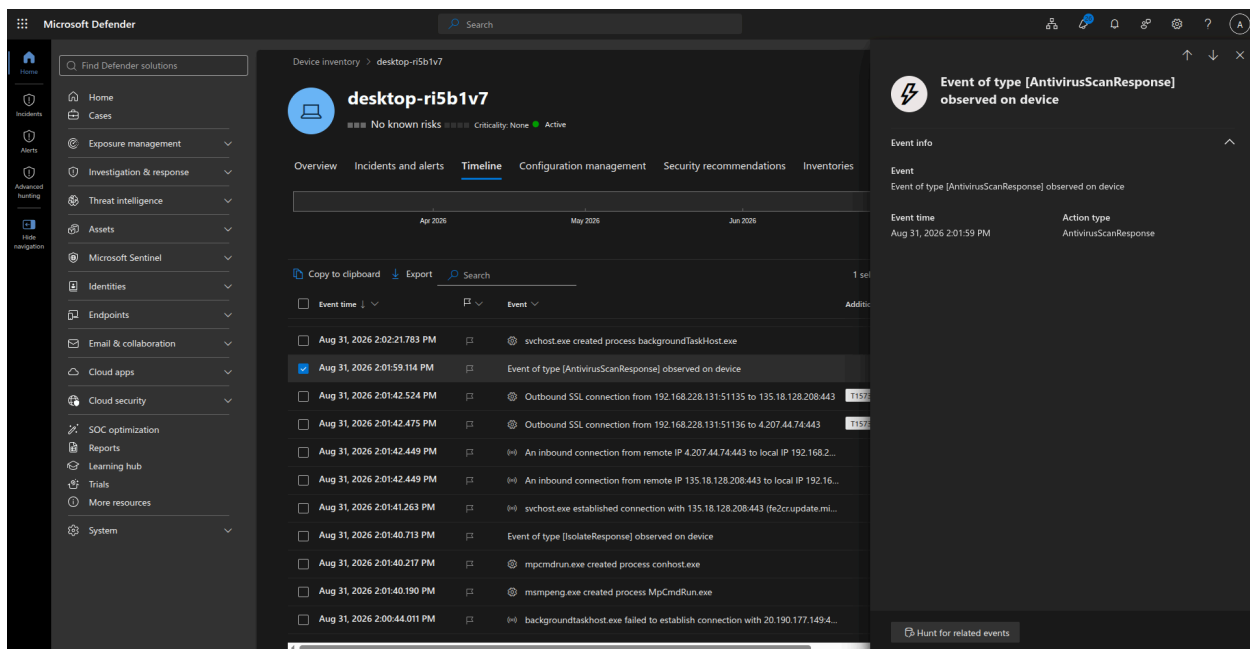
Event info

Event: Event of type [IsolateResponse] observed on device

Event time: Aug 31, 2026 2:01:40 PM

Action type: IsolateResponse

Hunt for related events



Issue I faced:

Summary of the behavior observed

Setup: A PowerShell script simulating ransomware creates ~60 `.txt` files and renames them to a suspicious extension (e.g. `.encrypted`), on a device onboarded to MDE.

What happened:

1. Initial test runs worked as expected — all 60 file creations and renames were fully visible in `DeviceFileEvents`, and the device timeline showed the aggregated "powershell.exe renamed 60 files" card tagged `T1486: Data Encrypted for Impact`.
2. After running the *same script with the same target extension* repeatedly over ~1 day of testing, `DeviceFileEvents` stopped logging most of the `FileRenamed` events for that pattern — only 1–2 rows showed up instead of 60 — even though:
 - The files were genuinely renamed on disk (confirmed locally).
 - `FileCreated` events for the same run logged completely and reliably (all 60).
 - The device timeline UI still correctly showed "60 files renamed" / T1486, sourced from MDE's own backend detection engine.
3. This detection card was *not* backed by a queryable event: it never appeared in `AlertInfo`, and a global `search *` for a hypothesized `FileRenameAggregatedByProcess` action type returned nothing anywhere in the workspace. So the portal-level "60 renamed" summary is generated and displayed by MDE's backend independent of what's exposed to Advanced Hunting.
4. **Root cause identified:** changing the target extension to something never used before (`.wncry`) caused full `FileRenamed` telemetry to resume immediately. Switching back to `.encrypted` also worked fine. This confirms MDE is **deduplicating/suppressing repeated `FileRenamed` events tied to a specific, previously-seen rename signature** (same process + same extension pattern reused many times on the same device) — not a general volume throttle, not a sensor fault, not a connector/ingestion outage.

Net conclusion: The raw rename telemetry (`DeviceFileEvents`) is reliable *the first several times* a given rename pattern occurs on a device, but becomes unreliable once that exact pattern (extension + process) has been repeated heavily during testing. `FileCreated` events did not show this suppression. The MDE portal's own ransomware behavioral detection is unaffected by this and continues firing correctly even when the underlying `FileRenamed` events are suppressed — but that detection is not currently exposed through any Advanced Hunting table available to you, so it can't be used directly as a Sentinel analytic-rule data source.

You might also encounter an error when the Logic App runs, because the "Add Comment" node in both the Device Actions and User Actions executes at the same time, which causes a conflict.

Step 5 — Validation and Final Result

After completing the detection rule, Logic App playbook, permissions, and automation rule, the complete ransomware response workflow was validated end-to-end.

The final workflow is:

```
Ransomware activity occurs
↓
MDE detects suspicious mass file renames
↓
Sentinel analytic rule identifies the behavior
↓
Sentinel creates a High/Critical incident
↓
Automation rule triggers the Logic App
↓
Extract host and account entities
↓
Resolve the MDE Device ID
↓
Isolate the affected device
↓
Start a full Defender Antivirus scan
↓
Disable User Account
↓
Reset the compromised user's password
↓
Revoke active sign-in sessions
↓
Add response results to the Sentinel incident
```

The playbook also contains fail-safe conditions. If a valid MDE Device ID cannot be resolved, the workflow does not attempt isolation against an unknown device. Instead, it records the failure in the incident and leaves the device for manual investigation.

This provides a safer automated-response model while still allowing the workflow to operate when Sentinel's host entity is incomplete during the initial automation-

rule execution.

Final Outcome

The completed implementation provides an automated ransomware-response workflow that connects:

- **Microsoft Defender for Endpoint** for endpoint detection, isolation, and antivirus scanning.
- **Microsoft Sentinel** for detection, incident creation, entity processing, and incident tracking.
- **Microsoft Graph Advanced Hunting** for resolving an MDE Device ID when Sentinel does not provide one.
- **Microsoft Entra ID / Microsoft Graph** for password reset and session revocation.
- **Logic Apps** for orchestrating the complete response.

The testing also demonstrated an important limitation when using `DeviceFileEvents` as the direct telemetry source for ransomware detection: repeated testing of the same rename pattern can result in reduced raw `FileRenamed` telemetry, while MDE's own behavioral ransomware detection can continue to identify the activity.

Therefore, the analytic rule should be treated as one layer of the detection strategy rather than the sole ransomware detection mechanism. In a production environment, it should be complemented by native Defender detections and additional signals covering persistence, credential access, lateral movement, and other stages of the attack chain.

Case 1 — Final Status

Detection:  Implemented

Sentinel Incident Creation:  Implemented

Automation Rule:  Implemented

MDE Device Isolation:  Implemented

Full Antivirus Scan:  Implemented

MDE Device ID Fallback Resolution:  Implemented

Account Disabled:  Implemented

Account Password Reset:  Implemented

Session Revocation:  Implemented

Incident Comments / Audit Trail:  Implemented

Managed Identity Permissions:  Implemented

End-to-End Testing:  Completed

Lessons Learned

The implementation highlighted several important considerations for Sentinel automation:

1. **Entity data may not be immediately complete** when an automation rule executes directly after incident creation.
2. **A fallback resolution mechanism** is therefore valuable for reliable automated containment.
3. **Containment actions should always validate their target identifier** before making API requests.
4. **Sequential execution matters:** the antivirus scan is started only after successful device isolation.
5. **Automation identities should follow least-privilege principles** and receive only the permissions required by the playbook.
6. **Raw telemetry and portal-level behavioral detections are not always equivalent.** A detection visible in the MDE portal may not necessarily exist as a directly queryable Advanced Hunting event.
7. **Testing should use controlled simulation assets** and should account for telemetry behavior observed during repeated testing.